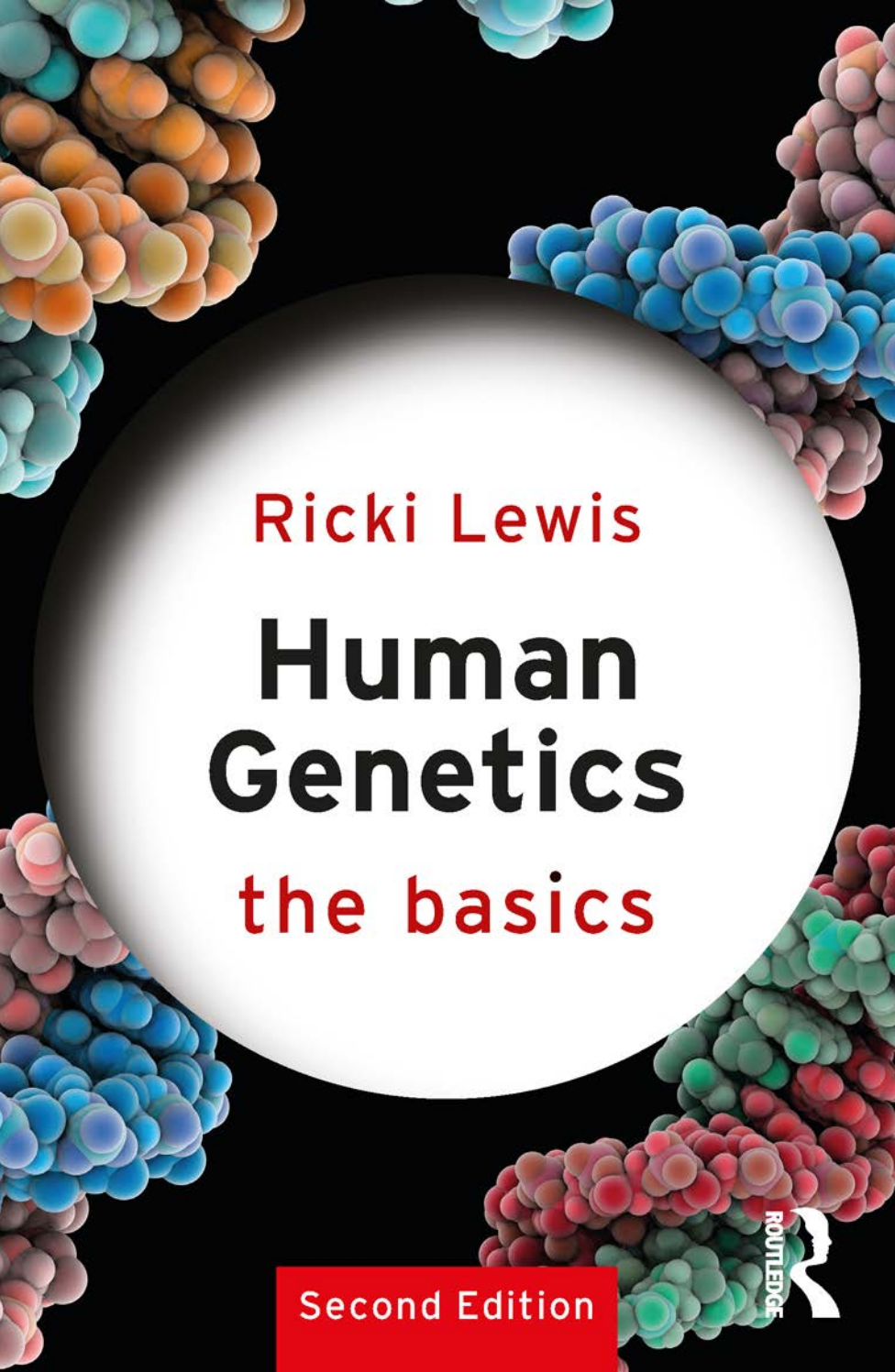




Ricki Lewis

Human Genetics

the basics

Second Edition



HUMAN GENETICS

THE BASICS

Human genetics has blossomed from an obscure biological science and explanation for rare disorders to a field that is profoundly altering health care for everyone. This thoroughly updated new edition of *Human Genetics: The Basics* provides a concise background of gene structure and function through the lens of real examples, from families living with inherited diseases to population-wide efforts in which millions of average people are learning about their genetic selves. The book raises compelling issues concerning:

- the role of genes in maintaining health and explaining sickness;
- genetic testing, gene therapy, and genome editing;
- the common ancestry of all humanity and how we are affecting our future.

Written in an engaging narrative manner, this concise introduction is an ideal starting point for anyone who wants to know more about genes, DNA, genomes, and the genetic ties that bind us all.

Ricki Lewis is Adjunct Professor of Medical Education at Alden March Bioethics Institute, Albany Medical Center, US. She has a PhD in Genetics, writes the DNA Science blog at the Public Library of Science, is a genetic counsellor, and is the author of several popular titles.

The Basics

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HUMAN GENETICS

THE BASICS

Second Edition

Ricki Lewis

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Figure 2.1

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Figure 5.1

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Figure 5.2

Ernesto del Aguila III, National Human Genome Research Institute (NHGRI)

ABBREVIATIONS

A	adenine
AAV	adeno-associated virus
ADA	adenosine deaminase
AIDS	acquired immune deficiency syndrome
AV	adenovirus
C	cytosine
CAR	chimeric antigen receptor
Cas9	CRISPR associated protein 9
CF	cystic fibrosis
CFTR	cystic fibrosis transmembrane regulator
CRISPRs	clustered regularly interspaced short palindromic repeats
DMD	Duchenne muscular dystrophy
DNA	deoxyribonucleic acid
DOE	Department of Energy
ER	endoplasmic reticulum
FH	familial hypercholesterolemia
G	guanine
GAN	giant axonal neuropathy
GWAS	genome-wide association study
HD	Huntington's disease
HDL	high-density lipoproteins
HH	hereditary hemochromatosis

HIV	human immunodeficiency virus
HNPCC	hereditary nonpolyposis colon cancer
iPS	induced pluripotent stem
LDL	low-density lipoproteins
LPLD	lipoprotein lipase deficiency
MRC	Medical Research Council
NHS	National Health Service
NIH	National Institutes of Health
OTC	ornithine transcarbamylase
PKU	phenylketonuria
PMI	Precision Medicine Initiative
RNA	ribonucleic acid
RPE	retinal pigment epithelium
SCID	severe combined immune deficiency
SNP	single nucleotide polymorphism
T	thymine
TALENs	transcription-activator-like effector nucleases
TB	tuberculosis
U	uracil
VLDL	very-low-density lipoproteins
WHO	World Health Organization
XIAP	X-linked inhibitor of apoptosis
ZFN	zinc finger nuclease

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FROM MENDEL'S PEAS TO DOUBLE HELICES

The Rosetta Stone is an ancient Egyptian artifact that made it possible to glean meaning from hieroglyphic writings. The content of the etchings on the stone tablet proved not as memorable as the connections of the symbols to each other. The stone, a statement from Ptolemy V set down in 196 BC, held instructions to erect statues in temples and spelled out a particular tax repeal—news of the day, more or less. But the tablet depicted the messages in three languages—two Egyptian and classical Greek. The lasting knowledge was to come in the comparison of the three languages, and in the ability to go from one to another.

FROM GENES TO GENOMES

Like the Rosetta Stone, the human **genome** is a set of instructions, also for building a temple of sorts—a human body. Two copies of our genome are tightly entwined inside each of our **cells**, which are the microscopic building blocks of our bodies. The three languages of the genome are the informational molecules **DNA**, **RNA**, and **protein**, presented in Figure 1.1 and Table 2.2. A **gene** is the information in a sequence of DNA building blocks that is transcribed (copied) into a molecule of RNA, which is translated into (used to build) a protein. The protein is responsible for, or contributes to, the

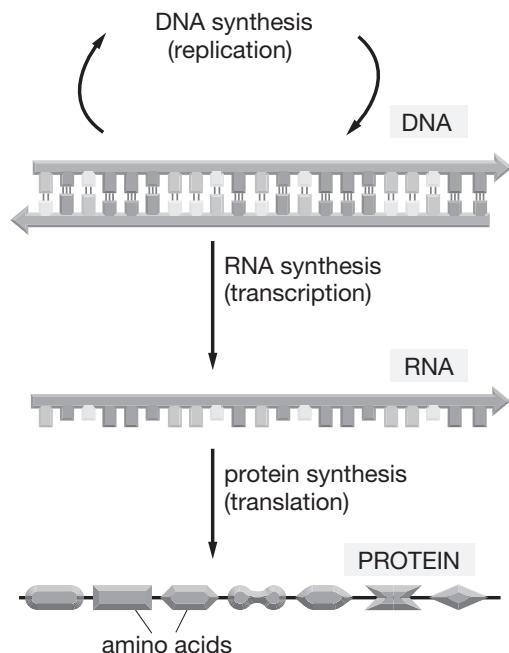


Figure 1.1 From DNA to RNA to protein.

Source: *Molecular Biology of the Cell*, 6th edition, Figure 1–4, p. 4

trait or illness associated with the gene. Although only about 1.5 percent of the 3.2 billion bases of a human genome encodes proteins, these DNA sequences are very important—abnormal or absent specific proteins account for about 85 percent of inherited diseases. The protein-encoding part of the genome is called the **exome**. If the genome is compared to the Internet in terms of amount of information, then the exome is a little like accessing only a tiny part of that information.

A cell can transcribe a gene's worth of DNA sequence information into RNA, and then translate the RNA into a specific sequence of protein building blocks, the **amino acids**. These two processes are appropriately called **transcription** and **translation**. Proteins provide basic characteristics, such as the ability of blood to clot or for the

body to extract energy from foods, as well as contribute to harder-to-define qualities such as talents and personality.

The DNA sequence of a particular gene can vary slightly from person to person. Most gene variants have no effect on characteristics, including health, like the meaning of a written sentence being clear even if it contains one or two typos. Just as a more serious keying error, such as removing words or changing an important letter—bloke to broke, for example—can greatly alter the meaning of a sentence, some changes in a gene's DNA sequence can harm health. These types of variants are traditionally termed **mutations**.

Different types of cells under different circumstances transcribe and translate different sets of genes. A fat-stuffed adipose cell, which stores energy, utilizes a very different gene set than a long, lean muscle cell, which expends energy to fuel contraction. The gene expression (activation) patterns of these two cell types—which gene sets they transcribe—would differ between a well-fed athlete and a person who is starving.

A human genome is also like the Rosetta Stone in that the bigger genetic picture may be more valuable than the many individual bits of information it contains. A genome-wide view of ourselves reveals that our DNA sequences are 99 percent alike, uniting humanity. Yet within that other less than 1 percent of our DNA sequences lie the intriguing differences that are responsible for our individuality and our illnesses.

The six chapters of this book explain what genes are, how they provide instructions for the functioning of the human body, and how they can reveal our past. Chapter 1 traces the history of the science of genetics, from thoughts on family resemblances to discovery of the underlying laws of inheritance, to cracking the genetic code and sequencing and deciphering human genomes. Chapter 2 paints the landscape of genetics from the DNA level to the population level through the example of a disease, cystic fibrosis. Chapter 3 explores how considering genes is making the practice of medicine both more precise and more personal, focusing on cardiovascular disease, cancer, and infectious disease. Genetic testing teamed with information science has shrunk times to diagnosis from years to minutes. It is the subject of Chapter 4. Adding, removing, or replacing genes is covered in Chapter 5. Finally, Chapter 6 is both a look back into our past and glimpses of our future, courtesy of our DNA. The "What Would You

Do?" feature ending each chapter presents dilemmas that arise from genetic research, information, applications, and technologies.

AGRICULTURE IS APPLIED GENETICS

Today, primary school students are familiar with the structure and function of DNA. However, the rules of inheritance were deduced over many years, from experiments based on observations of trait variations. It wasn't until 1865 that "the monk in the garden" Gregor Mendel carried out his famous experiments on pea plants that revealed the two basic laws of heredity, and decades before his contribution was acknowledged. However, the science of genetics actually began millennia earlier, when people learned how to control the reproduction of the species they needed to survive.

Agriculture provided the numbers from which researchers eventually deduced the laws of inheritance. The controlled breeding that is the backbone of farming, with its intense selection of coveted traits and removal of undesirable ones, is actually a huge genetic experiment.

From 15,000 to 8,000 years ago, agriculture arose independently in different parts of the world. The time of its origin in different places depended upon climate, geography, the movements of people who "hunted and gathered," and their growing ability to control their environment. Agriculture grew out of astute observations. Imagine a group of nomads settling overnight in an area with edible plants. They ate, perhaps took food with them when they left, and inadvertently left seeds behind from tasty fruits and grains. A year later, the people returned and noted the same types of plants growing. At some point, someone must have observed a seedling sprouting a tiny stem and linked the seeds of one season to the bounty of the next. Then, people began leaving seeds on purpose, saving the seeds from the most robust plants each season to plant the crops the next year. Domestication of wild animals such as sheep and goats paralleled the control over plant breeding. Eventually, people must have realized that if they could grow crops, they needn't move to find food. Settlements eased living and provided some safety.

The first farmers were insightful observers of natural variation. Consider the origin of corn, also known as maize and *Zea mays*, in southern Mexico about 9,000 years ago. People began, unknowingly

at first, fashioning the seeds of a wild grass that still grows there, called teosinte, into corn.

Corn and its ancestor, teosinte, do not look much alike. In contrast to a corn plant's majestic height, with its seeds tucked into paired protective husks and its stalk firm enough to withstand a strong wind, teosinte is stout and branched, each outgrowth holding a small, ragged ear. Corn kernels are packed with protein, oil, and starch, in a soft covering that teeth can easily tear. In contrast, teosinte kernels are small, trapped in a hard fruitcase, and form two scraggly rows in a puny cob. A corn husk hugs the succulent kernels in their neat rows; pebble-like teosinte seeds simply drop and scatter. Teosinte's tough seeds, though, traverse an animal's digestive tract unscathed, plopping from the anus into hopefully fertile ground, where they start a new plant.

Corn could not have evolved naturally, because its seeds would never have survived the journey through the digestive tracts of herbivores. So how has corn, with its shrouded seeds, persisted? We helped it.

Archeological evidence indicates that people once grated teosinte seeds into flour. In doing so, someone must have come upon an unusual plant that had larger or softer kernels. Perhaps from knowing how to cultivate other crops from early agriculture in Mexico—yams, squash, cotton, peanuts, and peppers—the people knew to save the seed and plant it to propagate the desirable traits. Over time, with continual selection at each generation for the most palatable and digestible kernels, teosinte became corn. However, in the process, the new crop sacrificed its self-sufficiency, because people would always have to remove the kernels from the husks and plant them.

Selecting seeds to perpetuate a valued trait is recognition of genetics. Today, we know that the differences between corn and teosinte are due to the actions of only a few genes. One, called *tb1*, controls body form—the gene variant in teosinte confers the shrub-like shape, whereas the variant in corn suppresses an ancestral tendency to grow laterally, resulting in the cornstalk. Another gene controls the pattern of deposition of silica (the main component of sand) and lignin (a carbohydrate) in the kernels, which underlies the differences in hardness between the seeds of corn and teosinte.

Comparing differences in gene sequences among modern cultivars of corn and teosinte and considering mutation rates and geographical

distribution suggest a single origin for corn about 9,000 years ago, in southern Mexico. Consistent with the DNA story are preserved ears that track the changes that accompanied corn's human-aided evolution from teosinte. Once corn had been cultivated, it rapidly replaced its forebear. Evidence of tools to process corn date from 8,700 years ago—tiny crevices in the tools bore starch characteristic of modern corn, but not of teosinte.

Agriculture began in different parts of the world within a few thousand years and transformed humanity. The hunter-gatherer lifestyle gave way to settlements, and people came to rely on fewer varieties of foods as crops abounded. So, too, were the seeds sowed for the birth of the field of genetics.

MENDEL'S EXPERIMENTS—THE MYSTERY OF THE MISSING TRAITS

The two basic laws of inheritance emerged from a mystery in gardens and farms: traits that vanish and then reappear. Specifically, crossing plants where each has a different version of a trait—flower color, for example—produces hybrids with flowers of only one of the colors. Cross the hybrids, or allow them to self-fertilize, and, in the next generation, both flower colors appear. How does a trait disappear in the hybrid generation, and then reappear when the hybrids are crossed or selfed? Gregor Mendel (1822–1884) used math to unmask the hidden traits.

MENDEL THE MAN

Johann Mendel was born in 1822 to parents who were serfs, in a tiny village in northern Moravia. Johann became “Gregor” in 1843, and Moravia became Czechoslovakia in 1945. Mendel was a good student, but attended university only briefly before entering the monastery of the Augustinian Friars at Saint Thomas in Brunn (now Brno), where he studied theology, history, Greek, law, and teaching. After being ordained in 1847, Mendel taught math and Greek at a local school, but failed his teacher certification exam because he couldn't classify mammals. For a short time, as a parish priest, he visited hospitalized patients, but became so squeamish, in those days before anesthesia, that he quickly ruled out a career in medicine.

Despite his ineptitude at classifying mammals, Mendel proved himself a gifted teacher, and the monks sent him to the University of Vienna. There he studied botany, physics, and math. Still without a degree, Mendel returned to Brunn in 1853 to teach natural history and physics, which he did for 14 years. Shortly after his arrival, he retook the teaching certification exam, this time failing due to a botany question. Ironically, he had started breeding plants in back of the monastery.

EXPERIMENTS REVEAL THE LAWS OF INHERITANCE

To investigate how traits vanish in hybrids, Mendel followed the inheritance of seven “differentiating characters,” each with two obvious expressions, in the garden pea. He was methodical, persistent, and perhaps somewhat obsessed. He hand-transferred pollen, to follow which plants transmitted which traits. He would kneel before the chosen plants, cut off the stalks that house the pollen, then collect and paint the pollen onto the recipient female plants using a small artist’s brush. He’d place a paper bag over the altered plant and scribble in a tattered notebook the trait variants he had crossed.

In the first set of experiments, Mendel crossed plants that exhibited alternate expressions of a trait, such as peas that were round or wrinkled, yellow or green, or stems that were tall or short. Much of the language of Mendelian (single-gene) inheritance came later, but introducing it here provides a way to follow the pea experiments. A variant of a gene is called an **allele**. The specific alleles in an individual constitute the **genotype**. The expression of the inherited trait is its **phenotype**—what it looks like.

Mendel’s initial crosses established which allele was **dominant** and which **recessive** for each pair. A dominant allele is expressed in the hybrid and a recessive allele is not. For example, he bred tall plants for many generations until only tall plants resulted, and he did the same for short plants. Then, he crossed a pure or “true-breeding” tall plant to a true-breeding short plant. The offspring were all hybrids, and tall. The dominant expression of a trait is the one that appears in the hybrid generation, whereas the recessive expression disappears in the hybrid generation, but reappears in the generation after the hybrids. Traits reappear, Mendel reasoned, because particles

transmit them, and the particles persist, even if the corresponding traits are not seen. For traits with only two alleles, the dominant one is designated with a capital letter and the recessive one with a small letter, both italicized.

Next, Mendel self-fertilized hybrids. This experiment is called a **monohybrid cross**, because it follows one trait. When Mendel catalogued the offspring of the self-crossed monohybrids, a pattern emerged: plants with the dominant trait outnumbered plants with the recessive trait three to one. For example, self-crossed tall hybrid plants yielded three tall offspring to every one short offspring (Figure 1.2 and Table 1.1). Experiments for each of the seven traits yielded similar ratios, suggesting a common mechanism of trait transmission. Mendel interpreted the 3:1 ratio in the second generation to represent the segregation of alleles for genes transmitted on different **chromosomes**, although chromosomes hadn't yet been identified.

If A is the allele for a tall plant and a is the allele for a short plant, then a monohybrid cross yields three tall plants for every one short plant. A genotype of Aa arises in two ways because either parent can provide the dominant (A) or the recessive (a) allele.

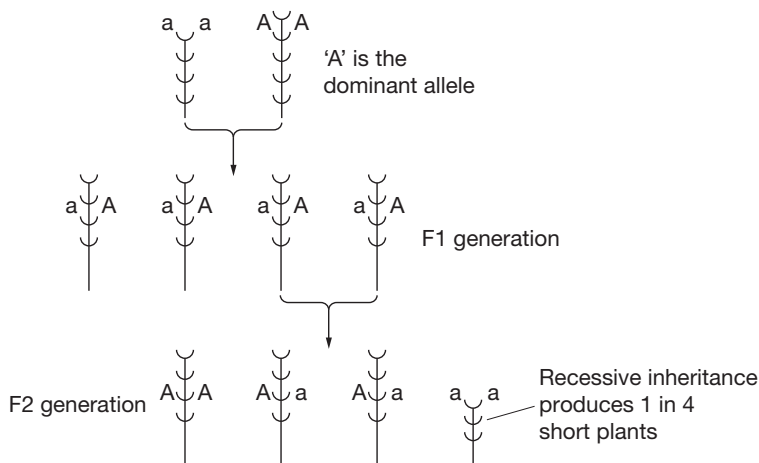


Figure 1.2 A monohybrid cross.

Source: Adapted from *Genetics for Healthcare Professionals*, Figure 19, p. 59

Table 1.1 Derivation of the 3:1 phenotypic ratio of a monohybrid cross

<i>Parents</i>	<i>Sex cells</i>	<i>Offspring combinations</i>	<i>Phenotype</i>
<i>Aa</i>	<i>A, a</i>	<i>AA</i> (one way)	Tall
		<i>Aa</i> (two ways)	Tall
		<i>aa</i> (one way)	Short

Phenotypic ratio: three tall:one short

Genotypic ratio: 1 *AA*:2 *Aa*:1 *aa*

Mendel continued the first set of experiments a generation further, which confirmed that the dominant allele masks the recessive allele. Crossing short offspring of the tall hybrids to each other produced only short plants. But, crossing the tall plants in the generation after the hybrids to the rarer short plants revealed that the tall plants themselves came in two varieties: one-third of them “bred true,” producing only more tall plants, but two-thirds produced tall to short offspring in the ratio 3:1, as if they themselves were hybrids. They were. This first of Mendel’s generalizations, which would be named the “law of segregation,” states that inherited characters are present in two copies, which separate as sperm and egg form, and then come together in predictable combinations. An assumption is that the two alleles are present in equal numbers and join at random.

New terminology eventually replaced Mendel’s “hybrid” and “true-breeding” (Table 1.2). The hybrids are called **heterozygotes** and have two different alleles of a particular gene (Figure 1.3). The true-breeding individuals that transmit the same phenotype generation after generation are called **homozygotes**. They have two identical alleles of a specific gene. One individual—pea or person—has only two alleles per gene, which can be the same or different. But because most genes have more than one allele, several types of heterozygotes are possible. Chapter 2 describes this situation vividly for the disease cystic fibrosis (CF), which varies greatly in severity due to differences among alleles of a gene called *CFTR*.

In a second set of experiments, Mendel followed the inheritance of two and then three traits at a time. He crossed plants that have only round, yellow peas with plants that have only wrinkled, green peas. The pods of the hybrid generation had only round, yellow peas. Round must be dominant to wrinkled, and yellow dominant to green,

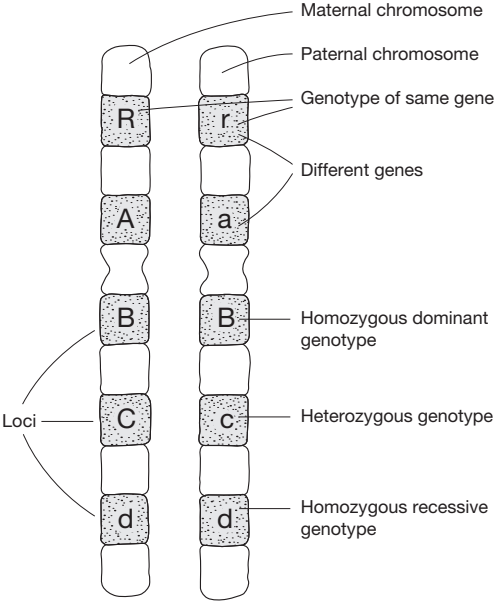


Figure 1.3 The relationship between genes and chromosomes. Each chromosome carries hundreds of genes.

Source: Adapted from *Introducing Genetics: From Mendel to Molecule*, Figure 2.2, p. 11

Table 1.2 Mendelian terminology

Designation	Genotype	Phenotype	Example
Heterozygote	<i>Aa</i>	Dominant	Round peas
Homozygous recessive	<i>aa</i>	Recessive	Wrinkled peas
Homozygous dominant	<i>AA</i>	Dominant	Round peas

he reasoned. Mendel assigned letters to the traits: *A* was round and *a* wrinkled; *B* was yellow and *b* green. The parents, therefore, were *AABB* and *aabb*, and the first-generation plants—called **dihybrids** because they were hybrid for two genes—were *AaBb*.

Self-fertilization of the dihybrids (*AaBb*) produced plants with pods containing four pea varieties. Most were round and yellow like the dihybrids (*A_B_*), but a few were the other three possible

Table 1.3 Derivation of the 9:3:3:1 ratio of a dihybrid cross

<i>Parents</i>	<i>Sex cells</i>	<i>Offspring combinations</i>
<i>AaBb</i>	<i>AB, Ab, aB, ab</i>	<i>AABB</i> (one way) <i>AaBB</i> (two ways) <i>AABb</i> (two ways) <i>AaBb</i> (four ways) Nine ways <i>AAbb</i> (one way) <i>Aabb</i> (two ways) Three ways <i>aaBB</i> (one way) <i>aaBb</i> (two ways) Three ways <i>aabb</i> (one way)

Genotypic ratio: 9 *A_B_*:3 *A_bb*:3 *aB_*:1 *aabb*

Phenotypic ratio: nine round, yellow:three round, green:three wrinkled, yellow:one wrinkled, green

(The different ways reflect the fact that a particular allele can come from either parent)

combinations: round and green (*A_bb*), wrinkled and yellow (*aaB_*), and wrinkled and green (*aabb*) (Table 1.3). After observing many crosses, Mendel deduced that the four classes of offspring from a dihybrid cross appear in an approximate 9:3:3:1 ratio. What would become known as Mendel's "law of independent assortment" explains it: "the relation of each pair of different characters in hybrid union is independent of the other differences in the two original parental stocks" (Mendel 1865: 3). The traits Mendel studied represented genes carried on different chromosomes. Had the genes been "linked" on the same chromosome, the results would have been different.

Over 8 years, Mendel analyzed 16,384 plants. He concluded, "hybrids form egg and pollen cells of different kinds, and that herein lies the reason of the variability of the offspring." His genius was that the chromosomes that separate the "characters," as sperm and egg form in meiosis, had not yet been described.

OVERLOOKED AND THEN RE-DISCOVERED

Mendel presented his work in talks in February and March of 1865 and published his paper in the proceedings of the Brunn Society for

the Study of Natural History (Mendel 1865: 3). A flattering local newspaper article was the only immediate reaction. Disappointed, he left his beloved garden in 1868 to focus on monastery administration. Meanwhile, Mendel's paper went to 120 libraries, including some in the United Kingdom and the United States. A few papers referenced, but misinterpreted, his findings. Fortunately, reprints of his paper passed, silently, among a few scientists.

It remains a mystery why Mendel's elegant experimental work went unnoticed at the same time that Charles Darwin's much longer and mostly observational *Origin of the Species* became an instant sensation. Perhaps it was math anxiety. H.G. Wells, Julian S. Huxley, and G.P. Wells wrote, in their 1929 textbook *The Science of Life*, that Mendel's presentation "dealt chiefly with peas and arithmetic, not the sort of things that cause excitement and clamour, and in the confused tumult of the nineteenth century evolution controversy, they passed unnoticed."

His genius unknown, Mendel died, according to Wells, Huxley, and Wells, in 1884, a perplexed and sad man. But the sparse citations and sharing of the paper would, at the turn of the century, come to the attention of three botanists who had independently discovered Mendel's laws. Preparing to publish their own work, Hugo De Vries (1848–1935), Carl Correns (1864–1933), and Erich von Tschermak (1871–1962) scouted the literature and came upon Mendel's paper in 1900. English embryologist William Bateson (1861–1926) translated the paper into English in 1901, drawing many more readers.

With recognition of Mendel's contributions, a new lexicon to describe heredity began to emerge. Danish botanist and geneticist Wilhelm Johannsen (1857–1927) contributed *genotype* and *phenotype* in 1903, when he called the units of inheritance "theoretical hereditary corpuscles." In 1909, he coined the phrase "genes," and Bateson contributed "genetics."

FROM PEAS TO PEOPLE

Experimental data confirming Mendel's laws quickly began to accumulate in experiments using animals, including mice, guinea pigs, chickens, rabbits, and fruit flies. Ratios were revealed in natural variation, from dwarfism to flower color to cattle horns and attached earlobes. Such single-gene traits became known as Mendelian traits.

Mendel's laws are vividly illustrated in families that have disorders caused by mutations in single genes. Several thousand such conditions are known, many of them very rare. A common scenario is that parents who unknowingly have mutations in the same gene have a child who inherits the associated disease. The parents are heterozygotes (carriers); the ill children are homozygous recessive. These families echo the three-to-one ratios seen in monohybrid crosses of peas, with each child having a 25 percent chance of inheriting the disease, a 50 percent chance of being a carrier, and a 25 percent chance of being free of the mutant allele.

The Polski family of Minneapolis, Minnesota in the United States illustrates both of Mendel's laws. Donna and Michael's teen daughter Mikayla has CF, which impairs her breathing and digestion. Donna and Michael do not have the disease, but they each passed a mutant allele to Mikayla. Their older son has been tested, and does not have a mutant allele. Should Donna and Michael have another child, he or she would face a 25 percent chance of inheriting CF. But the family also illustrates Mendel's second law, because Mikayla has *two* inherited diseases—hemochromatosis as well as CF. Because of the hemochromatosis, her blood retains too much iron from food. The genes for the two diseases are on different chromosomes. The chance that a child in the Polski family inherits both diseases is the product of the chance of inheriting either disease: $1/4$ multiplied by $1/4$, or 1 in 16. Note that this is the same probability as in the final genotype entry, *aabb*, in Table 1.3. Peas and people follow the same rules of inheritance. (Chapter 2 returns to the Polski family.)

Mikayla inherited two genetic diseases due to two rare events happening, like lightning striking twice. The chance of inheriting more than one single-gene disease increases if parents are blood relatives (Mikayla's are not), because the parents may have inherited copies of the same mutant alleles from a shared ancestor, such as a mutual great-great-grandparent. This situation happens when small groups of people leave one country to begin life in another, usually to escape persecution or poverty. For example, the Swiss Anabaptists, reformed Protestants who left Lutheranism and Calvinism beginning in the early 1500s to come to North America, lived among themselves, even while spreading across the continent as the Plain populations that include the Old Order Amish and Mennonite peoples. As the Plain populations remained geographically isolated

and married only among themselves, over time recessive alleles for the same genes combined, leading to a higher prevalence of more than 100 single-gene diseases than is seen in other groups. Some families have more than one inherited disease.

Today, more families of all backgrounds are becoming aware that they have or carry more than one genetic disease not due to chance or isolated ancestors, but simply because with exome and genome sequencing, we can now look at more genetic mutations. About 5 percent of people who have their DNA sequenced to explain specific symptoms discover that they actually have two, three, or even four distinct Mendelian conditions.

At the same time as Mendel's laws were rediscovered, cell biologists (then called cytologists) were closing in on the chromosomes. Yet a third thread, the chemical nature of the gene, would soon entwine with the other two to weave the fabric of the fledgling field of genetics.

THE CHROMOSOME CONNECTION

A dozen years after Mendel published his paper, researchers elsewhere were discovering the chromosomes that would explain how traits and genes are transmitted. Chromosomes appear as dark-staining, rod-shaped structures in cell nuclei when the cell is dividing, but at other times are so unwound that they cannot be visualized. Both chromosomes and genes come in pairs and separate at each generation, then unite in random combinations. Genes are part of chromosomes—but that was not initially obvious.

COLORED THREADS IN CELLS

The earliest known sketches of chromosomes came from German biologist Walter Flemming (1843–1915). In 1878, he observed tangled chromosomes in cells from salamander gills and fins and named the mysterious substance *chromatin*. He did not know of Mendel's work. German anatomist Wilhelm von Waldeyer-Hartz (1836–1921) originated the term *chromosome* in 1888.

By the turn of the nineteenth century, the colored threads of chromatin were known to divide lengthwise during both types of cell

division (mitosis and meiosis) and pass in constant number to daughter cells. **Mitosis** is the division of body (somatic) cells. **Meiosis** forms the sex cells (gametes)—eggs and sperm. Body cells have two sets of the 23 chromosome types and are said to be **diploid**. Sex cells have only one copy of each chromosome and are said to be **haploid**. The joining of sperm and egg at fertilization restores the diploid chromosome number.

In 1902 and 1903, Walter S. Sutton (1877–1916), a graduate and medical student at Columbia College in New York City, and Theodor Boveri (1862–1915), in Germany, independently connected chromosomes to inherited traits. Both men observed the doubling and parting of chromosome pairs as sex cells formed during meiosis, as well as the maintenance of chromosome number as cells divided by mitosis. Sutton looked at cells from a giant grasshopper, while Boveri investigated sea urchins and a worm that infects horses. Mendel, Sutton, and Boveri had each deduced that, to maintain the correct amount of genetic information from generation to generation, each parent's contribution must be halved. (It is a little like two people moving in together each bringing only half of their stuff.) Sutton also noted that the number of traits exceeded the number of chromosomes—therefore, a chromosome must carry the units for several traits. Evidence for Sutton's idea required a return to Mendel's approach of setting up crosses and observing the offspring.

From 1905 to 1908, Bateson and fellow British geneticist R.C. Punnett (1875–1967) followed two traits in the sweet pea that did not yield Mendelian ratios, because their genes were on the same chromosome. They crossed true-breeding plants with purple flowers and long pollen grains to true-breeding plants with red flowers and round pollen grains. When they crossed the dihybrids, instead of the expected 9:3:3:1 ratio, most of the offspring instead resembled a parent. The few exceptions arose, they deduced, when something disrupted the physical relationship of genes on a chromosome.

Bateson and Punnett didn't know that their observations of linked genes explained Sutton's idea that each chromosome carried several genes. The facts fell into place in room 613 in Schermerhorn Hall at Columbia College, in yet another experimental organism, the fruit fly *Drosophila melanogaster*.

THE FLY ROOM

The messy and stinky “Fly Room” was the domain of Thomas Hunt Morgan (1866–1945). Early in the twentieth century, he headed a new program in experimental zoology. Excited by Mendel’s findings, Morgan grew impatient with the long time it took to work with rodents. Fruit flies were an enticing substitute: they offered 30 generations a year, and required only mashed bananas and milk bottles.

The Fly Room had eight cluttered and crammed desks. Cockroaches skittered across the floor. But the sense of excitement and cooperation that permeated the place would become a model of how to run an academic science lab. Every surface was festooned with small, cardboard-stoppered glass milk bottles, each buzzing with hundreds of flies. Insect escapees would often light on the researchers’ hair, and the air bore the distinctive aroma of the overripe bananas smashed into milk bottle bottoms. Female flies would lay their eggs onto this goo, and out would hatch the first of three waves of larvae that would mature within it. The worm-like creatures with their perpetual motion jaws would eat constantly, finally secreting cocoons and, a few days later, emerging and unfolding as full-fledged flies. Half a dozen future Nobel laureates were among “Morgan’s boys,” and the Fly Room spawned an academic pedigree of the founders of modern genetics in the twentieth century.

An easy-to-raise animal with many offspring was not enough to study genetics, however. Morgan needed variants, akin to Mendel’s short and tall pea plants. It took a few years of breeding flies to find interesting new mutants, for it was not yet known how to help nature along and use chemicals and radiation to induce mutations.

One early spring day in 1910, Morgan discovered a white-eyed male fly alone among his red-eyed (normal or “wild type”) brethren. Morgan mated the precious odd-eyed male to his virgin sisters to track trait transmission, much as Mendel hand-pollinated plants. All of the white-eyed male’s offspring had red eyes. Were white eyes another trait hiding among hybrids? To find out, Morgan crossed the red-eyed hybrids with each other and then scrutinized the offspring. If he didn’t consider gender, then red eyes outnumbered white three to one—a neat Mendelian ratio. However, he quickly noticed that all of the female flies had red eyes, whereas the male flies were split—half white-eyed, half red-eyed.

In further experiments Morgan discovered two other traits with the same peculiar inheritance pattern—yellow body and under-developed wings. Aware that males had an X and a Y sex chromosome, whereas females had two X chromosomes, Morgan proposed that the genes for the traits appearing only in males were carried on the X and were expressed in males because they lacked a second X that would mask the traits. Males could be white-eyed or red-eyed, depending upon which X chromosome they inherited from their red-eyed, hybrid mothers. Morgan had described X-linked inheritance, the specific pattern of trait transmission for genes on the X chromosome.

The “fly pickers” in room 613 continued to catalog new mutants. They set up crosses to see which pairs of traits assorted independently—carried on different chromosomes—and which traits tended to be inherited together. The trait pairs fell into four groups. Was it coincidence that the fly has four sets of chromosomes? No. Genes linked on the same chromosome are inherited together because the chromosomes transmit them together, like passengers on a train. The rare flies with mixed up, or recombined, trait combinations, compared with their parents, arose when paired chromosomes physically crossed over, swapping parts. This happens during meiosis.

An undergraduate researcher in the Fly Room, Alfred Henry Sturtevant (1891–1970), was intrigued by the idea of crossing over disrupting linkage. As a child growing up on a farm in Alabama, Sturtevant had drawn pedigrees of his father’s horses. When he learned about Mendel’s laws in Morgan’s class, he became excited at applying them to the traits he had followed in horses. That led him to join the lab group.

Sturtevant thought about five traits linked on the fly’s X chromosome. One night, he was struck with an idea so compelling that he cast his homework aside and stayed up until the wee hours, sketching. He took his mentor’s vision of genes linked on a chromosome, but occasionally separated by crossing over, a step farther: what if the frequency of the recombinant class of offspring reflects relative distance of the genes along the chromosome? Genes at opposite ends of a chromosome would have a large recombinant class and represent a large physical distance, whereas genes close together on the chromosome would generate few recombinants, simply because there was less space for a crossover to occur. It was like considering stores

on opposite sides of a street. There are more sites at which to cross between stores at either end of the block than there are between two stores in the middle of the block.

At age 19, Sturtevant drew the first map depicting genes on a chromosome (Sturtevant 1913: 43). Little did he know that it would become the foundation for all future genetic maps, and inspire the idea of sequencing genomes.

HUMAN CHROMOSOMES

While some researchers were demonstrating Mendel's laws, others were examining chromosomes, not, at first, realizing that the two fields of study were closely linked. The colored tangles of chromatin were so difficult to visualize under a microscope that these early cell biologists counted from 30 to 80 of them in human cells. The first close estimate came in 1923, when American zoologist Theophilus Painter published sketches of what he thought were the 48 human chromosomes (Painter 1923: 247). Painter might not have been mistaken (the correct number is 46). He used a sample of testicular tissue from a patient at a mental hospital who had been castrated to quell his violent behavior. The man's cells might have had extra chromosomes that caused his behavior. Painter also showed that human males and females have the same chromosome number—for a time it had been thought that males have one X but no Y, because the Y is so small.

Visualizing chromosomes accurately awaited improved technique, which happened gradually. First, researchers learned how to use an extract from the crocus plant, called colchicine, to stop cells while dividing. At this time, the chromatin is most condensed, appearing as rods. In 1951, a technician stumbled upon how to untangle chromosomes while trying to culture white blood cells by bathing them in a salt solution that wasn't concentrated enough. With the interiors of the cells saltier than their surroundings, water rushed in, swelling them and spreading the chromosomes. The effect was like water added to a small swimming pool moving floating toys away from each other. Two years later, researchers began to draw up cells into a pipette, splash them down into a droplet of stain, and then gently add a glass coverslip, spreading the chromosomes at one depth.

These experiments eventually revealed that the human somatic (body) cell chromosome number is 46. Other work showed that gametes (sperm and egg) each have 23 chromosomes.

Staining chromosomes improved, along with ways to separate them. At first, stains colored all chromosomes alike, and charts depicted them in size groups. In 1959, researchers associated the first syndrome with an abnormal chromosome number—the 47 chromosomes of a person with Down syndrome. In the 1970s, newer stains created banding patterns distinct to each chromosome type. Shortly after, researchers labeled individual chromosomes, first with radioactive chemicals and later with fluorescent chemicals, which bound to specific DNA sequences.

Each human somatic cell has 23 pairs of chromosomes, including 22 pairs of **autosomes** (non-sex chromosomes) and 2 **sex chromosomes** (the X and the Y).

Human chromosomes are displayed in size-order charts called **karyotypes**. The chromosomes are numbered 1 through 22, from largest to smallest, plus the sex chromosomes. Abnormal numbers of chromosomes or their parts usually affect health. Extra chromosomal material is better tolerated than missing material, and this is why one form of Down syndrome, caused by an extra chromosome 21, is the most common chromosomal abnormality. The chromosome is the smallest, and therefore individuals with an extra copy can live to adulthood. Extra or missing copies of the other chromosomes usually halt prenatal development early. Table 1.4 lists types of chromosome problems.

Table 1.4 Types of chromosome abnormality

<i>Description</i>	<i>Technical term</i>
An extra entire set of chromosomes	Polyploidy
An extra individual chromosome	Trisomy
A missing chromosome	Monosomy
An extra part of a chromosome	Duplication
A missing part of a chromosome	Deletion
A part of one chromosome moved to another	Translocation
A part of a chromosome turned around	Inversion

IN PURSUIT OF THE GENETIC MATERIAL

Human genetics began with observations of trait transmission in families and shifted to the level of the chromosome as technology improved. The next and most fundamental level of genetic information is deoxyribonucleic acid (DNA). The clues to understanding the biochemical nature of the gene came from diverse places—soiled bandages, darkened diapers, bread mold, microbes, and molecular models.

IDENTIFYING THE PARTS OF DNA

The first description of the genetic material came in 1871, when Swiss biochemist Friedrich Miescher (1844–1895) developed a technique to separate nuclei from cells (Miescher 1871: 441). He collected pus on bandages—a fortuitous source, because pus is mostly white blood cells. Had he chosen a bloodstain to probe, he might not have obtained much DNA because red blood cells lack nuclei, where DNA resides. Miescher named the phosphorus-rich, acidic substance he discovered *nuclein*. The soiled bandages also yielded abundant protein, which for many years would be the prime candidate for the genetic material.

Russian-born biochemist Phoebus Levene (1869–1940) further described DNA with hundreds of experiments in his laboratory at Rockefeller University in New York City. His discovery of the sugars ribose, in 1909, and deoxyribose, in 1929, defined two types of nucleic acids. Throughout the 1930s, he identified the nitrogen-containing bases that provide the information in the nucleic acids: **adenine** (A), **guanine** (G), and **cytosine** (C) in both types, and **thymine** (T) in DNA and uracil (U) in RNA. He also confirmed the presence of phosphates in the nucleic acids. Levene deduced that DNA is built of units, which he named **nucleotides**, which consist of a single deoxyribose sugar, a phosphate group, and a base. Figure 1.4 depicts DNA at several levels, beginning with the building block, the nucleotide.

Levene correctly identified the details of DNA and RNA structure, but he erred in envisioning the bigger picture too simply, with equal numbers of the different base types. This idea led people to conclude, at first, that DNA was too lacking in information content to be a genetic material. Levene, like Mendel, died before his work on DNA could be appreciated, but it would prove pivotal in deducing the molecule's structure.

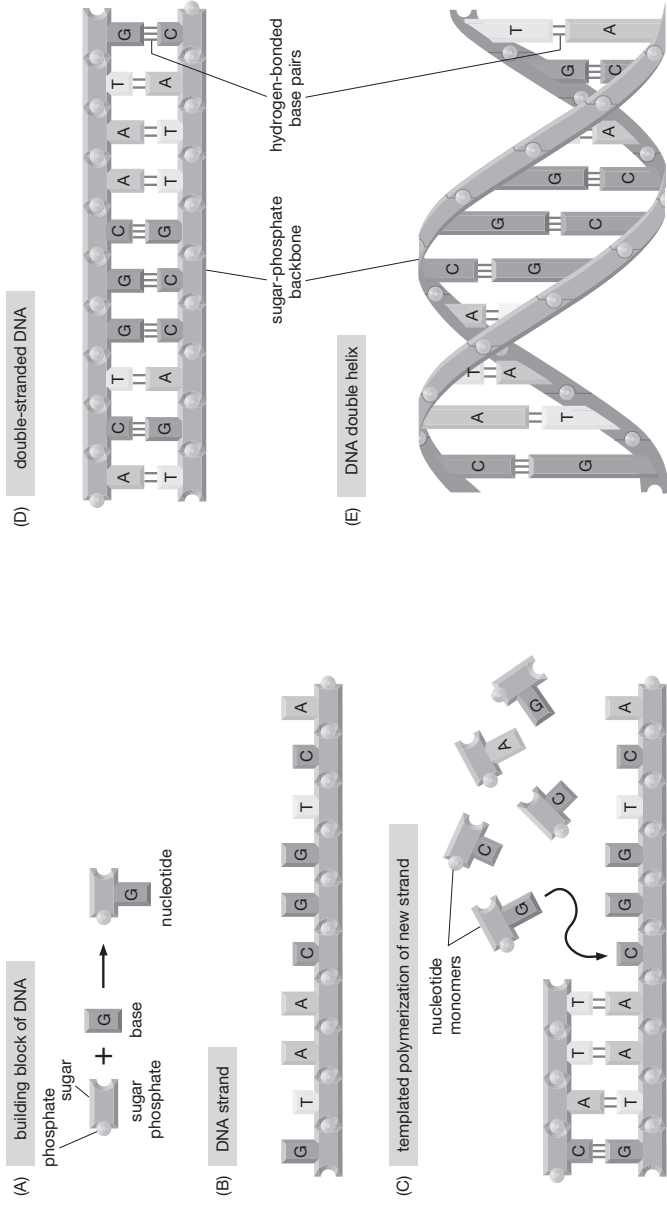


Figure 1.4 DNA and its building blocks. DNA, the hereditary molecule, is a very long molecule built of units, called nucleotides, that consist of a nitrogenous base, a phosphate group, and deoxyribose sugar.

Source: *Molecular Biology of the Cell*, 6th edition, Figure 1–2, p. 3

ONE GENE, ONE PROTEIN

Between the times that Miescher and Levene described the composition of the genetic material, a physician with a dubious bedside manner contributed a key clue to its function.

Sir Archibald Garrod (1857–1936) had a great interest in his patients' urine. Shortly before the turn of the twentieth century, as a visiting physician at a children's hospital, Garrod examined a 3-month-old, Thomas P., whose nappies (diapers) had a peculiar, dark bluish-black stain. The color was a sign of alkaptonuria, in which a substance called homogentisic acid builds up in joints and darkens the skin in telltale places. Young Thomas appeared none the worse for his odd urine. Then two siblings were born and also blackened their nappies. Garrod noticed that the parents were first cousins.

Garrod had recently read about Mendel's work. Walking home from the hospital one night, pieces of the puzzle of his young patients began to assemble. Alkaptonuria was then considered an infection, which might be why it affected siblings. But it could instead be a recessive trait, Garrod reasoned, as the parents were unaffected. *His brainstorm:* the siblings had the condition because their parents had each contributed the same mutation, inherited from their mothers, who were sisters, who in turn had inherited it from the same parent. Garrod soon found other examples (Garrod 1902: 1616) of alkaptonuria in families with cousin–cousin marriages, and credited Mendel's laws for his epiphany. The carrier parents were the "hybrid generation," and the offspring with the darkened nappies were the homozygous recessive offspring class.

Garrod was the first to link heredity to biochemistry, in what he named "inborn errors of metabolism." In the absence of an enzyme's activity, the chemical that it normally breaks down, builds up. In alkaptonuria, homogentisic acid accumulates. Chapter 4 tells a similar tale of foul nappies resulting from another inborn error, phenylketonuria (PKU). Garrod died in 1936, and, like Mendel's and Levene's, his work would not be appreciated in his lifetime. In 1941, Garrod's idea resurfaced as "one gene–one enzyme," which grew out of experiments using yet another "model" organism, the bread mold *Neurospora crassa*.

George Beadle (1903–1989) and Edward Tatum (1909–1975) chose the orange-hued bread mold for the same reason that geneticists worked with fruit flies—the organism was easy to raise, and

mutations were obvious (Beadle and Tatum 1941: 499). The trait they followed was nutritional. Wild type *Neurospora* would grow on a certain medium (food) that contained simple nutrients, and its metabolism would synthesize more complex nutrients. Mutants, however, would perish on this paltry diet, because they lacked an enzyme (a type of protein that speeds biochemical reactions) necessary to build the more complex nutrients. Beadle and Tatum studied mutant bread mold that could only grow on supplemented medium. If, for example, mold only grew in the presence of vitamin B6, then the mutation disrupted an enzyme required for cells to synthesize vitamin B6. Beadle and Tatum concluded that one gene specifies one enzyme—which Garrod had shown, in humans, nearly 40 years earlier.

Genes are the instructions for all types of biological protein, not just enzymes. In 1949, biochemist Linus Pauling (1901–1994) identified the protein that is abnormal in sickle cell disease (Pauling et al. 1949: 543). He was working at the California Institute of Technology on the hemoglobin molecule. Hemoglobin carries oxygen in the blood. It is built of four wound-up chains of amino acids called globins, two termed alpha and two beta. Each globin coils around a small organic (carbon-containing) chemical group that, in turn, shields an iron atom, creating a molecule that looks a little like four marshmallows stuck together.

Pauling discovered that beta globin from people with sickle cell disease, when placed on a gel-like material subject to an electric field (a technique called electrophoresis), migrated to a different site than beta globin from a person who did not have the disease. Furthermore, people who were carriers of the disease—inferred because their children were affected but they were not—had beta globin proteins that moved to *two* sites on the gel—the wild type position *and* the sickle cell position. Pauling had vividly unveiled the “disappearing trait” in the hybrid. At the molecular level, it didn’t really disappear. In 1956, chemist Vernon Ingram added that sickle cell beta globin differs from wild type by a single DNA building block. But a great deal had happened in those 7 intervening years.

DETHRONING PROTEIN

The next chapter in the story of the discovery of the genetic material demonstrated that it is DNA, and not protein. A series of beautiful experiments led to the answer.

Frederick Griffith (1871–1941) was a medical officer at the British Ministry of Health just after World War I. He was investigating the cause of many deaths during the great influenza pandemic of 1918, which was not due to the flu virus directly, but instead to bacterial infection of the lungs. Intrigued by finding more than one strain of the bacterium *Streptococcus pneumoniae* in a single patient, Griffith wondered if the two types of bacterium could swap parts. To test his idea, he set up experiments using mice (Griffith 1928: 113).

The two strains of bacteria Griffith looked at had different outside surfaces. “Smooth” (type S) bacteria were enclosed in a carbohydrate capsule that shielded them from the mouse’s immune system. They were virulent, killing mice within two days of infection. In contrast, “rough” (type R) bacteria lacked the smooth, sugary exterior and succumbed to the mouse’s immune system—these bacteria were avirulent. Griffith showed that the ability to kill could be passed from type S bacteria to type R bacteria, and type R could transmit the trait when they divided.

Griffith heated type S bacteria, killing them. When injected alone, they could no longer sicken mice. But when he injected the heat-killed type S bacteria along with live type R bacteria, the mice died. Blood from the dead mice was swarming with the usually unadorned R bacteria wrapped in smooth, sugary capsules! Had something in the type S bacteria, which Griffith named the “transforming principle,” entered the normally tame R bacteria, making them deadly? Again, Griffith’s contribution was not appreciated in his time. He was killed while working in his laboratory in 1941, during the London blitz.

The search was on for Griffith’s transforming principle. In 1944, a trio of physicians/geneticists at Rockefeller University in the US, Oswald Avery (1877–1955), Colin MacLeod (1909–1972), and Maclyn McCarty (1911–2005), repeated Griffith’s experiments, but with a twist. They added either an enzyme that destroys protein or an enzyme that destroys DNA. Type S bacteria were able to transform type R into killers when their protein was destroyed, but not when their DNA was destroyed. Therefore, the transforming principle was DNA. To confirm the results, they transferred DNA from heat-killed type S bacteria with type R bacteria into mice. Not only did the mice die of pneumonia, but their bodies yielded live type S bacteria (Avery et al. 1944: 137).

Despite the clear evidence that DNA was the genetic material, doubt persisted, because its four-base alphabet seemed insufficiently informational in this time before combinatorial thinking. In 1951, biochemist Erwin Chargaff (1905–2002), at Columbia College, provided a key piece of information. Probing the DNA of various species, he discovered that the four bases were *not* present in equal numbers. Instead, the proportions of A and T matched, as did those of C and G.

DEDUCING THE DOUBLE HELIX

In the spring of 1951, 23-year-old James Watson, fresh PhD in hand, was doing a year's research in Copenhagen when he heard Maurice Wilkins (1916–2004), a biophysicist at King's College in London, present his work on a technique called X-ray diffraction, which bounces X-rays off of a crystal. The photographic image made by capturing the deflection of the X-rays reveals the crystal's shape, like sonar uses sound waves to reconstruct the shapes of sunken ships. The X-ray diffraction pattern from a hunk of calf thymus, rich in DNA, indicated a regular, repeating structure within, Wilkins reported (1953: 738).

Watson soon took a position at the Cavendish Laboratory at the University of Cambridge, where he met 35-year-old Francis Crick (1916–2004). Crick was a physicist-turned-molecular-biologist, a common career trajectory in those early days of the latter field. The pair set to work on determining the three-dimensional structures of proteins, but were much more keen on DNA, especially Wilkins' X-ray diffraction views. A young physical chemist, Rosalind Franklin (1920–1958), had joined Wilkins and had produced dozens of X-ray diffraction images of DNA, in both a dry and a wet form. She accurately measured the dimensions of the molecule (Franklin and Gosling 1953: 740). However, when she presented her findings at a seminar to which Wilkins had brought Watson, she did not include the "B" form of DNA, which was swollen with water molecules.

Energized by the X-ray diffraction results, Watson and Crick asked the machine shop at the Cavendish to supply colored balls to represent atoms and sticks to represent chemical bonds, and they built models of the molecule that fit the data. First, they tried a triple helix. (A single helix is impossible—it would simply unwind.) Franklin took

a look and pointed out that their model was inside out—the bases should be inside the sugar–phosphate backbone.

The next clue came in the spring of 1952, when Watson discovered Chargaff's paper of a year earlier, in which he'd described the equal proportions of A and T, and G and C. When Crick read the paper, he immediately envisioned base pairs. If correct, then DNA could not only encode information, but would contain instructions and a mechanism for its duplication! The molecule would be, in essence, a self-perpetuating code of life.

Meanwhile, evidence was finally building *against* protein as the genetic material, from two researchers at the Cold Spring Harbor Laboratory on New York's Long Island, Alfred Hershey (1908–1997) and Martha Chase (1927–2003) (Hershey and Chase 1952: 39). In "blender experiments," they infected *E. coli*, a common bacterium, with a virus, T2. When a virus—a tiny bit of nucleic acid wrapped in a protein coat—infests a cell, it injects only its DNA or RNA, leaving the protein coat outside. The viral DNA or RNA then commandeers the protein synthesis machinery in the cell, instructing it to make more viruses.

Hershey and Chase grew viruses on two types of medium: one contained radioactive phosphorus, found in nucleic acids but not amino acids, and one contained radioactive sulfur, found in amino acids but not nucleic acids. Next, they infected bacterial cells. Then, they poured the material into a blender and turned it on, not creating a smoothie, but shaking free the viral protein coats clinging to the infected bacteria. Spinning the material in a centrifuge separated the viral coats from the infected cells, which sunk to the bottom of the tube. The viral coats floated to the top of the centrifuge tube and were easy to scoop off and analyze.

Where was the radioactivity in the two batches? If radioactive sulfur was in the viral protein coats but not in the infected bacterial cells, and radioactive phosphorus was in the cells but not in the protein coats, then the genetic material was DNA. This is, in fact, what Hershey and Chase saw. They had shown, definitively, that not only is the genetic material DNA, but that it is *not* protein. The blender experiments catalyzed the race to figure out DNA's three-dimensional shape, or conformation.

At about this time, Watson saw an X-ray diffraction image taken by Franklin's graduate student that hadn't made it into her

presentation the year before. "Photo 51" showed the waterlogged B form of DNA. Its X-shape indicated a helix cross section. So Watson and Crick built more molecular models, incorporating the new information: DNA is a helical-shaped acid, with a sugar-phosphate backbone on the outside and equal numbers of A and T and G and C pairs on the inside.

The machine shop could not keep up with the demands of the model builders, and so, in the morning of the last Saturday in February 1953, Watson sat at his desk, arranging and rearranging cardboard cut-outs he'd made of the bases, sugars, and phosphates. The only shape that accommodated all the experimental findings was a double helix. Watson saw that arranging the base pairs side by side, rather than in short stacks as Crick had suggested, produced a molecule sleek enough to curl up inside a cell's **nucleus**.

Crick arrived within the hour, looked at Watson's cardboard creation, and knew instantly that the implications were profound. The elegant, symmetrical DNA double helix could do three key things: carry information, replicate, and change (Watson and Crick 1953b: 964). At lunch in the pub next door, Crick famously announced that he and Watson had discovered the "secret of life." The prestigious journal *Nature* published their paper just weeks later (Watson and Crick 1953a: 737).

REVEALING REPLICATION

Like a film whose ending suggests a sequel, the famed Watson and Crick paper concluded with, "*It has not escaped our notice that the specific pairing we have postulated immediately suggests a possible copying mechanism for the genetic material.*" But exactly how does the DNA base sequence copy itself?

In a second *Nature* paper, Watson and Crick proposed that the double helix separates, and each exposed half pulls in new nucleotides to build a new double helix—a little like a row of dancing partners separating and choosing new mates. This mechanism of DNA replication is called "semi-conservative," because each double helix (row of dance partners) conserves half of itself from a previous double helix. But how could an experiment show this?

Two young researchers took up the charge to demonstrate DNA replication (Meselson and Stahl 1958: 671). In 1954, Matthew

Meselson met Franklin Stahl at a summer course at Woods Hole, on Cape Cod, Massachusetts. The instructors: Watson and Crick. Meselson and Stahl met again at the California Institute of Technology, where they would conduct what many have called “the most beautiful experiment in biology.” The work not only demonstrated semi-conservative DNA replication, but ruled out the other two possibilities—that one double helix creates another (conservative replication), or that a double helix shatters and rebuilds itself into two using spare parts (dispersive replication).

Meselson and Stahl marked newly replicated DNA with radioactive nitrogen in its bases to follow its fate. The radioactive DNA was denser, making it sink further than non-radioactive DNA in a centrifuge tube. By growing bacteria on radioactive medium, then switching the cells to non-radioactive medium, Meselson and Stahl deduced, from the patterns of heavy and light cells over a few generations, that DNA replication is semi-conservative, and not due to either of the other two mechanisms. Their findings were repeated in other species, with DNA marked in different ways.

CRACKING THE GENETIC CODE

While Meselson and Stahl tackled DNA replication, others were investigating how DNA guides a cell to manufacture protein molecules. How does a 4-letter DNA alphabet encode a 20-letter amino acid alphabet? How is DNA recycled, rather than consumed?

The first step to understanding how DNA encodes protein was to orient events in the cell. Experiments that broke open cells showed that proteins are synthesized, even when the nuclei are removed, on structures called **ribosomes**. What carries the DNA instructions to build a protein from amino acids out of the nucleus and to the ribosomes? Crick, who was very good at imagining microscopic events, suggested that a small molecule ferries the information to the cytoplasm, where amino acid building blocks are strung together. What is this molecule? And what, exactly, *is* the genetic code, the correspondence between the DNA information, the carrier molecule information, and the ultimate string of amino acids?

Logic provided some clues. The first question: *How many DNA letters specify an amino acid?* Three nucleotides are the minimum. One letter obviously would encode only four bits of information

(A, C, T, and G), two could combine in only 16 ways (AA, CC, GG, CC, AT, AG, AC, TA, GA, CA, CT, CG, CA, TC, GC, AC), and 20 were required to specify the 20 types of amino acids in cells (many more than this are possible chemically). In 1961, Crick led experiments to show the three-letter minimum using a virus called T4, whose entire base sequence was known (Crick et al. 1961: 1227). By treating the virus with a chemical that removes nucleotides, the researchers found that deleting one or two that were next to each other had a devastating effect on the encoded protein, but that deleting three nucleotides in a row was not as severe.

Logic also guided experiments to answer the second question: *Is the genetic code overlapping?* If so, then certain amino acids would always follow others in a protein's sequence. When researchers deciphered the amino acid sequences of several proteins, they realized that this was not the case. The code is non-overlapping.

Experiments answered other questions. While some researchers were identifying the three most abundant types of RNA molecule in cells—**messenger RNA** (mRNA), **ribosomal RNA** (rRNA), and **transfer RNA** (tRNA)—two young researchers at the National Cancer Institute in Bethesda, Maryland, Marshall Nirenberg and Johann Matthaei, were working with a soup of ribosomes, enzymes, various RNAs, and amino acids, called a "cell-free system." They synthesized short RNA molecules that consisted solely of the RNA base uracil (chemically similar to the DNA base thymine), and added it to the cell-free system. What would happen? Out of the soup came snippets of protein consisting entirely of the amino acid phenylalanine—and the number of phenylalanines was exactly one-third the number of RNA nucleotides, confirming the triplet nature of the code. This experiment revealed the first RNA word, or **codon**, of the RNA alphabet that is the genetic code: *UUU = phenylalanine*.

Repeating the experiment with different short RNAs filled in the rest of the code. First came AAA (lysine) and CCC (proline). Then came "co-polymers," such as UGUGUG . . . , which, in the test tube soup system, directed synthesis of alternating cysteines and valines, two other amino acids. But which was which? Was UGU cysteine and GUG valine, or vice versa? Analysis of a different co-polymer, UGUUUUUGUUUU . . . , provided the answer. It specified phenylalanines alternating with cysteines. As UUU encodes phenylalanine, then UGU must encode cysteine, and therefore GUG must encode

The Genetic Code					
1st position (5' end)	2nd Position				3rd Position (3' end)
↓ U	U Phe Phe Leu Leu	C Ser Ser Ser Ser	A Tyr Tyr STOP STOP	G Cys Cys STOP Trp	↓ U C A G
C	Leu Leu Leu Leu	Pro Pro Pro Pro	His His Gln Gln	Arg Arg Arg Arg	U C A G
A	Ile Ile Ile Met	Thr Thr Thr Thr	Asn Asn Lys Lys	Ser Ser Arg Arg	U C A G
G	Val Val Val Val	Ala Ala Ala Ala	Asp Asp Glu Glu	Gly Gly Gly Gly	U C A G

Figure 1.5 The genetic code. The same mRNA codons specify the same amino acids in all life on earth.

Source: Adapted from *Molecular Biology of the Cell*, 6th edition, Figure T:1

valine. In this manner, the researchers matched 64 mRNA codons to their amino acid counterparts (Crick et al. 1961: 1227) (Figure 1.5).

By 1965, the team had cracked the entire genetic code. A critical characteristic of the genetic code is that it is the same in all organisms, as well as viruses. The same codons specify the same amino acids, whether in a tulip or a tiger. This “universality” of the genetic code is profound evidence that all life descends from shared ancestors.

GENE EXPRESSION AND EPIGENETICS

A genome is a dynamic instruction set, with some genes being transcribed and translated while others aren't, in any given circumstance.

Many genes are termed “housekeeping,” because their functions are required all the time, to sustain metabolism and life. Others are transcribed in particular situations, such as producing proteins necessary to respond to an injury.

A layer of biochemical control oversees accessing the genome. Small molecules (methyl [CH₃] groups, acetyl groups, and phosphates) bind to DNA as it winds around proteins called histones in ways that turn certain individual genes on or off, depending on circumstances. Proteins called *transcription factors* combine in ways that determine which sets of genes are expressed in a certain circumstance. Changes in gene expression are termed **epigenetics** (“epi” means “outside”) because they do not change the actual sequence of chemical building blocks of the DNA. Epigenetic changes are not passed to the next generation; genetic changes (mutation) are. The media frequently misstates *epigenetic* change as *genetic* change, such as the headline “eating fructose alters hundreds of genes.” Indulging in sweets alters gene expression, not the genes themselves.

SEQUENCING HUMAN GENOMES

In the 1950s and 1960s, experimental pathways converged to discover and describe the DNA double helix. Then in the 1970s, research directions diverged. Some investigators focused on the details of protein synthesis, some mapped genes to chromosomes, and others invented the biotechnologies that would eventually make sequencing human genomes feasible.

The idea to sequence “the” human genome was first seriously discussed in the mid-1980s. The international effort to do so began in 1990, and the first human genome sequences were deciphered and published by 2003. However, several biotechnologies preceded genome sequencing: recombinant DNA technology, DNA sequencing, and gene mapping.

BIOTECHNOLOGIES THAT MADE GENOME SEQUENCING POSSIBLE

The first modern biotechnology was **recombinant DNA technology**, which combined (“spliced”) DNA from different species. This

intervention was possible because of the universal nature of the genetic code, which enabled such combinations as a human gene transcribed and translated in a cell of a bacterium.

Careful thought preceded the first “gene splicing” experiments, and it set a precedent of oversight for genome editing today, the subject of Chapter 5. In early 1975, 140 molecular biologists met at Asilomar, California, to discuss the safety of recombinant DNA technology. The researchers drew up plans to contain such organisms in the lab, including ways to engineer their creations with genes that would kill them should they “escape.” Rather than a route to producing “triple-headed purple monsters,” as one researcher joked, recombinant DNA technology instead became a way to mass-produce human proteins useful as drugs, without needing human donors. This would become especially important in the 1980s, when AIDS made use of donated blood unsafe. The first recombinant drug, human insulin, became available to people with diabetes in 1982. It had previously come from pigs or cows, and some people with diabetes were allergic to these sources of insulin. Today, a few dozen drugs are manufactured using recombinant DNA technology. These drugs treat a variety of conditions, including CF, hypertension, anemia, cancer, and clotting disorders.

The 1970s were also the years when sequencing DNA became possible. Fredrick Sanger, at the Medical Research Council (MRC) Laboratory for Molecular Biology in Cambridge, UK, and Alan Maxam and Walter Gilbert, at Harvard University, invented different ways to do this. These approaches, and others that would follow, used the same logic: cut several copies of the same long DNA molecule into pieces that differ in size by one base from the same end, with that base labeled in some way (radioactivity, a chemical modification, or fluorescence). Aligning the pieces by size reveals the sequence, read in the order of the overhanging bases. DNA sequencing was automated by 1982 and commercialized by 1987. In the mid-1980s, a researcher could sequence up to 1,000 DNA bases a day; today, a human genome can be sequenced in under a day. Newer techniques use microfluidics, in which small sequences of DNA are attached to tiny beads, and other nanotechnology-based methods. However, Sanger sequencing remains the gold standard, and is still used to check results for accuracy and to fill in gaps.

Mapping genes to specific chromosomes also flourished in the 1970s and 1980s. Often, a gene hunt would start with an individual who had an unusual syndrome and a specific chromosome glitch, such as a missing or moved part. Linkage data were important, too, echoing Bateson and Punnett's work on pea plant flower color and pollen grain shape. By the mid-1980s, human chromosome maps had become so crowded with genetic landmarks that several researchers realized that sequencing a human genome was within reach.

HUMAN GENOME SEQUENCING: AN IDEA WHOSE TIME HAD COME

In the 1970s, when sequencing a single gene in a simple bacterium might take up an entire 4-to-5-year PhD project, determining the DNA sequence of an entire human genome was unimaginable. Today, more than a million people have had their genomes sequenced, and that number is projected to hit one billion by 2025. Much of the sequencing is part of nation-wide projects, such as the UK 100,000 Genomes Project and other efforts discussed in the first section of Chapter 3.

The idea to sequence a human genome arose in at least three places, in the US, at about the same time. Late in 1984, Robert Sinsheimer, chancellor of the University of California at Santa Cruz, assembled a group to consider sequencing the human genome as a fundraising goal. In late 1985 at the Salk Institute, Renato Dulbecco proposed sequencing the human genome to learn about the origins of cancer. The third founder, Charles DeLisi, was director of the US Department of Energy's (DOE) Health and Environmental Research Programs. The DOE had a long-standing interest in the effect of radiation on the human genome, because the agency developed and delivered the atomic bomb and was monitoring mutations in survivors of the blasts.

Not everyone thought sequencing the human genome was a great idea. Would such "big" science steal precious funds from "little" science, such as identifying the genes behind rare disorders? Would it divert funds from fighting AIDS? Sequencing the genome was compared to climbing Mt. Everest just because it is there. Some scientists scoffed at the value of gobs of data, compared with the

traditional research route of discovering mechanisms. Ironically, today the science of genomics is all about big data.

In an effort reminiscent of the scrutiny of the safety of recombinant DNA research, the genetics community debated the challenges and promises of sequencing the genome. In the UK, the MRC, Imperial Cancer Research Foundation, and the Wellcome Trust were the main players, and, in France, this role fell to a joint venture, Genethon. In the US, the National Research Council of the National Academy of Sciences took the lead, and both DOE and the National Institutes of Health (NIH) funded the beginnings of a 15-year human genome project in 1988. The National Human Genome Research Institute, part of the NIH, formed a year later, with James Watson as director, and the project shifted from DOE to NIH. Similar efforts were underway in the USSR, Italy, Japan, Canada, and Latin America. The Human Genome Organization emerged to oversee the ten major sequencing centers and other academic labs and companies staking out parts of the genome.

Participants in the various projects shared concerns about the ultimate use of DNA information, especially in light of the legacy of eugenics that tainted the field of genetics earlier in the century. Five percent of the NIH budget went toward investigating ethical, legal, and social issues. Genome research throughout the world also called for the sequencing of genomes of model organisms (Table 1.5), and improving sequencing and data analysis. By 1991, companies and a few not-for-profit organizations had joined the genome sequencing effort.

Further technological advances laid the groundwork for the post-genome era, when the mountains of data would require analysis. At Stanford University in 1991, Patrick Brown invented the “gene chip,” or **DNA microarray**. A chip reveals the mRNAs, and therefore the proteins, made in a particular cell under particular conditions, by copying the mRNA into DNA called complementary, or cDNA. Microarrays revolutionized the study of gene expression and interaction.

The availability of human genome sequences has been a boon to researchers; to clinicians, who now have more diagnostic tests for Mendelian disorders; and to drug developers, who have many new targets to pursue (Venter et al. 2001: 1304; International Human

Table 1.5 Genome projects

<i>Species</i>	<i>Information revealed</i>
Cat	Model for HIV research
Chicken	Influenza origin and transmission Clues to dinosaurs (bird ancestors)
Chimpanzee	What makes us human
Coelacanth (fish)	Conquering land habitats
Cow	How breeds differ How to improve meat and milk
Dog	Breeds resulted from extreme artificial selection Human disease model (arthritis, cancer, epilepsy, heart disease)
Honeybee	Biological underpinnings of insect societies Honey production
Horse	New diagnostics and treatments for racehorses Breeding strategies to avoid unwanted traits
Mice and rats	Have versions of nearly all genes that cause disease in humans
Mosses	Conquering land habitats
Pig	Human disease model (heart disease, obesity) Organ donors
Poplar	Improved wood products
Rice	Improved nutrition as dietary staple
White rot fungus	Clean hazardous waste sites

Genome Sequencing Consortium 2001: 860). Perhaps unanticipated is the use of human genome sequences to unveil normal variation, the differences and distinctions that make us each unique. That analysis will continue for some time to come, because we now know that there is not “a” human genome, as we thought in the 1980s, but many variations—where the field of genetics actually started.

WHAT WOULD YOU DO? *ANIMALS IN RESEARCH*

Species other than us have been essential in genetics research. Gregor Mendel worked out the two laws of inheritance using pea plants. The earliest views of chromosomes were from salamanders, grasshoppers, and sea urchins. The first genetic map represented thousands of fruit fly lives. Experiments that deciphered DNA's structure and revealed its function used viruses, bacteria, mice, and bread mold. We could never have sequenced a human genome without the contributions of a diversity of other organisms. To learn how CF affects babies, and because mice do not develop the disease, researchers genetically modified pigs to have CF. Experiments on those pigs revealed much of what we know about the beginnings of this illness, discussed in Chapter 2.

Many people become upset at the idea of using animals in biomedical research, while some think that animals should be used to develop medical treatments only when alternatives do not exist. Another aspect of the animal research dilemma is “the Bambi effect”—that is, people tend to object more to research that uses mammals than to research that uses other types of animal. Perhaps it is because we love our furry pets.

How do you feel about the use of animals in research? Did reading this chapter change your view? List factors that you think are important in deciding if animals should be used in an experiment. Would you feel differently about a particular experiment that is conducted on eels or cockroaches versus kittens or if an experiment was to test a treatment for a disease that affects someone you know? What are some alternatives to using animals in research?

SUGGESTED READING

The chapter cites the key papers in the history of genetics, but many books also recount the experiments and events that led to our understanding of the genetic material. Robin Marantz Henig's *The Monk in the Garden* (2000; New York: Houghton Mifflin Co.) and Vítězslav Orel's *Gregor Mendel the First Geneticist* (1996; Oxford University Press) colorfully cover Mendel's life and experiments. A.H. Sturtevant recounts the origins of gene mapping in *A History of Genetics* (1965; Cold Spring Harbor Press). *DNA Discovery: The Annotated and Illustrated Double Helix*, by James D. Watson, Alexander Gann, and Jan Witkowski (2012; Simon and Schuster) describes the deciphering of DNA's structure, as does Brenda Maddox's *Rosalind Franklin: The Dark Lady of DNA* (2003; HarperCollins). Donald S. Fredrickson, MD, discusses the role of the scientific community in policing itself in *The Recombinant DNA Controversy: A Memoir* (2001; ASM Press). Victor K. McElheny covers sequencing the first human genomes in *Drawing the Map of Life: Inside the Human Genome Project* (2012; A Merloyd Lawrence Book). *The Gene: An Intimate History* by Siddhartha Mukherjee (2016; Simon and Schuster) relates the history of genetics. *Human Genetics: Concepts and Applications* by Ricki Lewis is a good basic human genetics textbook for non-science majors (2017; McGraw-Hill Education).

HOW GENES WORK

The 20,000 or so genes in the human genome encode a great diversity of proteins, which carry out many activities. Blood clotting, muscle contraction, digestion of food and absorption of its nutrients, and cell-to-cell communication are a few of the vital functions of proteins in a human body. Yet, a single pattern of information flow underlies the making of proteins from genetic instructions. Understanding what a gene does when it is working, and what happens when it is not, can suggest ways to treat genetic and other types of diseases.

Inherited illnesses affect us at several levels. The initial glitch, a mutation, occurs in a gene. A single mutation affecting one of the two copies of a specific gene is sufficient to cause a dominant illness, whereas mutations in both copies are required to cause a recessive illness. The genetic change alters a specific protein in a way that affects cells and then the tissues and organs of which they are a part. The individual is born with or develops symptoms, and then the illness may reverberate through a family and a population as others inherit it.

This chapter examines the levels of genetics through the lens of a single-gene disease, cystic fibrosis (CF) (Table 2.1). The disease is inherited from both parents, who are carriers or have the disease, and it affects males and females. Mutations that cause CF are in a gene, *CFTR*, that encodes a protein called the cystic fibrosis

Table 2.1 Levels of genetics

<i>Biological level</i>	<i>CF as an example</i>
Molecular	Mutation in the gene that encodes a cell-surface protein that forms channels for chloride and bicarbonate ions prevents the protein from reaching the cell surface or staying there long enough to function.
Cell	Disrupted ion transport draws water into cells lining the airways and ducts in the pancreas, dehydrating secretions as abnormal mucus collects. Also affected are the tubes that carry sperm, the cervix, and cells of salivary and sweat glands.
Tissue	Disrupted ion transport in linings (epithelium) and secretory tissue of digestive organs impairs breathing and clogs the pancreas.
Organ	Stringy mucus plugs the airways, intestines, pancreas, and gallbladder.
Individual	Difficulty gaining weight, inflammation and infection of airways, impaired digestion, diabetes mellitus, small sinuses, small gallbladder, infertility.
Family	CF is inherited as autosomal recessive (two parents are carriers or have CF).
Population	Modern geographical distribution of CF may reflect protection against tuberculosis.

transmembrane conductance regulator. Between the last edition of this book and this one, drugs have become available that help people with certain *CFTR* mutations by altering the protein in ways that enable it to function. We begin with considering the history of this intriguing inherited illness.

A BRIEF HISTORY OF CF

The third season premiere of BBC's television memoir *Call the Midwife* featured a distraught young mother. One-month-old Ian was losing weight and was fussy, writhing from abdominal pain. The baby was filling nappies with foul-smelling loose stools so fast that

even the father had to help, something unheard of in the poor, working class neighborhood of London's East End in the late 1950s and early 1960s. The tips of Ian's fingers were swollen from poor circulation, and he dripped, coughed, and sneezed nearly constantly, his little lungs wracked with unending respiratory infections. Both Ian and his toddler brother Martin choked frequently on thick phlegm.

Many viewers probably recognized the classic signs of CF, but the TV show's nurse/midwives, nuns, and lone doctor didn't. Everyone was deeply puzzled, until the clues started to appear.

First the young father told Nurse Jenny, the narrator, that his brother had died at age 4, and no one ever knew why. Did the 4-year-old brother and now Ian and Martin have the same condition? Then Sister Monica Joan, whom everyone considered to be demented, kissed the baby on the brow, commenting on the salty taste. Suddenly, she ran upstairs to the book collection that she spent her days meticulously cataloging, found something, and came back down with a volume, the German Children's Songs and Games of Switzerland, opened to a particular page. With a sly smile, she quoted:

Woe to that child which when kissed on the forehead tastes salty. He is bewitched and soon must die.

No one paid any attention to the silly saying from "Queen Anne's time," the 1600s. The young mother blamed herself. Perhaps her depression was making her neglectful, the good doctor, resting and enjoying a smoke, suggested as Nurse Jenny forcefully thumped on the baby's chest to dislodge the stifling mucus. Finally, the supposedly daft nun thrust the book into the hands of the doubting doc. Grudgingly, he read it and the light bulb went off. The boys had CF! The case was real, based on a sister and brother of an actress who portrays one of the midwives.

The nursery rhyme warning of the salty sweat of a "bewitched" child came from a professor of anatomy in Leiden, the Netherlands, who had recognized, in 1596, that such a child was likely to waste away and suffocate, and often had a swollen and rock-hard pancreas. The professor was also familiar with the stinky nappies.

During the 1930s, several clinicians named the combination of symptoms that clogged the lungs and pancreas "cystic fibromatosis

with bronchiectasis” and “cystic fibrosis of the pancreas.” One of them, Dorothy Andersen, a physician at Babies Hospital of Columbia Presbyterian Medical School in New York City, deduced in 1944 that the condition is autosomal recessive (Andersen 1944: 100), inherited from carrier parents. Five years later, when a sizzling summer’s end caused heat prostration among patients with “CF of the pancreas,” a young colleague, Paul di Sant’Agnese, discovered that the suffering children had three to five times the normal amount of salt on their skin (Kessler and Andersen 1951: 648; di Sant’Agnese et al. 1953: 549).

Researchers tried to turn the validated folklore of the salty kiss into a diagnostic test. Putting patients in big bags to get them to sweat enough to measure the salt could raise temperatures to deadly levels, so physicians developed the “sweat test,” still used today to evaluate response to drugs used to treat CF. In a sweat test, a chemical that induces sweating is applied to a small patch of skin on an arm, and an electrode delivers a weak electrical current to the area, which stimulates sweating. A warm or tingly feeling ensues for about five minutes. Then, sweat is collected for 30 minutes, on a piece of filter paper or gauze, and sent to a laboratory for analysis.

In the 1960s and 1970s, researchers closed in on a unifying factor behind the symptoms of CF. In these days before gene searches, secretions and excretions from patients—saliva, sweat, mucus, tears, and stools—were applied to all sorts of experimental systems, including fringed oyster cells, rabbit tracheas, rat salivary glands, and an obscure marine worm used as fish bait. What effects did secretions from people with CF symptoms have? Experimental findings suggested that CF disrupts what goes into and out of cells that secrete.

In 1981, researchers found the predicted salt disturbance in patients’ nasal passages (Knowles et al. 1981: 1489). Table salt (sodium chloride, NaCl) is dissolved in body fluids as equal numbers of sodium (Na^+) and chloride (Cl^-) ions (charged atoms). When CFTR is normal, Cl^- flows in and out of the cell, as water leaves by osmosis, thinning secretions. But when CFTR is abnormal, Cl^- can’t exit the cell. Water flow becomes disrupted, and the mucus outside the cell thickens as it dries out.

Suddenly, symptoms made sense: electrolyte (salt) imbalances set up a chain reaction by altering the pH (acidity/alkalinity) inside and

outside lining and secretory cells in certain places in the body. Cells lining the tubes and spaces of the respiratory system draw water in from secretions. Mucus changes chemically and thickens, beginning in the bronchial tubes and spreading downward into the lungs. Over time, the bronchi widen and become scarred and infected. In the pancreas, digestive juices dry out, cutting off enzyme delivery to the small intestine. The blocked enzymes begin to digest the pancreas itself, while tiny cysts and fats infiltrate the intestine, blocking the folds-within-folds where digested nutrients normally enter the bloodstream. Malnutrition results. In about 15% of newborns with CF, fat, mucus, and undigested proteins plug the small intestine, creating a condition called meconium ileus. Sweat is salty, saliva stringy. In some patients, the gallbladder fills with a green, jello-like substance.

Doctors discovered the various blockages of CF during autopsies, but finding the gene finally explained the pathology. The gene mapping that became possible in the 1980s narrowed down the search to about 2 million DNA bases on the long arm of the seventh largest chromosome (Knowlton et al. 1985: 380; Tsui et al. 1985: 1054). A multinational team led by Lap-Chee Tsui, MD, from The Hospital for Sick Children in Toronto, and Francis Collins, MD, PhD, at the University of Michigan, reported the discovery of the most common mutation, *delta F508*, in the *CFTR* gene, in cells from the sweat glands of patients, in 1989 (Riordan et al. 1989: 1066). Researchers have been identifying and describing variants of the gene ever since. The Hospital for Sick Children maintains a database that lists and describes the 2,000 mutations in the *CFTR* gene.

MIKAYLA'S STORY

Today, with prenatal and newborn screening for CF, diagnosis no longer takes months or even years. "We found out when I was about five months pregnant that Mikayla would have CF," recalls Donna Polski of Minneapolis, Minnesota, whose daughter was born on November 19, 2002.

I was having amniocentesis because I was over 35, looking for conditions related to the mother's age. The genetic counselor suggested CF testing because of our northern European background. We didn't have CF in the

family, so we weren't sure we really needed to take the test. Although recommended, it wasn't yet routine to test for CF in pregnancy. The insurance company approved, so I had a blood draw to "rule out" being a carrier, and was shocked to find out that the test came back positive. Next, my husband had a blood test that identified he was a carrier too. So the lab doing the amnio checked the amniotic fluid, and the next day, we got the phone call that our baby would have CF. We were really shocked.

Donna and her husband Michael were right to be surprised—CF affects only one in every 2,500 non-Hispanic Caucasian newborns in the US, where 30,000 people have CF. About 11,000 people have CF in the UK. The global figure is about 70,000, and millions of people are carriers. Mikayla inherited two copies of *delta F508*, known as a "double delta" genotype. Donna and Michael each have one copy of the allele.

Mikayla's medical team acted quickly, knowing that the pancreas and lungs can begin shutting down even before birth. Ultrasound scans of Donna's uterus already showed blockage in Mikayla's intestines. Doctors delivered the baby early, and performed surgery the next day to open up her small intestine, but it ruptured and had to be repaired. Mikayla spent her first month in the neonatal intensive care unit. Donna recalls,

They started big guns antibiotics right away, and pancreatic enzymes to help her digest fats. She came home mid-December. Two weeks later, she developed aspiration pneumonia and MRSA (methicillin-resistant *Staphylococcus aureus* infection), and so she was hospitalized for another month, and given intravenous antibiotics.

The hospital was to become a familiar place for the little girl, who contracted many viral and bacterial infections. With time, the visits became less frequent, perhaps because of the "bronchial drainage" treatments Donna and Michael performed twice a day. They would forcefully tap on their daughter's lungs, holding her in different positions, to loosen the sticky mucus that relentlessly built up.

Evan Scully, a young man living with CF in Ireland who is also a double delta, describes the lung-thumping regimen using an analogy that his parents originated.

CF is like a ketchup bottle. The ketchup won't come out unless you turn it upside down and belt the bottom of it until it comes out. It's like CF. If you don't do anything with your lungs, the "ketchup" is just going to stay in your lungs.

By the time Mikayla was 16 months old, she had grown enough to switch to using a vibrating vest instead of the twice-daily pounding. By the time she started primary school, she was accustomed to the daily routine (Figure 2.1). At night, a feeding tube provided extra calories. After she awakened, she wore the vibrating vest for 30 minutes, while using her laptop or watching TV. At the same time, she used a nebulizer to inhale antibiotics, bronchodilators, mucolytics to break up the thick mucus, steroids to quell inflammation, and hypertonic saline to draw fluid into the secretions. She received pancreatic enzymes through the feeding tube at night, or Donna opened pills and shook the enzymes onto a soft food such as applesauce. These enzymes are vital for digesting the fat-soluble vitamins. "We try to pack in the calories. She gets lots of butter, whipped cream, and high fat foods," said Donna. Mikayla went to



Figure 2.1 Mikayla Polski giving herself a nebulizer treatment for CF.

school, and then repeated the treatment regimen before dinner. When ill, Mikayla did her “vest/nebs” up to four times a day.

Outside the home, CF means constant awareness of pathogens. Studies with pigs genetically modified to have CF, so that researchers can examine affected tissues from the very first days after birth, indicate that bacterial infections begin immediately. At first, diverse bacteria infect the lungs, but the airways change shape and narrow as the disease progresses, forming nooks and crannies that attract microorganisms that rarely inhabit healthy lungs, such as *Pseudomonas aeruginosa*. Donna describes how vigilance against infection can affect a child.

Pseudomonas bacteria live in soil and stagnant water, like old hoses. We let Mikayla run through the sprinkler, but only if the hose is new. At pools, if the chlorine is too high, it affects her lungs, but if it is too low, she can get *Pseudomonas* or mold in her lungs. We have to be careful around bonfires, because of the smoke and particulates.

The dedication to Mikayla’s care has paid off, for she has made enormous strides. In 2010, she said, “I almost feel like a normal kid. I am having so much fun,” skating, swimming, and riding horses and jet skis. But her father, Mike, recalls when the situation was much more dire. “There were times we didn’t know if she would make it through the night—but she fought and she did.”

Today, Mikayla’s activities continue to overshadow her CF. Her Facebook page shows her doing headstands in the snow, performing in dance contests, and playing a flute. Recent sinus surgery has helped, and she isn’t taking as many medications as she did as a young child, although she needs an Albuterol inhaler for heavy aerobic activity. She may qualify for the new CF drugs if her lung function improves sufficiently.

With luck and perseverance in preventing symptoms, Mikayla may live several decades longer. This wasn’t always so. In 1955, most youngsters with CF didn’t survive to enter kindergarten, succumbing to malnutrition more often than dying from failing lungs. Today the median survival age for all people with CF is 37 for females and 40 for males, reflecting hormonal differences. But for children born during the past few years, median survival age is 54 for females and 58 for males! The brighter picture is due to the new drugs that correct

the misfolding of CFTR protein, but also to improvements in diagnosis that are finding people with milder cases, or who have variants of the disease that do not affect the lungs and therefore escaped diagnosis years earlier.

INSIDE THE CELL

Sprinkled among *CFTR*'s 250,000 DNA bases in a person who does not have CF are 6,129 bases, distributed in 27 segments, that encode the 1,480 amino acids of CFTR protein. On both copies of Mikayla's chromosome 7, however, the *CFTR* gene is missing the same three DNA bases in a row, which encode the 508th amino acid in the sequence of 1,480. That amino acid is phenylalanine, the first piece to the puzzle of the genetic code discussed in Chapter 1. The name of her mutation—*delta F508*—means "deleted at position 508." (Italics denote a gene; roman letters denote a protein.) Mikayla's double delta mutations are in each of the trillions of cells of her body, but cause her symptoms by affecting specific types of cell that use CFTR protein. Other mutations shorten the protein or substitute specific amino acids.

MAKING A PROTEIN

Imagine a cell lining a bronchial tube leading to one of Mikayla's lungs. The cell arose when another lining cell divided, replicating its DNA and passing on the faulty instructions missing the crucial trio of bases in the *CFTR* gene. The gene is transcribed into messenger RNA (mRNA), which is snipped and stitched into its final 6,126 nucleotides. The spliced mRNA then exits the nucleus through a nuclear pore, entering the gel-like **cytoplasm** where structures called **organelles** compartmentalize the cell (Figure 2.2). Some organelles function as molecular machines, with specific jobs. The mitochondrion, for example, houses the energy reactions of the cell. Some organelles sequester biochemicals that might harm other cell parts. A lysosome, for example, is a sac of enzymes that dismantle debris. If the enzymes were free in the cytoplasm, they would digest the cell!

Several types of organelle form a protein-processing network. The components of the network fold proteins, add sugars or fats to some of them, and then hustle them on their way. Some proteins work

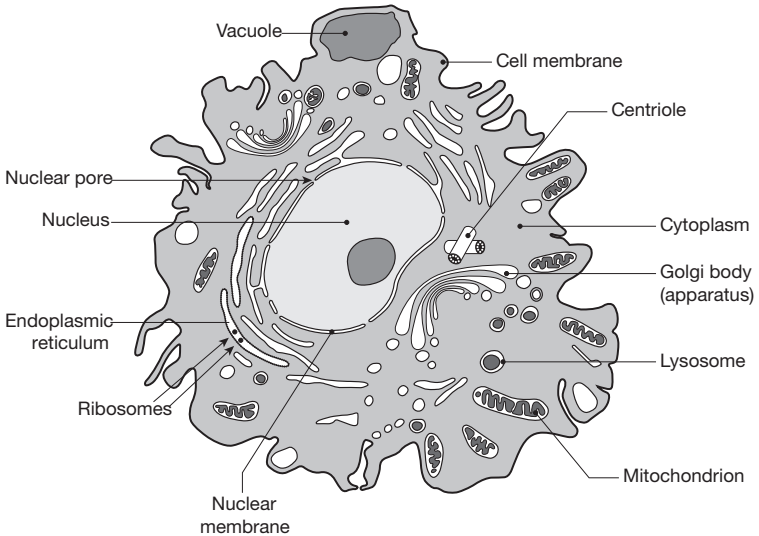


Figure 2.2 The anatomy of a cell. Organelles are specialized structures that divide the work of the cell, increasing efficiency and protecting delicate cell parts.

Source: Adapted from *Applied Genetics in Healthcare*, Figure 2.15, p. 28

inside the cell or in the cell membrane that defines the cell. Other proteins are packaged into oily bubbles, called vesicles, for jettisoning from the cell as a secretion. The secretory network consists of the endoplasmic reticulum (ER), ribosomes, and one or a few Golgi bodies. The same complementary base pairing that lies behind DNA replication, discussed in Chapter 1, also guides protein synthesis. A binds with T (in DNA) and U (in RNA), and G binds with C (in DNA and in RNA).

Production of CFTR illustrates the steps in how cells manufacture proteins. The process begins as the mRNA molecule exits the nucleus and contacts ribosomes that stud the ribbon-like ER. Ribosomes, made of ribosomal RNA (rRNA) and proteins, are staging areas for the assembly of a protein. First, an rRNA pairs with a few bases at the start of an mRNA in a way that exposes the mRNA's entire base sequence. Then, molecules of a third type of RNA, transfer RNA

(tRNA), form complementary base pairs with the mRNA. Transfer RNAs are small, fold into a cloverleaf shape, and come in several different varieties, each type bonded to one of the 20 types of amino acid. The key to tRNA's role as a "connector" is its specificity: each of three bases in a row in the mRNA—a codon—attracts the complementary three bases of part of a tRNA called its anticodon. A tRNA with a particular anticodon always carries the same type of amino acid.

Constructing a protein is a little like manufacturing an appliance on an assembly line. The protein must not only have the correct components added in the correct order, but it must fold into a specific three-dimensional form, or conformation. Consider the section of CFTR mRNA that is the site of Mikayla's *delta F508* mutation:

...AUCUUUGGU...

In the normal (wild type) *CFTR* allele, these three mRNA codons attract tRNAs that bring in the amino acids isoleucine (AUC), phenylalanine (UUU), and glycine (GGU). (Figure 1.5 has the codon-amino acid correspondences of the genetic code). The *delta F508* mutation does not have the UUU. Normally, ribosomes deliver CFTR into the labyrinth of the ER, where the proteins move in the winding channels toward the cell's outer surface (cell membrane). In Mikayla's cells, this transport halts, because the one absent amino acid disrupts the folding of CFTR proteins into their final form. This is a major problem.

PROTEIN FOLDING

Proteins must fold a certain way to be active. This is so important that cells have several mechanisms to police themselves and either discard or refold misfolded proteins. "Chaperone" proteins, for example, peruse forming proteins for misshapen local regions and correct them. If misfolding persists, another protein, called ubiquitin, attaches and herds the doomed protein towards a spool-like structure called a proteasome. The errant protein is threaded through the proteasome and, like food being pushed through a garbage disposal, is torn into pieces. The double *delta F508* CFTR proteins that Mikayla makes are dismantled before they even leave the ER to reach the Golgi

Table 2.2 The events of protein synthesis: Making CFTR

-
1. Cell receives signals to produce CFTR proteins.
 2. *CFTR* gene is transcribed into mRNA, which is spliced into a final form and enters the cytoplasm.
 3. CFTR mRNAs bind ribosomes. tRNAs bring in the appropriate amino acids, building CFTR protein, which enters the ER.
 4. Ubiquitin yanks misfolded CFTR proteins from the ER and sends them to proteasomes for destruction.
 5. Normal CFTR proteins continue on to a Golgi body, picking up sugars (glycosylation).
 6. CFTR proteins are shuttled in vesicles to the cell membrane, where they embed and form ion channels when ATP binds and splits.
-

body. In her parents' cells, only half of the CFTR proteins are taken apart, because her mother and father each have one wild type allele.

A normal CFTR protein continues through the ER, emerges in a vesicle, and glides to the next stop, the glistening stacks of a Golgi body. Here, the protein picks up sugars; CFTR is actually a *glyco*-protein. Sugared CFTR protein is then shuttled to the cell membrane, where it nestles into the double lipid layer that separates the cell's insides from the extracellular environment. When the protein binds ATP, the biological energy molecule, it contorts, forming a channel, like a lump of dough being shaped into a doughnut. Then ATP splits, releasing energy that powers the transit of chloride ions into and out of the cell. CFTR also regulates other ions, including sodium (Na^+) and bicarbonate (HCO_3^{-1}). Table 2.2 summarizes the cellular events that go awry in CF.

DIFFERENT MUTATIONS, DIFFERENT EFFECTS

Because a gene carries information, it can vary, just as letters in a sentence vary when keying errors occur. The Cystic Fibrosis Mutation Database (www.genet.sickkids.on.ca/app) lists more than 2,000 variants in the *CFTR* gene, but only half of them affect health, and exactly how they do so is understood for only 127 mutations. Mutations can disrupt a protein in several ways: altering the length or composition of the amino acid sequence; adding sugars or fats; or changing production rate, transport, or stability.

Knowing a person's exact genotype for *CFTR*—that is, which alleles are present—is clinically important, because new drugs are effective for some mutations but not others. Nearly 90 percent of people with CF have at least 1 *delta F508* mutation, and only 11 other mutations are present in more than 1 percent of CF patients worldwide. The first drug that treats CF by affecting the protein, and not just ameliorating symptoms like breaking up mucus or dilating airways, is ivacaftor (Kalydeco), which became available in 2012 to treat the 4 to 5 percent of patients who have a mutation called Gly551Asp (it changes a glycine to an aspartic acid at position 551 in the protein). In these individuals, *CFTR* reaches the cell membrane, but can't unfold to form the ion channel. The protein can't function, like a car going down a driveway and then stalling. Results of taking the drug can be dramatic: less-salty sweat, noticeable weight gain, and improved breathing. Some patients report that an early sign that the drug is working is improvement in the odor of their flatulence!

Ivacaftor alone does not help the double deltas like Mikayla, in whom misfolding causes 98 percent of the protein to be taken apart in the ER. Most of the few *CFTR* proteins that reach the cell surface don't open, and the scant few that do are recycled too quickly to have an effect. If a patient has one *delta F508* mutation and another, different one, sometimes ivacaftor can get more proteins to the cell surface. However, a second drug, lumacaftor, works in a different way and allows more *CFTR* proteins to reach the cell surface and stay there longer. Teaming the two drugs into one (marketed as Orkambi) can help some double-deltas, and still other drugs are in development.

THE INDIVIDUAL LEVEL: TISSUES AND ORGANS

Mutations affect cells, which affect the tissues and organs of which they are a part, producing the distinctive signs and symptoms of a genetic syndrome. For some conditions, the connection between mutation and health is obvious. A person missing a clotting factor gene bleeds and bruises easily. A boy with Duchenne muscular dystrophy (DMD), missing a protein his muscle cells need to withstand contraction, grows weaker.

Developing a new treatment for a genetic disease is challenging, because while the compound may correct or compensate for an

abnormality at the molecular level, the drug must make a difference to the patient. In the US, the Food and Drug Administration at first did not approve two drugs that treat DMD because even though they enable cells to make dystrophin, improvement at the level of the patient, measured as distance walked in 6 minutes, did not exceed the variability seen in a natural history study, which observes the extent and pace of worsening symptoms over time. Study design is important. The DMD clinical trials in the US might not have continued long enough, nor treated enough cells in the boys, nor compared the boys to equivalent controls, to see significant improvement in walking. One of the drugs was approved in the UK.

When a new CF drug works, response at the individual level is fast and unmistakable. Don Bowler has chronicled his young son Dean's experience with ivacaftor on YouTube. In the first post, from February 2012, Dean tries to open the bottle containing the pills but is too weak to do so. Just 2 months later, he says, "I feel like I don't have CF anymore" and then demonstrates his basketball skills. In that short time, Dean gained 9 kilograms and his lung function increased from 59 to 80 percent. He has continued to do well.

In addition to the new drugs to treat CF, the UK Cystic Fibrosis Gene Therapy Consortium is conducting clinical trials to introduce healthy versions of *CFTR* into the lungs of patients in fatty bubbles called liposomes. So far, the inhaled treatment stabilized some patients, while those given placebo deteriorated. Gene therapy to treat CF has been difficult because of the rapid turnover of cells lining the airways. The advantage of gene therapy is that patients with any mutation can be treated, but it is not as effective as the new drugs.

THE PHENOTYPE

An illness is **pleiotropic** if signs and symptoms—the mutant phenotype—affect different body parts. CF is a classic example, because abnormal *CFTR* protein affects the composition of several body fluids, including sweat, saliva, mucus, tears, and digestive enzymes. A direct symptom may have secondary consequences. For example, blocked delivery of digestive enzymes to the small intestine causes vitamin deficiencies, malnutrition, abdominal pain, and fatty, greasy bowel movements. CF is best known for hampering respiration, but less well known is that in about 99 percent of men

with the disease, thick mucus blocks the sperm-carrying tubes, which degenerate. Fathering a child becomes impossible. In women, CF thickens the secretions around the cervix. For some *CFTR* mutant genotypes, infertility may be the only symptom.

The definition of CF continues to evolve as more mutations and mutation combinations are identified and their phenotypes described. A person with severe asthma, or frequent bronchitis or sinusitis, may in fact have CF. Some investigators suggest that CF be considered two diseases, distinguished by whether or not the lungs are affected. At least nine *CFTR* mutations spare the lungs, but cause male infertility, pancreatitis, and/or sinusitis. In the past, a diagnosis for CF would never have been considered for a man with these symptoms, but with working lungs.

OUTSIDE INFLUENCES

Inheriting two mutations in the *CFTR* gene can cause CF, but daily life with the disease can vary greatly among individuals, in severity, symptoms, and rate of progression, due to the actions of other genes and environmental influences. This makes sense. A child with CF will have more severe respiratory problems if exposed to air pollution or pathogens, or if she lives in a climate with sustained high heat that impairs breathing. Two patients may have respiratory infections, but the species of bacteria inhabiting one person's lungs may be very different from the bacterial communities in another. When symptoms worsen, the species of bacteria common in people with CF tend to die as rare species come to dominate, such as *Streptococcus anginosus*. Testing sputum for DNA sequences unique to specific bacterial species can guide physicians in choosing appropriate antibiotics. More than 60 species of bacteria reside in the sinuses, airways, and lungs of CF patients. Because of the ever-present pathogenic bacteria and the narrowing and distortion of airways (bronchiectasis) that happens as CF progresses, common infectious diseases that are generally mild among healthy people, such as influenza, can be deadly in those with the inherited disease.

The environmental component of CF empowers many people with CF to use exercise to ease symptoms. One young woman finally began to have days without feeling as if she had a severe cold when she began running—"With every step, I'm beating CF," she says.

When Evan Scully, the young man from Ireland who is a double delta like Mikayla, was 6 months old near the end of 1986, his “failure to thrive” and coughing led to a diagnosis of CF. Doctors told his shocked parents that he wouldn’t survive to see his tenth birthday. “Obviously they were wrong,” says Evan today, who has never been sick enough to require hospitalization. He credits his parents’ strategy of listening to his body for both his healthy lifestyle and his career as a sports therapist, in which he helps athletes avoid and recover from injury through nutrition and exercise. Evan eats vegetarian during the week and meat at weekends, and is so active that he was running 120 miles a week when he ran competitively for Ireland. “I do a 3-day exercise cycle: an hour of running on day 1, gym workout on day 2, and yoga and movement on day 3.” He has reduced the number and doses of medications he takes, and his lung function has steadily improved. Ironically, Mikayla’s lung function is too poor to qualify to take the new drugs, but Evan’s is so good that he has declined taking them.

Still, a healthy lifestyle doesn’t make a genetic disease simply vanish, so Evan must be vigilant at keeping up his routine. In 2015, he contracted three infections after traveling the world while working with an Olympic athlete. He missed 10 days of running and didn’t do any airway clearance exercises. Plus, he was exposed to air-conditioning, which can trigger chest inflammation. Evan was scared.

I started doing everything possible to clear it because I had a feeling I was on my way to hospital for the first time. I knew I needed to get back out there. My first run back was 1 mile. It took me 12 minutes and I was dying doing it!

Soon, though, he broke 40 minutes in a 10-kilometer run, his goal. “I’m prone to infections, but because I don’t let mucus lay in my body, it doesn’t give the infections a place to house. A lot of it is down to mental strength and positivity.”

Genes other than *CFTR* affect the symptoms and severity of CF. Meconium ileus, the blocked intestine discovered in Mikayla before she was born, occurs if an individual inherits a mutation in a gene called *CFM1* along with *CFTR* mutations. Mutations in a gene whose protein product interacts with CFTR can cause CF-like symptoms. This is the case for a protein called filamin A, which attaches CFTR

proteins in the cell membrane to the protein network beneath it. As more human genomes are sequenced and annotated, more such “modifier genes” are being discovered. Identifying them can aid treatment selection.

THE FAMILY LEVEL

Remember Mendel? The laws he derived from following the inheritance of traits in peas whose genes are on different chromosomes apply to any species whose cells have two copies of each chromosome, a state termed diploid. Tall plants that, when crossed, give rise to tall and short offspring in the ratio 3:1 illustrate the same pattern of trait transmission as Donna and Michael Polski, who each carry a *CFTR* mutation. If Donna and Michael were able to have many offspring, as pea plants do, the chances are that about a fourth of them would inherit CF, according to Mendel’s law of segregation. Each child they conceive faces a one in four chance of inheriting CF, like Mikayla, a one in two chance of being a carrier, and a one in four chance of being wild type. Mikayla’s older brother has no *CFTR* mutations, but Evan has several siblings who are carriers.

INHERITANCE PATTERNS

The inheritance pattern seen among the tall and short pea plants, and in a family with CF, is **autosomal recessive**. “Autosomal” refers to the location of the gene on a non-sex chromosome, or **autosome**, which does not carry genes that determine sex. Recessive refers to an allele (gene variant) that is only expressed when inherited in two copies, one from each parent. In a small family, a homozygous recessive individual may not appear for several generations, as carriers silently pass on the mutant allele. An autosomal recessive disease or trait may be more obvious in larger families. One woman, for example, was diagnosed with mild CF as a college student, when an astute physician tested her upon learning of her history of frequent sinusitis and bronchitis. She convinced her six siblings to be tested, and three of them had CF!

Geneticists use a diagram called a **pedigree** to trace traits and illnesses in families (Figure 2.3). It uses squares to represent males and circles to represent females, with lines to show the relationships

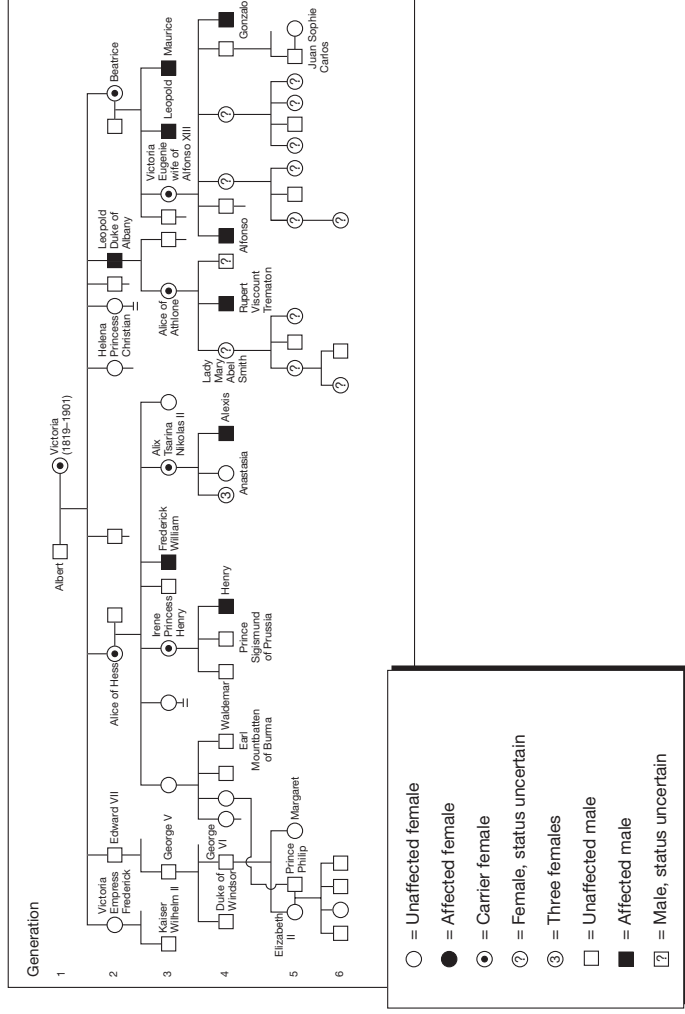


Figure 2.3 A famous pedigree. This pedigree shows the inheritance of the clotting disorder, hemophilia B, from Queen Victoria of the British royal family to other royal families.

Source: *Introducing Genetics: From Mendel to Molecule*, Figure 6.8, p. 87

of family members. Vertical lines depict generations, horizontal lines emanating from the centers of the squares and circles indicate parents, and horizontal lines above the shapes represent sibling relationships. Heterozygotes are depicted as half-filled-in shapes, and individuals who express a mutant phenotype as filled-in shapes. Additional labels are used to follow multiple traits.

Autosomal recessive is one mode of inheritance, which is the pattern of trait transmission (Table 2.3). A second mode of inheritance is **autosomal dominant**, in which only one mutant allele need be inherited to express the trait, and it is present in both sexes. In families in which one parent is affected, each child has a one in two chance of inheriting the condition.

Traits inherited from genes on the sex chromosomes are termed X-linked or Y-linked. A female has two X chromosomes, and a male has one X chromosome and one Y chromosome. A male inherits an X-linked trait or condition from his mother, and his Y chromosome from his father. The bleeding disorder hemophilia B (factor IX deficiency) is a classic example of an X-linked recessive condition. A son who inherits an X-linked recessive mutation is affected because he does not have a second X chromosome to mask the mutant allele.

X-linked dominant conditions are quite rare and are so severe that affected males do not survive to be born, and females are severely affected. An example is incontinentia pigmenti. It causes swirls of skin pigmentation and other symptoms in females, but males do not survive to be born. Y-linkage is rarely seen, because the Y chromosome has very few genes.

Table 2.3 Modes of inheritance

<i>Mode</i>	<i>Description</i>
Autosomal recessive	An allele on a non-sex chromosome must be present in two copies to affect the phenotype.
Autosomal dominant	An allele on a non-sex chromosome must be present in one copy to affect the phenotype.
X-linked recessive	A male expresses a trait whose gene is on the X chromosome because he does not have a second allele to mask it; a female requires two copies of the allele to express the trait.

Whether an allele is dominant or recessive depends somewhat on what we can measure or detect. A carrier of an inborn error of metabolism may be healthy but have only half the normal amount of the biochemical that the implicated enzyme would normally break down. Sometimes, a carrier has mild symptoms of the associated illness, a situation called incomplete dominance. This is the case for CF. Some carriers are prone to asthma, perform below average on lung function tests, and make half the normal amount of sweat (Dahl et al. 2005: 1911).

FROM GENOTYPE TO PHENOTYPE

Phenotypes of the same disease can vary, even within a family (Table 2.4). Two terms describe this variability: **penetrance** and **expressivity**. Penetrance refers to the percentage of individuals with a particular genotype who express the associated phenotype. It is all-or-none, and for a pleiotropic condition such as CF, penetrance applies to specific symptoms. Penetrance is high, for example, for impaired breathing and digestion, meaning that most people with CF have these symptoms. Less common manifestations, such as diabetes and blocked intestines, have lower penetrance.

Expressivity refers to the severity of a symptom. It is low in a person with CF who has occasional bronchitis, but high in someone who is often hospitalized with severe lung infections. Nearly all single-gene traits and conditions are “incompletely penetrant and variably expressive,” which means that not all the people who inherit a particular genotype develop the associated disease, and, for those who do, severity varies. Add pleiotropy, and single-gene inheritance in a family can be confusing, as different individuals have different

Table 2.4 Aspects of gene expression

<i>Term</i>	<i>Definition</i>
Pleiotropy	The association of several symptoms with a genotype.
Penetrance	The percentage of individuals with the same genotype who express the phenotype.
Expressivity	Variability in symptom severity among people with the same genotype.

experiences. CF may manifest in a large family as a child frequently hospitalized with pneumonia, a sibling with sinus infections, and a male cousin with infertility—even doctors might not realize they have the same disease.

THE POPULATION

The level of the population is where genetics intersects history, anthropology, and human behavior. The distributions of alleles in different societies hold telling clues to the narratives of humankind, past, present, and future.

HOW POPULATIONS CHANGE

At the population level, genetic information takes the form of allele frequencies. All of the alleles in a population constitute the **gene pool**. Because allele frequencies are almost always changing, gene pools are in constant flux. When two populations change enough genetically that they can no longer interbreed, they become separate species.

Five processes change allele frequencies: mutation, migration, genetic drift, non-random mating, and natural selection (Table 2.5). These factors are not equally important. Mutation, for example, is very rare. More commonly, migrants introduce novel alleles into populations. A city such as New York or London is continually being remade, in a genetic sense, as people bring their distinctive alleles from different parts of the world.

In **genetic drift**, part of a population becomes genetically distinct when its members reproduce only among themselves, even if they are within a larger group. A dramatic form of genetic drift, called a founder effect, occurs when a few people leave one settlement to begin another. The inhabitants of the Faroe Islands of Denmark reveal a classic founder effect—all people with CF are “double deltas,” like Mikayla and Evan, and are possibly all descended from the same individual! A founder effect also happens when people marry only people of the same ancestral group, such as members of religious communities or immigrants from the same part of a country, within a large city.

Table 2.5 Forces behind population change (evolution)

<i>Force</i>	<i>Definition</i>
Mutation	A new allele forms, such as a DNA base substitution.
Migration	Individuals bring new alleles into a population, or remove alleles.
Genetic drift	A subpopulation sequesters certain allele combinations from the larger group.
Non-random mating	Some individuals contribute more alleles to the next generation than others.
Natural selection	A phenotype confers a reproductive advantage or disadvantage.

Another type of genetic drift, called a population bottleneck, occurs when a disaster kills most members of a population, leaving a few behind to replenish the gene pool. The genetic structure of the new group is unlikely to be exactly the same as the first. Due to chance sampling, some gene variants may have been eliminated while others become more common. A global population bottleneck is a classic science fiction theme. To escape a doomed Earth, a handful of people board spaceships, or hide in huge arks or underground, or are shielded by a lucky accident, until the environmental upheaval abates. Table 6.3 lists science fiction plots that borrow from population genetics.

In non-random mating, one individual contributes more to the next generation than others by having many offspring, altering allele frequencies. Non-random mating is actually a goal in agriculture, where one genetic variety of crop or one bull with valuable sperm becomes the sole source of genes, to perpetuate traits that are valuable to us. In humans obviously males can have a greater influence than females via non-random mating.

Natural selection is the most powerful force that molds population genetics, and therefore evolution, which is why it is the subject of Charles Darwin's *Origin of the Species*. Natural selection refers to the increased representation, in a population, of an inherited trait that makes an individual more likely to successfully reproduce. Darwin's phrase "survival of the fittest" refers to reproductive success, not physical prowess.

WHY CF IS COMMON IN SOME PLACES,
RARE IN OTHERS

When Donna Polski told the genetic counselor that her and Michael’s families came from Norway, Sweden, and Finland, the counselor was alerted. She knew that both the incidence (number of new cases each year) and prevalence (percentage of affected individuals in the population at a given time) of CF are highest in northern Europe, where about 2 percent of the population carries a mutant CF allele. CF is much more common in some parts of the world than others (Table 2.6).

The principles of population genetics explain why CF and other inherited diseases affect different populations to different degrees. Changes in allele frequencies in one direction, such as from west to east, indicate long-ago migrations and where and when people stopped and had children with native residents, introducing new alleles while replacing some alleles in the established community. This directional change in allele frequency is called a **cline**. The *delta*

Table 2.6 Incidence of CF in selected populations

Population	Incidence
Albania	1/555
Australia	1/2,021
Basque	1/4,500
China	1/90,000
Faroe Islands	1/1,775
Iceland	1/8,344
India	1/40,000
Italy	1/4,238
Japan	1/350,000
Jordan	1/2,560
Sweden	1/5,600
UK	1/2,415
US:	
Caucasian non-Hispanic	1/2,500
Hispanic	1/13,500
African-American	1/15,100
Asian-American	1/35,000
Ashkenazi Jewish	1/2,271

F508 CF allele follows a clear cline, falling in frequency from northwest to southeast in Europe, parallel to known migration patterns.

Once a disease-causing allele is introduced into a population, its specific effect on reproduction influences whether it increases or decreases in frequency. However, in some instances, an inherited disease persists because in addition to the associated symptoms, it offers an advantage. Such a benefit may be why CF is much more common among Europeans and those of European ancestry than other groups. The reason is not genetics, but protection against an infectious disease—tuberculosis (TB). The strongest clue to CF's prevalence is the predominance of the *delta F508* mutation in many populations. Globally the mutation is present in 75 percent of people known to have CF. It is unlikely that the exact same allele would arise anew from mutation, repeatedly, just by chance, in different places at different times. Nor could a founder effect account for the gradual nature of the cline. A more plausible explanation for the prevalence of the mutation is a variation of natural selection called **balancing selection**, in which carriers are better able to survive because they resist a second illness. The superior health of carriers keeps the disease-causing allele in the population.

The classic example of balancing selection is between sickle cell disease, which is inherited, and malaria, which is an infectious disease that causes cycles of high fever as the pathogen curls up inside human red blood cells. Sickle-shaped cells keep the pathogens out, and carriers apparently have enough of the spiky cells to keep malaria away, but not enough to disrupt circulation, as sickle cell disease does. Geographical maps of where sickle cell disease and malaria are most prevalent coincide, in Africa, revealing the balancing selection. Sickle cell carriers are healthier than others in places where malaria kills, and have more children, perpetuating the mutation. Chapter 4 discusses sickle cell disease further.

How does CF protect against TB? Balancing selection arises from altered processing of an enzyme called arylsulfatase B. In CF, the abnormal ion channels change the acidity in the cell in a way that affects folding of the enzyme into its active form. However, researchers followed two dead ends before arriving at the TB connection. The first was cholera, a water-borne infection responsible for outbreaks that rapidly kill thousands.

Being a CF carrier might protect against a diarrheal disease because CFTR protein affects the composition of body fluids, and diarrhea is a massive loss of water from the digestive tract. Might CF carriers resist cholera? The bacterium that causes the diarrheal disease, *Vibrio cholerae*, under certain conditions produces a toxin that binds to normal CFTR ion channels in intestinal lining cells. This releases water in a diarrheal rush that can cause circulatory collapse, and death, in just hours or days. As *delta F508* carriers do not have the normal number of CFTR proteins opening out through their cell membranes, perhaps in them cholera toxin cannot bind enough cells to cause diarrhea. However, unlike the elegant sickle cell/malaria tale, geographical maps of cholera and CF do not align, and cholera arrived in Europe too recently—in 1831—to have had such a powerful effect in selecting *delta F508* alleles (Quinton 2007: 212–215).

The diarrhea idea persisted, and researchers followed a second dead end. Could CF carriers resist a different condition, typhoid fever, caused by the bacterium *Salmonella typhi* (Pier et al. 1998: 79)? Again, the bacterial toxin binds CFTR ion channels, but the maps of geographic distribution of the two diseases do not align. In fact, some populations riddled with typhoid fever, such as Indonesia, have no *delta F508* mutations at all.

Researchers went back to brainstorming why and how CF carriers were at an advantage. Balancing selection must satisfy three criteria:

1. a plausible molecular mechanism;
2. clinical correlation between the genotype and the infectious illness;
3. similar geographic distribution of the two diseases.

The last criterion rules out cholera and typhoid fever, but TB fulfills all three.

- *Molecular mechanism*: People with CF lack arylsulfatase B activity; carriers have half the normal amount. The TB bacterium, *Mycobacterium tuberculosis*, requires the enzyme. Therefore, the bacteria can't survive in a human body without the enzyme.

- *Clinical correlation between genotype and infectious disease:* CF carriers have lower TB mortality rates, and people with CF rarely get TB.
- *Similar geographic distributions:* The longer TB has been in a region, the greater the proportion of CF alleles that are *delta F508*.

TB has a long enough legacy to have fueled the increased prevalence of the protective *delta F508* allele. Present for at least 15,000 years, TB became very deadly from 1600 to the late 1900s, killing 20–25 percent of the European population. Growing cities fueled the spread of TB. Today's geographical distribution of *delta F508* tracks with the infectious disease: where TB is most prevalent, so are CF carriers. TB did not reach India or Africa until the late 1800s, and here *delta F508* is rarer. Before the twentieth century and the discovery of antibiotics, TB was a deadly, systemic illness that killed people before they could reproduce. The infectious disease, therefore, would have preferentially removed people *without* protective CF mutations from the population, leaving the legacy of CF today.

Clues to the time and place of origin of different CF mutations lie within the DNA sequences elsewhere in the gene, and in surrounding regions on chromosome 7. A gene and its neighboring DNA sequences constitute a **haplotype**. All *delta F508* chromosomes have the same haplotype, which points to a single origin of that mutation, because a gene and its neighbors are usually transmitted together on their shared chromosome. A modern population group where the *delta F508* haplotype is the most prevalent is likely an original population. Several investigations point to the Baluchis, an ethnic group from western Pakistan that emigrated to the United Arab Emirates, eventually reaching Europe (Saleheen and Frossard 2008: 157), as the population in which *delta F508* originated. Today, their descendants live in Iran, Afghanistan, and Pakistan. The *delta F508* haplotype exploded in European populations under the selective pressure of TB. Other CF mutations have other origins—such as ancient Phoenicia and among the Celtic tribes.

WHAT WOULD YOU DO? *THE HIGH COST OF NEW DRUGS*

When Kalydeco, the very successful drug to treat CF, was approved in 2012, it cost £182,000 (\$270,579). The National Health Service (NHS) initially refused to cover it, but did so after the Cystic Fibrosis Trust sent a request with 12,000 signatures to the agency and Vertex Pharmaceuticals, the manufacturer, asking to lower the price. By 2013, the NHS provided the drug to people with the “Celtic” mutation G551D—about 4 percent of the UK’s 9,000 people with CF. By 2015, coverage extended to 10 mutations.

In the US, the Cystic Fibrosis Foundation, which funded development of Kalydeco, sold the royalty rights, which generated enough money to help many patients and fund further research. But for some of the 75,000 people in North America, Europe, and Australia with CF, paying for costly new drugs is still a problem, and patient activism has been key in getting funding.

Madi Vanstone’s life has turned around on Kalydeco—but she very nearly didn’t get it. Before June 2014, when the Ontario Drug Benefit Formulary announced that they would cover the cost of the drug for people with treatable mutations, her family had to raise \$60,000 a year to supplement what Madi’s dad Glenn’s health insurance would cover, plus a grant from the drug company. Madi was then 12 years old, and upset at the thought of discontinuing the drug that had finally given her a near-normal life. Her Mother Beth wrote on the fundraising website,

Madi has been on Kalydeco since August 2013 and has never felt better. She now looks forward to an amazing long healthy life as long as she remains on this drug. Madi has been advocating hard for herself and others who need coverage for Kalydeco.

Madi goes to school and loves to ski and ride horses. Without the drug, she would be enduring frequent hospitalizations and would eventually have needed a lung transplant to survive. Now, she has a future.

Many new drugs to treat genetic diseases have been approved over the past few years. What criteria do you think insurers should use to prioritize which patients receive a very expensive new drug?

SUGGESTED READING

This chapter has explored the levels of genetics through the lens of a disease with a simple inheritance pattern: the autosomal recessive condition cystic fibrosis. Understanding the mechanism of how a change in a gene alters secretions in many parts of the body led directly to development of successful drug therapies. To learn about the many genetic mutations in the *CFTR* gene, see the CF Mutation Database at www.genet.sickkids.on.ca/app. The websites for the Cystic Fibrosis Trust (www.cysticfibrosis.org.uk) in the UK and the Cystic Fibrosis Foundation (<http://cff.org/>) in the US have terrific updated information on living with the disease. For a look at what life was like for a child with CF before many of the current treatments were developed, read the book *Alex: The Life of a Child*, by Frank DeFord (Viking Press, 1983), which chronicles the declining health of the author's 8-year-old. Two articles in the *New England Journal of Medicine* detail how the new CF drugs work (Davis, P.B. [2015] "Another beginning for cystic fibrosis therapy," *New England Journal of Medicine*, 373: 274–276; and Stoltz, D.A., Meyerholz, D.K., and Welsh, M.J. [2015] "Origins of cystic fibrosis lung disease," *New England Journal of Medicine*, 372: 351–362).

GENES AND HEALTH

Until we could sequence human genomes, the effects of genes on health were considered one gene, and one disease, at a time, such as cystic fibrosis. Instead of looking at illness and deducing what went wrong, scrutinizing entire genomes is leading to better understanding of what it means to be well. This chapter opens with a look at what we are learning from sequencing human genomes, and then considers the three most common classes of disease worldwide: cardiovascular disease (the heart, blood, and blood vessels), cancer, and infectious disease.

A GENOMIC VIEW OF HEALTH

Much of what we know about good health comes from those who live long and well. Several projects are sequencing the genomes of centenarians (over age 100) and supercentenarians (over 110). The UK has more than 14,000 centenarians and the US more than 54,000. Supercentenarians worldwide that we know of number fewer than 100. What are their secrets?

THE OLDEST OLD

Those who live the longest have been lucky. They inherited gene variants that protect them from the diseases that claim most lives at

much younger ages—cancers and heart disease—never suffering from them, developing them very late in life, or surviving them. Some very elderly people have inherited gene variants that protect their brains from the protein buildups that cause dementia (Schwarz et al. 2016: 74). These clearer-thinking elders are more available to help their grandchildren and great-grandchildren, giving them an advantage—this is natural selection at work.

Centenarian studies are following people in New England, Hawaii, and Okinawa, and some efforts are more specific, such as investigating Japanese-American men and Ashkenazi Jewish people. These studies are separating the genetic from the environmental influences on longevity. More than 130 genes are known to affect how long a person lives. These genes affect lipid (fat) metabolism, the cell cycle, nutrient usage, levels of antioxidant enzymes, functioning of the immune system, DNA repair, insulin secretion and glucose metabolism, and response to stress. On the environmental side, exercising, not smoking, and maintaining a healthy weight promote long life. Those who live the longest tend to have had good food, excellent health care, and devoted family members.

Individual cases of extreme longevity are fascinating. Consider Hendrikje van Andel-Schipper. At 115 years and 62 days when she died on August 30, 2005, “Hennie” was, at the time, one of 72 super-centenarians known in the entire world, 68 of them female. She died of stomach cancer, and had survived breast cancer 15 years earlier. Hennie spent her final years in a nursing home but was healthy and very alert, if frail, to the end.

The list of conditions that Hennie did *not* have was long. Her brain was oddly devoid of the plaques and tangles that are the hallmark of aging and of Alzheimer disease, and neither were her arteries clogged. Hennie’s wit and humor right up until the end indicated that mental decline is not an inevitability of aging. Hennie attributed her remarkable longevity to eating raw herring and drinking orange juice every day.

Researchers at Vrije University Medical Center in Amsterdam, led by Henne Holstege, PhD, whose neuroscientist father had studied Hennie, decided to look beyond the herring and OJ, for clues to the woman’s astounding good health in her genome (Holstege et al. 2014: 733). They sequenced the genomes in Hennie’s blood and brain cells, reasoning that white blood cells would have undergone many

mutations over that long lifetime because they divide frequently, which is when mistakes in copying DNA happen. In contrast, brain cells (neurons) never divide and so should represent the state of Hennie's genome at birth.

Hennie was genetically fortunate. Her white blood cells indeed bore some 450 mutations, but none of them were dangerous, and they all descended from just two stem cells in her bone marrow. As expected, her brain neurons had no mutations. The lesson: many gene variants do not harm health. That is, being "normal" takes many genetic guises.

THE DEAD—WOULD GENOME INFORMATION HAVE HELPED?

A clever application of genomics is to look back in time, to assess whether treatments, and therefore diagnoses, were correct. The Genomic Postmortem Research Project, being conducted at the Marshfield Clinic Research Foundation in Wisconsin, is sequencing the genomes of 300 dead people and comparing the data to information in their long-term electronic medical records.

Would knowledge of the information encoded in the DNA of the deceased have changed their health care? The researchers are sorting through more than 27 million gene variants from the corpses, seeking those that could have been "actionable"—that is, useful for preventing, detecting, treating, and monitoring specific diseases. To start, they are considering common chronic diseases such as age-related macular degeneration, heart attack, type 2 diabetes, cataracts, and glaucoma; cancers of the prostate, breast, and lung; susceptibility to bacterial infections and flu; metabolism of iron; and how people would have responded to certain drugs. Results should be interesting!

THE UK 100,000 GENOMES PROJECT

In late 2012, Parliament launched Genomics England, an endeavor that includes the "100,000 Genomes Project," which aims to sequence the genomes of 100,000 individuals by 2017. The 100,000 include 25,000 people who have cancer, 17,000 who have single-gene diseases and their blood relatives, and healthy individuals. Key to the effort

is the availability of health information in National Health Service (NHS) records.

The website for Genomics England compares large-scale genome sequencing to building the railways during the Victorian era. Affordable train transport made life easier for many people in England, but had the indirect effect of boosting the entire travel industry, pumping funds into many unanticipated sectors of the economy. Everyone benefited from the efforts of the railway workers. The same will likely be true of amassing and analyzing 100,000 human genome sequences.

The four parts of the 100,000 Genomes Project are:

- To ensure that all participants give informed consent.
- To establish a Genomic Medicine Service at the NHS, which will include precise diagnoses made possible with genome sequence data as well as pharmacogenomics information—which drugs will help a particular patient.
- Fueling insight into the workings of the human body and discovering new aspects of physiology that impact health—the research component.
- To continue to build the UK's genomics industry.

In the US, the similar Precision Medicine Initiative (PMI) began in 2015. It is cataloging individual variations in DNA sequences, environment, and lifestyle and their effects on health and health care, also using electronic medical records. The PMI is also considering biomarkers other than DNA, data from mobile health apps and technologies that enable individuals to track their own bodily responses to immediate environmental changes such as exercise, online communities based on health, genetic ancestry information, the **microbiome** (microbial components of our bodies), and geographical data to pinpoint large-scale environmental influences.

CAUSES OF DEATH—AN ORGANIZING FRAMEWORK

Medical specialties are traditionally based on organ systems. A gastroenterologist focuses on digestion, an endocrinologist on hormonal disorders, a dermatologist on skin. Medical specialists also define their

patient base by age, from pediatrics to adolescent medicine to internal medicine (adults) to gerontology. In contrast, the conditions that a medical geneticist encounters are more likely to span multiple organ systems than the problems a traditional medical specialist confronts.

Within any one medical specialty, single-gene (Mendelian) disorders are rare, and so health care providers may not have encountered some of them. Much more common are multifactorial conditions, caused by one or more genes plus environmental influences (Figure 3.1). One gene may have a large impact, and others very little, or many genes can each contribute to a small degree to a trait.

The most common causes of death reflect economics as well as biology. According to the World Health Organization (WHO), the top three causes of death for high-income nations are heart disease,

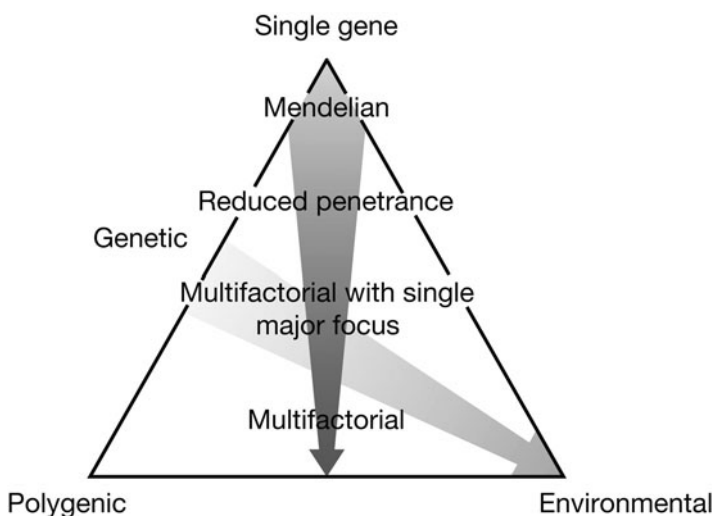


Figure 3.1 Genes/environment. Genes never function alone; the environment exerts many effects on gene expression. Penetrance refers to the fact that not everyone with a particular genotype expresses the associated phenotype. A multifactorial trait is determined by at least one gene plus environmental factors. A trait determined by more than one gene, with or without environmental input, is called polygenic.

Source: Adapted from *Applied Genetics in Healthcare*, Figure 13.2, p. 225

stroke, and cancers. In low-income countries, the top three causes are infectious: lower respiratory infections, Human Immunodeficiency Virus (HIV)/Acquired Immune Deficiency Syndrome (AIDS), and diarrheal diseases. Of the more than 12 million children who die each year worldwide, 99 percent live in low-to-middle-income nations, and more than half of them had any of just five infectious diseases: respiratory infections, measles, diarrhea, malaria, and HIV/AIDS. These diseases are common in poor nations, but very rare in wealthy ones.

Each of the three main classes of disease—cardiovascular, cancer, and infectious—presents unique challenges to the body, and genes influence them all. The cardiovascular system is a pump (the heart) and a network of pipes (blood vessels) that distribute an oxygen-bearing fluid (blood) to cells. Cancer stems from an imbalance between rates of cell division and rates of cell death. Infectious diseases are the consequence of our sharing the planet with microbes. Genes orchestrate our immune defenses against the pathogens that use their genomes to inhabit our bodies.

The following sections present examples of these three major types of disease at three levels: single genes, multifactorial influences, and genome-wide.

CARDIOVASCULAR DISEASE

The cardiovascular system is vast and its regulation complex. Diseases result from mechanical failure of the heart or disturbance of its electrical activity; deposition of fatty plaque on blood vessel linings or weakening in their walls; or blood that is too thin or too prone to clot, or moves through the labyrinth of blood vessels under too much pressure. Hundreds of genes contribute to cardiovascular health, including at least 50 that maintain blood pressure. Single-gene diseases of the cardiovascular system are rare. More common disorders reflect the actions, or failures, of several genes, each contributing a small degree of risk, coupled with environmental risk factors such as poor diet, smoking, and lack of exercise.

Table 3.1 lists single-gene disorders of the heart, blood, and blood vessels. Familial hypercholesterolemia (FH) is significant because it inspired development of statin drugs, which millions of people take who do not have inherited cardiovascular disease to maintain healthy

Table 3.1 Single-gene disorders of the cardiovascular system

<i>Disease</i>	<i>Defect</i>
Blood clotting:	
Hemophilia	Bleeding from deficient clotting factors
Von Willebrand disease	Bruising from deficient plasma protein
Blood composition:	
Erythrocytosis	Increased number of red blood cells
Fanconi anemia	Decreased number of red blood cells
Blood vessels and circulation:	
Marfan syndrome	Absent or deficient fibrillin (a connective tissue protein) weakens wall of largest artery, which may burst
Sickle cell disease	Abnormal beta globin bends red blood cells into sickle shapes, blocking circulation
Hypertrophic cardiomyopathy	Abnormal cardiac muscle enlarges heart and impairs pumping
Long QT syndrome	Malfunctioning potassium channels in heart muscle disrupt heartbeat
Lipid profiles:	
Familial hypercholesterolemia	Too few LDL receptors cause very high serum LDL cholesterol
Hyperlipidemia	Elevated blood serum LDL, VLDL, and triglycerides
Hypertriglyceridemia	Elevated blood serum triglycerides
Lipoprotein lipase deficiency	Elevated blood serum LDL and triglycerides

levels of blood serum cholesterol. FH, caused by a mutation in any of three genes, is extremely high blood serum cholesterol.

A RARE DISEASE—FAMILIAL HYPERCHOLESTEROLEMIA

In the 1980s, two people with FH made headlines—Stormie Jones and Jim Fixx. Stormie was born in 1978 in Texas. The first sign of a problem was the yellowish bumps on the baby's buttocks, which the pediatrician dismissed as something that would go away. But by age 5, Stormie had the odd, yellow deposits in the crooks of her

elbows, knuckles, and knees, and between her toes. The bumps were pure cholesterol and Stormie's blood serum cholesterol was nine times normal, her arteries dangerously blocked, as if she was elderly. Stormie inherited two mutations in a gene that gave her the severe form of FH. She was one in a million.

The little girl spent much of her time in hospital. By age 6, Stormie had suffered two heart attacks and had severe chest pain. At age 7, triple bypass surgery replaced the arteries in her heart, and she received a new heart valve. Then she underwent the first heart and liver transplant ever performed—she needed the liver because it is the site of the body's own cholesterol production, and Stormie's liver made too much. She needed the heart because her own was too weak to have survived a liver transplant. After spending several months in or near Children's Hospital of Pittsburgh, where she had the transplant, Stormie returned home. She lived more normally, even returning to school, until she died at age 13 because her body rejected the transplanted heart.

At about the same time that Stormie Jones received her historic transplant, a runner credited with starting the jogging craze of the mid 1980s died of a heart attack. Jim Fixx's *The Complete Book of Running* had encouraged many people to give up cigarettes and fatty diets and to start running, and the fact that the author had become an icon of healthful living made his sudden death in 1984, at age 52, shocking. Fixx had rarely seen a doctor, so little was known of his medical history, but he had once been a two-pack-a-day smoker. When physicians learned that his father had died at age 43 of a heart attack and had suffered his first at age 35, they realized that Jim Fixx likely had the less severe form of FH, having inherited one mutation. The autopsy supported this idea: Fixx's coronary arteries were almost completely blocked with plaque. In contrast to the rarity of homozygotes such as Stormie Jones, one in 500 people is a heterozygote for FH, like Jim Fixx, facing an elevated risk of a heart attack in early adulthood, unless they can control their cholesterol levels.

THE STATIN STORY

Cholesterol becomes a problem when it accumulates in the bloodstream, rather than being used inside cells. Our bodies require

cholesterol, a lipid, to synthesize steroid hormones (such as estrogen, testosterone, and cortisol), bile, and vitamin D, and it is a major component of cell membranes. The liver manufactures cholesterol, and we get it in food. When cholesterol is so abundant that it cannot enter cells fast enough, it backs up and builds up as sticky plaque on the interior surfaces of the arteries. Blood must then squeeze through an ever-narrowing tunnel as it circulates, raising pressure as artery walls strain to contain the torrent. The plaque buildup and blockage is atherosclerosis, which is the primary cause of coronary artery disease. (The coronary arteries serve the heart.)

Cholesterol is transported in the blood and lymph aboard lipoproteins, which are molecules consisting of a fat (lipid) attached to a protein. Bulk distinguishes different classes of lipoproteins: very-low-density lipoproteins (VLDL), low-density lipoproteins (LDL), and high-density lipoproteins (HDL). The genetic connection is that genes encode the protein portions of these lipoproteins and also encode the receptors on cell surfaces that bind to the lipoproteins, taking cholesterol into cells. The research that unraveled the steps of the transport and delivery of cholesterol to cells led directly to development of the statins. These drugs block the enzyme required for the liver to make cholesterol, sharply lowering blood serum cholesterol through a non-dietary intervention.

In 1856, German pathologist Rudolph Virchow (1821–1902) was the first to realize that cholesterol buildup in artery walls causes atherosclerosis. A century later, studies began to corroborate his work at the molecular level, correlating elevated LDL to increased risk of developing coronary artery disease (Keys 1980). Cholesterol became the enemy, and the first cholesterol-lowering drugs soon appeared: thyroid hormone, estrogen, the vitamin niacin, the antibiotic neomycin, various plant derivatives, and a compound called cholestyramine. These drugs worked in different ways, and most had unpleasant side effects. But investigating how they work led to development of the statins.

In 1959, researchers in Germany discovered that cholestyramine binds bile acids and excretes them in feces. The liver senses the diminishing supply of bile acids and starts to replace them. Making bile requires cholesterol, and to synthesize cholesterol, liver cells use an enzyme called HMG-CoA reductase. As the bile acid level falls, the cell makes more of the enzyme (Tennet et al. 1960: 469). Animal

studies showed that a cholesterol-free diet forces the liver to make its own cholesterol, and HMG-CoA reductase enables it to do so (Siperstein and Fagan 1966: 602). Would blocking the enzyme lower blood serum cholesterol by countering the liver's contribution? By 1970, it was clear that this approach, coupled with a low-fat diet, should lower cholesterol. The strategy sounded safe, because the substance that accumulates when the enzyme is blocked is easily excreted in the urine.

The statin story opened with the discovery from several research groups of a molecule from a fungus that inhibits HMG-CoA reductase. It was named mevastatin. At the same time, Michael S. Brown and Joseph L. Goldstein at the University of Texas were examining cells from patients with the severe form of FH. The researchers discovered that the defect behind the disorder was in the cell membrane receptor for LDL, encoded by a gene on chromosome 19 (Brown and Goldstein 1976: 150). People with severe disease, such as Stormie Jones, were homozygous recessive. People who suffer heart attacks at early ages and have relatives who do too, like Jim Fixx, may be heterozygotes (carriers) for this gene. Brown and Goldstein won the 1985 Nobel Prize in Physiology or Medicine for the discovery, because it shifted thinking about elevated cholesterol from a blood problem to an LDL receptor problem. "Brown and Goldstein's discoveries have led to new principles for treatment, and prevention, of atherosclerosis," wrote the Nobel committee.

Discovery of the LDL receptor explained how cholesterol forms plaque on artery linings—both in FH and in people who have high cholesterol from their diets. The LDL receptor is a protein that forms a coated indentation in the cell membrane. When an LDL particle fits into the receptor and is taken inside the cell, organelles called lysosomes degrade the protein parts, releasing the cholesterol cargo. In liver cells, some of the liberated cholesterol blocks the activity of HMG-CoA reductase, indicating "dinner had enough cholesterol, no need to make any more." If all is well, cholesterol not taken into cells stays in the bloodstream and binds to HDL, which carries it to the liver or out of the body. If the blood has too few HDL particles, or if the LDL particles cannot escort enough cholesterol inside cells, cholesterol builds up in the bloodstream. Hence, LDL is the "bad" cholesterol and HDL the "good."

In full-blown FH, the LDL receptor distorts, making it unable to effectively bind LDL, and LDL cholesterol in the bloodstream soars (Figure 3.2). Other genes (*APOB* and *PCSK9*) determine the number of LDL receptors. People who can eat a fatty and starchy diet yet have healthy cholesterol profiles likely inherited gene variants that lead them to naturally make many LDL receptors. Likewise, people who watch their diets very carefully yet have high cholesterol may have inherited gene variants that lead to too few LDL receptors.

Development of mevastatin continued. It didn't work in mice and rats, because rodents can make cholesterol even if the targeted enzyme is blocked. Chickens proved a better animal model because of the high cholesterol levels in eggs. Mevastatin lowered cholesterol significantly in hens, dogs, and monkeys. Effects on cells from FH patients indicated that the drug should work in people.

In 1976, a 17-year-old girl with severe FH who had a cholesterol level of 1,000 mg/dl was given mevastatin. After 2 weeks of

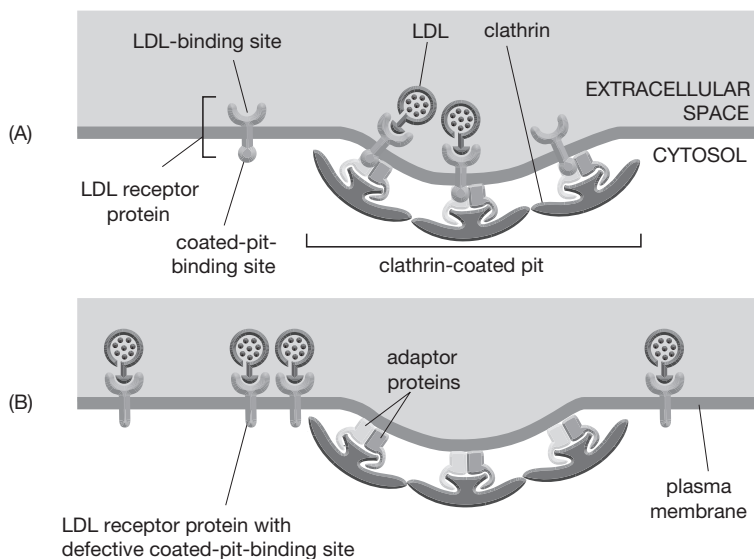


Figure 3.2 LDL receptor. LDL particles escort cholesterol inside cells for use and must bind to LDL receptors to do so.

Source: *Molecular Biology of the Cell*, 5th edition, Figure 13–51, p. 791

treatment, her cholesterol was down to 800. Several research groups tested mevastatin on patients who had severe FH, milder FH, and other forms of high cholesterol, at different dosages. The drug worked well, even in people who had no symptoms of cardiovascular disease.

Meanwhile, it was becoming clear that mevastatin was not the only way to block HMG-CoA reductase. Brown and Goldstein discovered a similar natural product from a different strain of the same fungus that had yielded mevastatin. In 1979, researchers at Merck Research Labs derived yet another HMG-CoA reductase inhibitor from “red yeast rice,” which is actually a fungus, *Aspergillus terreus*. The Chinese had used red yeast rice for centuries to treat a variety of ills, including diarrhea, indigestion, spleen problems, and poor circulation. The Merck compound became lovastatin, the first statin approved in the US, in 1987.

Other statins followed, and today, they are the bestselling drugs worldwide. Statins are the biggest item in the UK NHS’s drug budget: 3 million citizens take them. In the US, 13 million people take statins. The world market exceeds 20 million and is poised to double, as studies suggest that even healthy people may benefit by controlling the level of cholesterol in the blood. However, at the same time, people vary in their responses to statins and in the likelihood of adverse effects, particularly muscle pain (myalgia) or even destruction (rhabdomyolysis).

A genomic view of statin use shows that the greater a person’s genetic risk of developing coronary artery disease, the more likely a statin is to help. In one study, researchers calculated a genetic risk score based on 27 genetic variants for nearly 50,000 individuals for coronary heart disease. Low, intermediate, and high genetic risk corresponded to lowered risk when using statins of 13 percent, 29 percent, and 48 percent, respectively (Mega et al. 2015: 2264).

Muscle pain associated with statin use has been recognized for many years. Earlier studies that found only 1 to 5 percent of patients affected excluded people with pain not associated with biochemical evidence of muscle breakdown, and left out people with muscle diseases that might be at higher risk for this complication. More inclusive studies place the risk of developing muscle pain at about 10 percent (Bitzur et al. 2013: S325). People who have certain variants in a gene called *SLCO1B1* are at elevated risk. This gene encodes a protein that shuttles statins into liver cells.

CANCER: THE CELL CYCLE OUT OF CONTROL

The shape of an organ is so distinctive that it is easily recognizable in different people, and at different ages. A kidney is clearly a kidney, whether in a baby, young adult, or elderly person. Organs maintain their shapes thanks to a highly regulated process, called the **cell cycle**, that determines when, and how often, a particular type of cell divides (Figure 3.3). A skin cell may divide every few days, yet a brain neuron, never.

Cells have a built-in “clock” that tells them when to divide. The clock is in chromosome tips, called **telomeres**. Here, repeats of a six-base DNA sequence are shed with each cell division, shortening the chromosomes like a fuse burning down. When the telomeres shrink to a certain point, division halts. Most human cells growing in laboratory culture divide up to about 50–60 times; similar limitations presumably operate in the body. The cell cycle also has built-in

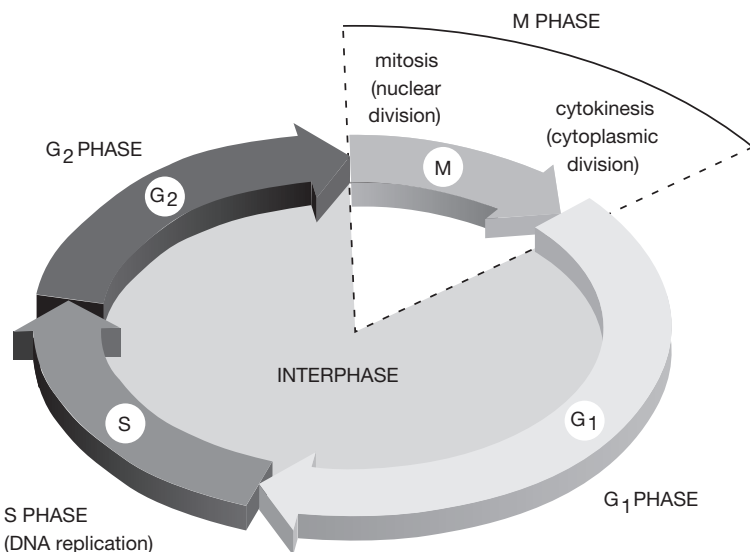


Figure 3.3 The cell cycle. Mitosis is when the cell divides. Interphase is a very active period, when the cell synthesizes many proteins.

Source: *Molecular Biology of the Cell*, 6th edition, Figure 17–4, p. 965

quality control measures. Some damage to DNA can be repaired, but, if the harm is too great, the cell dies by a process called apoptosis.

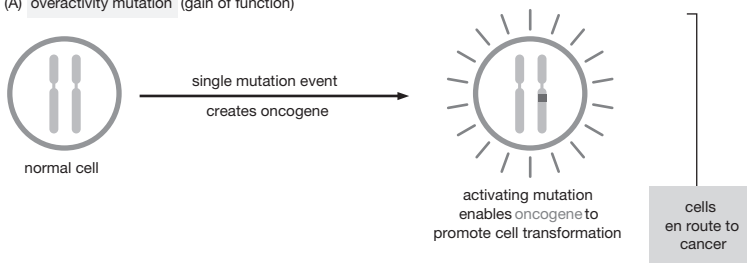
When control over how often cells divide is disturbed or lifted, cancer can result. Cancer cells have distinctive characteristics. They are less specialized than the cells from which they derive, and more likely to pile up on each other, forming lumps. Cancer cells invade surrounding tissue. They also secrete factors that coax nearby blood vessels to extend toward and then into them, a process called angiogenesis. Cancer cells enter the circulatory and lymphatic systems, spreading, or metastasizing, to distant parts of the body. Most importantly, cancer cells are “immortal,” dividing continually. This accelerated and immortal cell cycle is how tumors and blood-borne cancers crowd out healthy tissues.

CANCER GENES

Genes are implicated in cancer in several ways. Only about 10 percent of cancers are inherited as a single-gene trait. When this happens, a person inherits a mutation in every cell that confers *susceptibility* to cancer, starting with the fertilized ovum. Cancer actually begins, though, in a somatic (“body”) cell, where a second mutation occurs in the same gene on the other copy of the chromosome from the first mutation. The cancer therefore results from “two hits”—a susceptibility mutation in every cell, termed a germline mutation, and a second mutation on the other chromosome in a somatic cell. The other 90 percent of cancers also arise from two genetic “hits,” but they are in the same somatic cell—that is, the mutations are only in the cancer cells. Mutations in somatic cells are usually caused by an environmental insult, such as exposure to radiation or to a cancer-causing chemical (a carcinogen).

Two types of genes cause cancer when they mutate (Figure 3.4). Proto-oncogenes are genes that normally control the cell cycle, but become cancer-causing **oncogenes** when they are overexpressed. Their products are called oncoproteins. In contrast, **tumor suppressor** genes normally prevent cancer by holding down the number or rate of cell divisions, but cancer results when these genes are inactivated or deleted. Cancer may begin as a general destabilizing of the genome, which permits mutations to occur in proto-oncogenes and tumor suppressor genes.

(A) overactivity mutation (gain of function)



(B) underactivity mutation (loss of function)

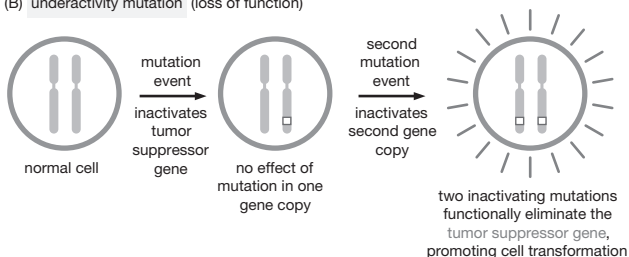


Figure 3.4 Oncogenes/tumor suppressors. Oncogenes cause cancer when they are overexpressed. Tumor suppressors cause cancer when they are deleted or are underexpressed. As a cancer progresses and spreads, many genes become involved.

Source: *Molecular Biology of the Cell*, 5th edition, Figure 20–27, p. 1232

Mutations in the same tumor suppressor genes or oncogenes may cause cancer in different body parts. In Li-Fraumeni syndrome, for example, family members who inherit a mutation in a gene called *p53* are at extremely high risk of developing several cancers. Half have at least one cancer by age 30, and 90 percent of them develop cancer by age 70. People with the syndrome have the *p53* mutation in every cell, and cancers develop in cells where somatic mutations affect the same gene in the second copy of the chromosome that includes *p53*. Only about 100 families in the world have the syndrome.

Although Li-Fraumeni syndrome, with its germline and somatic mutations in *p53*, is very rare, having two somatic mutations in the gene is fairly common. This may happen, for example, in a person who develops lung cancer after many years of smoking. Mutations

in *p53* also cause cancers of the thyroid, spleen, liver, esophagus, colon, bladder, and blood. The gene normally functions as a “cell cycle checkpoint,” determining whether a cell with damaged DNA dies or is repaired. With a double mutation, *p53* protein fails to condemn irretrievably damaged cells to death, and, instead, the cells become cancerous.

Oncogenes and tumor suppressor genes are called the “drivers” of cancer, because they directly cause or propel a cancer. Other mutations occur in tumor cells too, but they are more or less bystanders and are termed “passengers.” Many cancers undergo a series of mutations of both types. For example, a form of colon cancer in advanced stages has 15 mutations in driver genes, and 61 in passenger genes—but those are just genes already described for that cancer. When researchers scanned the genomes of the colon cancer cells from 18,000 patients, they discovered 848 genes that had somatic mutations! This means that many combinations of genetic changes cause and then fuel this common type of cancer, and we have not identified them all. In addition, many oncogenes and tumor suppressor genes are not unique to a particular type of cancer. A “passenger” gene in colon cancer, for example, is a “driver” gene in a glioma, which affects brain cells other than neurons (neuroglia).

Cancer can happen in many ways, because gene functions overlap. One mutation is typically not sufficient to trigger disease, because functioning genes can compensate. When enough mutations accumulate, cancer becomes inevitable. Smoking cigarettes for many years, for example, increases the risk of developing lung and other cancers by causing somatic mutations and, possibly, accelerating the mutation rate.

A SINGLE-GENE VIEW: BREAST CANCER

Different genes, when mutant, can cause cancers in the same body part. Different types of blood cancer (leukemia) cells can even look alike, yet have different genetic drivers. Breast cancer is a good example of a cancer that is often discussed in the media as if it is one disease, especially when a celebrity such as Angelina Jolie speaks out about her specific type, caused by a mutation in the breast cancer susceptibility gene 1 (*BRCA1*). Actually, mutations in more than twenty genes can raise the risk of developing, or directly cause,

breast cancer. These cancers often arise from mutations that disrupt DNA repair in a way that enables other mutations to directly cause the disease.

Like the *p53* cancers, breast cancers more commonly arise from two somatic mutations, but we have learned about the disease process from the rarer, germline forms. About 5 percent of breast cancers are inherited, and about a fifth of these are caused by mutations in tumor suppressor genes called breast cancer predisposition genes 1 and 2 (*BRCA1* and *BRCA2*). They were discovered in the early 1990s in families in which several members developed cancer at young ages, by researcher Mary Claire King at the University of Washington, Seattle (Hall et al. 1990: 1684).

Like the rare, single-gene disorders, *BRCA1* breast cancer is much more common in certain populations. It is most prevalent among the Ashkenazi Jewish people, whose ancestors came from middle or eastern Europe. About 2 percent of this population has the mutation, compared with one in 1,000 in the UK and one in more than 800 in the general US population. Even if a woman inherits a *BRCA1* mutation, she may never develop an associated cancer. For a woman with four Ashkenazi grandparents, the risk of developing cancer, if she inherits a *BRCA1* mutation, is 87 percent, but for a woman in the general US population, it is only 10 percent, and in the UK even less. These differences are due to influences of other genes and environmental factors.

Rarely, males can develop breast cancer. A man with a *BRCA2* germline mutation faces a 6 percent risk of developing breast cancer. This may seem low, but it is 100 times the risk for men in the general population. A man who has a *BRCA1* or *BRCA2* mutation, whether he develops breast cancer or not, still passes it to each of his offspring with a likelihood of 50 percent. Mutant *BRCA* genes can cause other types of cancer. The same *BRCA1* mutation may cause breast cancer in one family member, ovarian cancer in another, and prostate cancer in yet another. *BRCA2* mutations can also cause cancers of the kidney, colon, skin, stomach, pancreas, prostate, or gallbladder.

Understanding DNA repair helps to explain how breast cancer begins. Normal *BRCA1*, *BRCA2*, and *p53* proteins form a complex in cell nuclei that peruses newly replicated DNA and identifies places where both strands of the double helix are cut, which could break the chromosome. A process called DNA repair ensues. Mutations that

affect any of these DNA repair proteins, plus others that interact with them, can cause breast cancer.

A MULTIFACTORIAL VIEW: COLON CANCER

In contrast to cancers that are driven by mutations in a single gene, hereditary colon cancer is the culmination of a series of mutations that transform normal lining tissue into harmless (benign) growths that become precancerous and then cancerous. The multi-year process begins with a general destabilizing of the genome that activates certain oncogenes and disables certain tumor suppressor genes. It may begin with a mutation in a gene that maintains chromosomal structure during cell division, or in a DNA repair gene. A germline mutation in a DNA repair gene causes hereditary non-polyposis colon cancer (HNPCC, or Lynch syndrome), which brings a lifetime colon cancer risk of 80 percent, with disease usually beginning before age 45.

Genome destabilization may also result from abnormal patterns of gene expression, rather than, or in addition to, mutations. Whereas mutations change the DNA base sequence, changes in gene expression alter which genes are turned on or off in cells lining the colon. Normally, small molecules called methyl groups silence parts of the genome by binding to certain DNA sequences, physically shielding them from being transcribed into RNA. (A methyl is a carbon atom bonded to three hydrogen atoms, written CH_3 .) In cancer, the control shifts over where methyl groups sit on the DNA. This may activate oncogenes, silence tumor suppressors, or do both. An altered methylation pattern is a type of **epigenetic** change, which means an alteration in gene expression, not in the DNA base sequence.

Once the genome is destabilized, characteristic mutations ensue in order, almost like a choreography of cancer (Figure 3.5). The first change is often in a gene called *APC*, which is a tumor suppressor. In cells where both copies of *APC* are inactivated—either from a germline mutation plus a somatic mutation, or two somatic mutations—the smooth lining grows upward, forming a stalk-like structure called a polyp. Further mutations turn the benign polyp into a small adenoma, which is a precancerous growth. A gene that acts later in the process is *p53*, which enlarges the adenoma. Oncogenes are involved, too. Their activation affects enzymes called

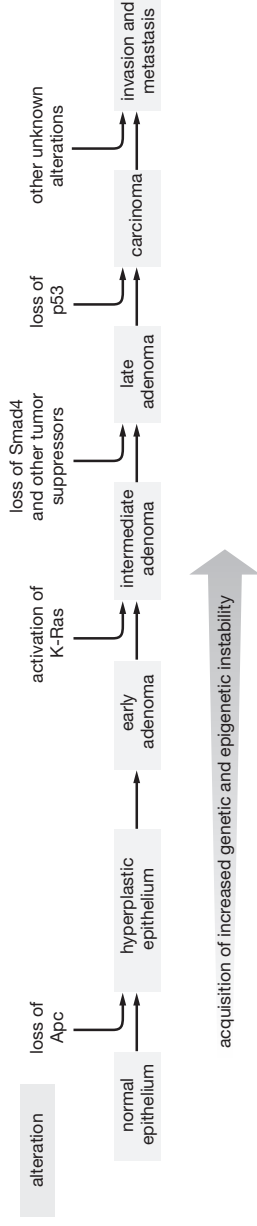


Figure 3-5 Colon cancer. Colon cancer progresses in a stepwise fashion. Oncogenes are turned on, and tumor suppressor genes are turned off.

Source: *Molecular Biology of the Cell*, 5th edition, Figure 20-48, p. 1255

kinases, which control the signals to divide that enter cells. Analysis of the DNA in cells in stool can detect these telltale mutations and identify patients who should be checked further with a lit instrument called an endoscope.

GENE-BASED CANCER TREATMENTS

Until we began to identify oncogenes and tumor suppressor genes in 1970, nearly all treatment for cancer followed a “slash and burn” approach: destroy the abnormal tissue, using surgery, radiation, or highly toxic drugs. Those three modalities are still in use, but treatments targeted to specific mutations have been so successful that in the future, cancers may be categorized and initial treatments selected based on genotype as well as or even instead of a body part.

The first targeted drug approved for cancer, in 1998 in the UK and US, was trastuzumab (Herceptin). It is a type of purified antibody called a monoclonal antibody that binds a cell-surface receptor, called HER2/neu, that is more abundant on breast cancer cells of a particularly aggressive and early-onset subtype. The bound drug blocks the cancer cells from signaling other cells to divide, halting the disease.

In 2001 came the blockbuster drug imatinib (Gleevec in the UK, Gleevec in the US), a small molecule that intercepts signals to divide by nestling into a cell-surface receptor that binds an enzyme called a tyrosine kinase. The initial clinical trial was cut short after only a few months when 53 of the 54 leukemia patients responded dramatically, and regulatory approval was astonishingly fast (Druker et al. 2001: 1031). Since then, imatinib has been found to be effective in several blood and bone marrow cancers as well as a gut cancer. Next came another monoclonal antibody, bevacizumab (Avastin), approved in the US in 2004 and for several cancers over the past few years in the UK, such as ovarian cancer. Avastin targets vascular endothelial growth factor A, cutting off the blood supply feeding various tumors, including those of the colon, breast, kidney, brain, cervix, and lung.

Because these first targeted cancer drugs work on disease in very different body parts, researchers are now testing new drugs according more to genotype than anatomy. “Basket studies” group cancers that are so rare that they have not been well studied, but that share specific

mutations. This was the case for the drug vemurafenib (Zelboraf), which blocks signals for cell division. The drug has been used since 2011 to treat a type of melanoma (skin cancer) in which cancer cells have a specific mutation (*V600*) in a gene called *BRAF*.

In a study from Memorial Sloan Kettering Cancer Center in New York City (Hyman et al. 2015: 726), 122 patients from all over the world tried the melanoma drug. They all had *BRAF V600* mutations, but they had a variety of cancers, including colorectal cancer, lung cancer, multiple myeloma, a form of thyroid cancer, a brain cancer, and the very rare conditions Erdheim–Chester disease, Langerhans'-cell histiocytosis, and cholangiocarcinoma. The drug helped some of the patients with lung and colon cancer. A woman with Erdheim–Chester disease had been in hospice care, yet felt fine just 2 weeks after starting the twice-daily pills and has done well for more than a year. The glioblastoma of a 59-year-old jazz pianist vanished following treatment with vemurafenib. But because the drug helped only some cancers, the researchers concluded that genes as well as body part should be considered in selecting treatments.

Oncologists are increasingly testing tumors for mutations, which accumulate as a cancer progresses in precise order. A more recent mutation will be in fewer of a tumor's cells than a mutation that has been present from the beginning of the disease. Programs such as the COSMIC Catalogue of Somatic Mutations in Cancer in the UK and the National Cancer Institute's Molecular Analysis for Therapy Choice (MATCH) trial and The Cancer Genome Atlas in the US are sequencing the genomes of cancer cells at different stages, and matching tumors to mutations on a genomic scale.

Traditionally, genetic testing for cancer has been done on cancer cells, from a sample (a biopsy) that is removed from a tumor. A new still-experimental technology detects DNA that tumors have shed into the bloodstream. If mutations in known oncogenes or tumor suppressor genes are found in such a "liquid biopsy" of a patient not diagnosed with cancer, then further testing can be done. If the patient already has cancer, then liquid biopsy can be used to monitor response to treatment. Before liquid biopsies can become part of standard care, however, researchers must determine how to tell whether a healthy person with cancer mutations in DNA in the circulation will actually go on to develop cancer. Studies on autopsies

reveal that many people had cancer and never knew it. We must learn to distinguish cancers that need not be treated from those that must.

José Baselga, MD, PhD, of Memorial Sloan Kettering Cancer Center and lead author of the melanoma basket study, predicts that genetics will play a larger role in cancer care over the next few years.

We are going to be able to look at mutations in the blood in a way that will be easy, fast, cheap, and reproducible. We will not only know the mutations, but also how the tumor is evolving to those mutations and the mechanisms for resistance to therapy. We cannot biopsy a tumor every 2 weeks or even months, but we can check the blood as much as we like. That will enable us to use a perfect combination of therapies.

THE MICROBIOME AND INFECTIOUS DISEASE

Infectious diseases are much more prevalent than single-gene disorders, cardiovascular diseases, or cancer. Nearly all of us have experienced an upper respiratory infection (a “cold”), influenza (the “flu”), acne, a cold sore, food poisoning, Athlete’s foot fungus, or a urinary tract infection. Common causes of death in many nations are AIDS, tuberculosis, and malaria, caused by a virus, bacterium, and protozoan, respectively.

Bacteria that cause disease are pathogens, along with more complex single-celled organisms such as amoebae; yeasts and other fungi; and “infectious particles” such as viruses, which are much simpler than cells. We contract infections because we live among many other species. Many bacteria, however, are normal residents of a healthy human body, and constitute our microbiome.

A human body consists of about 30 trillion human cells, and 39 million resident bacteria. All people share a core set of common bacterial species, plus others from our personal environments. Different body parts house distinct microbiomes, yet also reflect a person’s experiences. The microbiome of a circumcised penis differs from the microbiome of an uncircumcised one, as the microbiome from a mother’s vagina differs from the one in a woman who has never had a child or been pregnant. Our mouths, skin creases, armpits, hair follicles, and internal organs have highly distinctive microbiomes. The disappearance of type 2 diabetes mellitus in people who have

had gastric bypass surgery is partly attributed to a dramatic change in the gut microbiome.

Along the 1.5 meters of loops of the colon live some 6,800 bacterial species. A widely reported use of microbiome manipulation is the fecal transplant, which delivers other people's excrement to re-establish the "gut" microbiome in a way that treats recalcitrant infections of *Clostridium difficile*, a painful and sometimes lethal condition that sweeps in after antibiotics have altered the gut microbiome. Fecal transplants may also help treat ulcerative colitis. They have been done in cattle (via enema) for a century, and on people, in various settings, since the late 1950s. Researchers are still studying microbiome manipulation under controlled conditions—do not try this yourself! (van Nood et al. 2013: 407).

One of the most intriguing microbiome studies is actually the first, from 2007. Researchers at Stanford University examined the feces in 14 babies' diapers at 26 times during their first year of life, the study inspired by the newborn of one of the researchers (Palmer et al. 2007: e177). The experiments showed that a newborn's intestines are clean—devoid of bacteria. But, within just a few weeks, each baby had acquired a distinctive "gut microbiome." However, by the first birthday, the infants had become much more similar in their intestinal inhabitants.

Today, human microbiomes are being routinely tracked and associated with diseases and responses to treatment. Many digestive supplements work by altering the microbiome.

THE IMMUNE SYSTEM

The immune system protects us against infections with a highly orchestrated series of general, and highly specific, responses to pathogens. The system consists of about two trillion cells, their protein products, and the organs that store the proteins. The genetic connection is that genes encode the two major types of immune system protein—**antibodies** and **cytokines**.

The immune response is based on recognition of molecules called, generally, **antigens**. These are typically proteins jutting from the surfaces of cells or viruses. Some antigens are linked to sugars, forming glycoproteins. Cell surface antigens form a topography of sorts, like the mountains and valleys of a landscape. Our own cells

have antigen landscapes that the immune system perceives as “self” and does not attack; it reacts against foreign, or “non-self,” antigen landscapes that are parts of pathogens.

The immune response creates a series of barriers that make it difficult for a pathogen to penetrate and colonize a human body. An infecting virus or bacterium must find a break in the skin or an exposed mucous membrane, and must overcome the likes of earwax, tears, forceful sneezes, and waving cilia, the tiny, hair-like extensions that sweep irritating particles and microbes up and out of the respiratory tract. Pathogens that survive these initial defenses then meet an acid bath in the stomach, and, if they survive, may be forcefully expelled in vomit or washed out of the body in an explosion of diarrhea. In fact, much of what makes us sick is actually the response of the immune system, not an action on the part of the pathogen.

If a pathogen breaches or bypasses these physical barriers, it must face the reddening and swelling of inflammation. The changes of the inflammatory response are part of an initial, generalized, “innate” immune response that makes the human body less habitable. In inflammation, a rush of blood plasma brings in biochemicals that attack pathogens (Table 3.2), while diluting their toxins. White blood cells and giant cells called macrophages arrive from the circulation, engulf pathogens and draw them inside, where they are destroyed. Cells that engulf and destroy pathogens, like macrophages, are called phagocytes; the engulfing is called phagocytosis.

Innate immunity takes just minutes. Backing it up is the “adaptive” immune response, which is slower but highly specific. Fleets of white blood cells called **lymphocytes**—the T and B cells—carry out adaptive immunity. The T cells pump out cytokines and activate other cells to join the fight, including the B cells, which secrete antibody proteins that engulf and destroy specific pathogens. The precursors of T and B cells are in the bone marrow, and the T and B cells themselves patrol in body fluids. Of particular importance are T cells called “helpers,” because they activate B cells as well as secrete cytokines. Types of T cell are distinguished by their surface features, called “cluster of differentiation” antigens, such as CD4 and CD8. T cells are also classified as “killers” or “suppressors.”

An adaptive immune response begins when a macrophage—a giant, wandering cell—displays an antigen from a pathogen on its

Table 3.2 Proteins of the immune response

Antibodies	Built of Y-shaped subunits, antibodies bind non-self antigens, signaling other parts of the immune response
Collectins	Detect non-self antigens on certain viruses, bacteria, and yeasts
Complement	Plasma proteins that destroy viruses, rupture bacteria, and attract phagocytes
Cytokines:	
Colony stimulating factors	Signal bone marrow to manufacture lymphocytes
Interferons	Detect virally infected cells
Interleukins	Cause fever, which kills some pathogens
Tumor necrosis factor	Destroys bacterial toxins

surface. This flag alerts a helper T cell. The T cell in turn binds and thereby activates a B cell that has surface receptors which can bind the pathogen antigen that the macrophage displays, like a key fitting into a lock. At the same time, the T cell secretes cytokines that signal the bound B cell to rapidly divide, producing two types of “daughter” cell: plasma cells that mass-produce antibodies that bind to the displayed antigen, and “memory” B cells that snap into action should the pathogen appear again, protecting against future infection. Therefore, the antibody response is targeted and provides memory. It is also diverse. B cells can respond to nearly any provocation, because their antibody repertoire is so large. This is possible because DNA sequences that encode parts of antibodies can recombine in many different ways to create a great diversity of antibodies. The antibody response is a little like being able to assemble many outfits from a set number of items of clothing.

VARIATION IN IMMUNE PROTECTION

Most of us do not think often about our immune systems. However, for people with inherited immune deficiencies, life is a daily fight

against infection. Yet some lucky individuals inherit single-gene *protection* against infection. Just as studying rare, inherited forms of cardiovascular disease led to the invention of drugs to treat common conditions, studying rare individuals with single-gene disorders of the immune system has taught us about normal immunity. Inherited immune deficiencies affect all components of the immune response, from the initial generalized reactions, to innate immunity, to the specialized adaptive response.

Chronic granulomatous disease cripples the innate immune response by impairing the ability of phagocytes to destroy engulfed pathogens. In the disease, phagocytes cannot manufacture an enzyme that they require to produce chemicals, called reactive oxygen species, which kill certain pathogens that infect the skin, lungs, gums, and limb bones (*Salmonella*, *Klebsiella*, and *Burkholderia cepacia*), and fungi (*Candida* and *Aspergillus*). The missing enzyme is built of four different parts, and so the disease can be inherited four ways. Three of the genes are on the X chromosome, which makes the condition much more common in males. Before antibiotic drugs and vaccines were available, affected boys typically died in early childhood. Today, they can receive a stem cell transplant that introduces a functional gene, or undergo experimental gene therapy, and live into adulthood.

Twenty distinctive immune deficiencies affect either T or B cells, or both. Eliminate B cells, and the antibody response fails. Eliminate T cells, and ultimately the entire adaptive immune response topples.

Different types of immune deficiency make the body prone to specific types of infections. In agammaglobulinemia, which is also X-linked, infections are bacterial. A boy is typically healthy for the first month or two of life, because he still has antibodies from his mother. Then, the bacterial infections begin—ear infections, pneumonia, conjunctivitis (“pink eye”), sinusitis, and diarrhea. The boy is missing an enzyme that signals B cells to divide and mature into the antibody-secreting plasma cells. In contrast, in “immuno-deficiency due to DOCK8 deficiency,” viral infections predominate. CD8 helper T cells fail due to mutations in a gene called *DOCK8*. The viral infections include warts from human papillomavirus, skin infections from certain pox viruses, and herpes simplex infections. So deficiency of B cells is associated with bacterial infections, and deficiency of T cells with viral infections.

Severe combined immune deficiencies (SCID) eliminate *both* T and B cells. The original “bubble boy” was a young man with a form of the condition called SCID-X1 named David Vetter, who was born in 1971 in the US. Owing to a mutation on his X chromosome, he could not make T cells, and so his B cells lay dormant. David, thoroughly without immunity, survived by living in a large plastic bubble. After he emerged to receive a bone marrow transplant from his sister, he developed lymphoma (a B cell cancer) from a virus (Epstein-Barr) transmitted in the bone marrow and died. The virus hadn’t made his sister sick because she had normal immunity—many of us have this virus. David was 13. Children with this and other forms of SCID have been successfully treated with gene therapy. Chapter 5 tells their stories.

Some lucky individuals inherit variants in immune system genes that enable them to resist certain infections that make most people very sick. This is the case for human immunodeficiency virus (HIV), which causes acquired immune deficiency syndrome (AIDS).

HIV spreads through body fluids. The virus demolishes protection against other infections and cancer. Infection with HIV begins as the virus binds to two linked receptor proteins, CD4 and CCR5, on the surface of helper T cells. In response, the cell engulfs the viral particle, taking it inside. The virus then commandeers the protein synthetic machinery, forcing the T cell to mass-produce the virus, as innate immunity topples. The T cell bursts, releasing viruses. HIV next latches onto CD4 helper T cells, again entering via CD4 and CCR5 co-receptors.

HIV infection is devastating because helper T cells are so important to the immune response. They normally release cytokines, as well as activate the antibody defense. Once T cell numbers start to fall, immunity severely falters. Bacterial diseases usually appear first, then viral infections and certain cancers. Symptoms start when the million or so new T cells that the bone marrow produces each day cannot keep pace with the number of new HIV particles exploding from infected T cells. The different anti-retroviral drugs that are combined to treat HIV/AIDS attack the viruses at several key points as they enter and take over T cells. The sooner treatment begins, the longer survival and the less likely the person is to infect others.

People naturally resist HIV infection in various ways. About 1 percent of Caucasians and fewer people in other groups have protective

mutations in the gene that encodes the CCR5 receptor that HIV must bind in order to enter cells. People who are missing 32 bases in both copies of the *CCR5* gene cannot become infected because the virus cannot enter their cells.

The first person in whom clinicians noticed this natural protection was initially called “the Berlin patient,” because he was in a hospital in Berlin to undergo a bone marrow stem cell transplant to treat leukemia. But the man, named Timothy Brown, also had AIDS. Fortunately for Mr. Brown, the transplant donor was homozygous for the *CCR5* mutation! HIV vanished from Mr. Brown’s body as the donor’s bone marrow stem cells took over (Hutter 2009: 692). Not only did he get better, but the way that he did so inspired experimental use of a gene editing technique (zinc fingers, see Chapter 5) to remove the *CCR5* gene from CD4 T cells in AIDS patients. Researchers let the cells multiply and then infused them into their donors. Within a week, the modified cells had become established in the circulation of some of the patients (Tebas et al. 2014: 901). The effect on HIV level is still being evaluated.

Several studies have identified other inherited quirks of immunity that protect people against HIV infection (Cohen 2009: 1476). A two-decade-long study of healthy female sex workers in Nairobi, Kenya, who had been repeatedly exposed to HIV on the job, revealed that they have unusual CD4 T cells that protect other T cells. Intravenous drug users in Vietnam, who never contracted HIV despite repeated exposure to the virus, have extra natural killer cells, which are part of the innate immune response. Healthy men in Stockholm, repeatedly exposed to HIV through oral sex with infected male partners, have anti-HIV antibodies in their saliva. More mysterious is how 5 percent of people with hemophilia who received HIV in blood transfusions in the early 1980s avoided infection.

Over time, these various protections against HIV infection will likely increase in many populations, tempering the AIDS epidemic. More immediately, discovering how these inherited quirks protect people can suggest new ways to fight HIV infection.

A GENOME-WIDE VIEW OF INFECTIOUS DISEASE— LEPROSY AND INFLUENZA

During an infection, two genomes interact—that of the person and that of the pathogen. Susceptibility to leprosy (a bacterial infection)

and to influenza (a viral infection) illustrates the roles of the human and pathogen genomes.

The bacterium *Mycobacterium leprae* causes leprosy, which affects the skin and the peripheral nerves. In a mild form, small, pale, numb areas dot the skin, but in a severe form, larger skin lesions full of bacteria lead to disfigurement. The advent of a triple antibiotic “cocktail” in the 1980s brought the number of global leprosy cases down from 5.2 million in 1985 to about 180,000 in 2014, according to WHO.

About 95 percent of people are naturally immune to leprosy. Many of the susceptible 5 percent have affected family members as well as blood relatives who have had children together, suggesting an inherited risk to contracting leprosy. The disease is difficult to study, because the bacteria do not grow in the laboratory, and only infect humans and armadillos. For these reasons, and because the bacteria are very uniform genetically, researchers from the Genome Institute of Singapore and 26 institutes in China looked to the human genome to explain leprosy susceptibility, rather than the hard-to-study bacteria.

The researchers used an approach called a **genome-wide association study** (GWAS) that looks for single-base positions in the genome where people with a particular trait or illness share a rare base. Such a site is called a **single nucleotide polymorphism**, or SNP (pronounced “snip”). For example, if the 117th base from one end of chromosome 17 is a T in 96 percent of a population, then the 4 percent who have a C instead have a SNP there. People can be distinguished by large sets of SNPs that cover the entire genome.

In the leprosy genome study, researchers scanned 2.4 million SNPs among more than 10,000 people with either form of leprosy as well as healthy controls. Analysis of the SNPs that the people with leprosy share led the researchers to seven genes whose variants contribute to risk of contracting leprosy (Zhang et al. 2009: 2609). Some of the genes cause inflammation, and the others enable T cells to recognize an antigen that is unique to the cell walls of the leprosy bacterium.

Much more common than leprosy is influenza. Every year, many of us queue up to get the flu “jab,” a vaccine delivered in an injection. A vaccine is a disabled form of a pathogen, or part of it, that “fools” the immune system into reacting as if infection were taking

place, signaling B cells to produce antibodies. The vaccine stimulates immune memory, so that if the person encounters the actual pathogen, the reaction against infection is so fast that the person never gets sick.

Vaccines date back to eleventh-century China, and a hideously disfiguring and often deadly infectious disease called smallpox. Healthy people in long-ago China rubbed smallpox crusts from infected people into their own intentionally abraded skin, to induce a mild case that would protect against contracting a more severe one. In 1796, British doctor Edward Jenner improved the method by using lesions from cowpox, a milder illness, after noticing that milkmaids who developed cowpox did not contract smallpox. Jenner's vaccine rid the world of smallpox because so many people were protected that eventually infection could not take hold and spread. This large-scale protection against infection is called "herd immunity."

Immunization against influenza presents a different challenge from smallpox, because the influenza virus changes so readily that vaccines must be reinvented yearly. Because genetics is responsible for the rapid evolution of flu viruses, we can apply knowledge of the influenza genome to predict epidemics. Influenza viruses infect only a few species, including birds, pigs, ferrets, and us. New flu virus variants arise each year in parts of Asia, where humans, birds, and pigs live in close proximity.

Three families of influenza virus are recognized—A, B, and C. Only A and B infect humans, with A causing most cases of the flu. Like many viruses, influenza A has a streamlined structure, compared with a cell. It consists of about 13,500 RNA bases in a protein coat. The RNA encodes just 11 types of protein. Human antibodies recognize and bind to two types of glycoprotein on the viral surface. One type of surface glycoprotein, hemagglutinin (H), attaches to molecules of sialic acid on our cell surfaces. An H molecule consists of a "stem" portion that binds to the cell and is the same in all variants, and a "head" portion that varies. The other type, neuraminidase (N), releases new viral particles from infected cells. Flu viruses can have any of 16 types of H and any of 9 types of N. The most common combinations of influenza viruses that infect humans are "H3N2" and "H1N1."

Influenza viruses change at two levels. The smaller change, a single mutation, is called "drift." The single mutation rate is high,

because the virus replicates its genetic material frequently, and the enzyme that it uses to do so is error-prone—and the errors persist. A greater level of change, called a “shift,” happens when two different flu viruses swap parts, creating a novel virus. The site of a shift is a pig’s throat, where a bird (avian) flu virus might recombine RNA with a human flu virus. A shift can generate a virus so unusual that the human immune system might not have antibodies to respond to it.

WHO convenes a global surveillance program twice yearly to identify the strains of flu in different geographical regions. Participants use the information to predict the predominant strains for the next season. The vaccine “recipe” they suggest is typically a mix of H1N1, H3N2, the rarer influenza B, and whatever else is circulating. However, several “universal” flu vaccines are being developed that protect against all strains of the virus. Instead of targeting the parts of the virus that vary and distinguish strains, the new vaccines recognize the hemagglutinin stems present in all flu viruses. Researchers genetically modify the stem parts because in natural viruses, they fall apart when the head parts are removed.

Until very recently, medical genetics dealt with the rarest of the rare, single-gene disorders. Most physicians knew little about these conditions, and getting a diagnosis would often take years. Today, not only is genetics finally receiving attention in medical training, but health care consumers are much more aware of their family histories and genetic tests. We now know that even the common ills of humankind, which are not single-gene disorders, nevertheless unfold as a change in gene expression that parallels the course of the illness. In short, genes affect all aspects of our health. The next chapter considers genetic testing.

WHAT WOULD YOU DO? *PERSONAL GENOME SEQUENCING*

Fast, affordable genome sequencing is now possible, but how useful is the information?

Knowing your genome sequence is a little like having a list of every component of an automobile. If your car breaks down, the circumstance

of the problem is probably more informative than having the parts list. So it is for the human genome and what it can, and cannot, reveal about health. Consider the three types of illness discussed in this chapter. To what degree do errant genes cause them?

Cardiovascular disease might be completely inborn, such as Stormie Jones and her sky-high cholesterol. However, much more often, it reflects a lifetime of unhealthy lifestyle choices, such as a poor diet and insufficient exercise. Lifestyle habits can also affect cancer risk, but many people who follow all of the rules get cancer. Still, inheriting a cancer is never absolute. Women who face the possible consequences of inheriting a *BRCA1* or *BRCA2* mutation wrestle with the “incomplete penetrance” of their genotypes. Even a woman with such a mutation in the highest risk group, Ashkenazi Jewish people, faces an 87 percent lifetime risk of developing an associated cancer—not 100 percent. For most people, the risk is in fact much lower. When is the cancer risk high enough to justify having breasts and ovaries removed to prevent the disease?

We have better control over infectious diseases than we do cardiovascular diseases or cancer, largely because of the success of vaccines. People who do not vaccinate their children against such diseases as measles and chickenpox compromise herd immunity, endangering vulnerable children and hampering efforts to eradicate disease. Infectious diseases are treated with antibiotic or antiviral drugs, or supportive care until the immune system can stop the infection—not typically by consulting a genome sequence.

Human genome information should be placed into a practical perspective. Given the current state of medicine, how valuable would knowing a patient’s genome sequence be to a physician in treating a cardiovascular disease, cancer, or infection? What are possible psychological problems that might arise from a genome sequence providing too much information, such as identifying serious diseases that do not have treatments? Finally, at the societal level, how should people—individuals as well as groups—be prioritized for having their genomes sequenced?

SUGGESTED READING

Background information on cancer, cardiovascular disease, and infectious disease can be found on many websites and in several excellent books. The British Heart Foundation (www.bhf.org.uk/heart-health) funds investigations into inherited heart conditions. The American Heart Association website (www.americanheart.org) under “diseases and conditions” and “healthy lifestyles” analyzes the environmental component to common disorders of the heart, blood, and blood vessels. Cancer Research UK (www.cancerresearchuk.org) and the National Cancer Institute’s Cancer Information Service (<http://cis.nci.nih.gov/>) provide statistics about cancer. *The Emperor of All Maladies*, by Siddhartha Mukherjee (2010, Simon and Schuster), is a recent history of cancer research, and *Her-2: The Making of Herceptin, a Revolutionary Treatment for Breast Cancer*, by Robert Bazell, chronicles development of one of the first gene-targeted cancer drugs. *Spillover: Animal Infections and the Next Human Pandemic*, by David Quammen (2012, W.W. Norton & Co.), traces the beginnings of new infectious diseases in humans. The Foundation for AIDS Research (www.amfar.org/) helps developing countries to promote HIV research, prevention, treatment, and education. National Health Service Choices (www.nhs.uk/conditions/flu/Pages/Introduction.aspx) and the Centers for Disease Control and Prevention (www.cdc.gov/flu/index.htm) provide frequently updated information on seasonal flu. *The Great Influenza: The Story of the Deadliest Pandemic in History*, by John M. Barry (2005, Penguin Books), is riveting reading.

GENETIC TESTING

The genome sequencing that is at the forefront of conversations about genetics these days is both different from and similar to traditional genetic testing. Today, millions of people are participating in research projects to sequence genomes and assess the utility of knowing the 3-billion-unit order of A, C, T, and G. Some genome projects have participants who share a medical condition, such as heart disease or autism. Others are following healthy people, to see if anything in their genomes predicts disease. The introduction to Chapter 3 describes three such programmes.

A gene or genome sequence alone is not useful—it is like a string of letters with no obvious meaning. The information in a genome must be analyzed to tell which DNA sequences have value in establishing a diagnosis, selecting treatments, or predicting symptoms for diseases not yet recognized. If a person's particular collection of symptoms is well known, then genome sequencing can lead to a diagnosis in a matter of minutes. If the illness doesn't fit into an established diagnosis, the person may have an "atypical presentation" of a known condition, or a disease so rare that it has yet to be described. Those diagnoses might take a little longer. But in the days before genome sequencing, diagnosis of rare diseases caused by mutations in single genes typically took years. It's no wonder such efforts became known as "diagnostic odysseys." The effort to assign

a name to a disease could be overwhelming. But that picture is changing.

Genome sequencing differs most starkly from conventional one-gene-at-a-time genetic testing in that it offers a huge data dump from which algorithms, researchers, or astute clinicians must extract relevant information. The single-gene tests of days past, however, provided important precedents for the social issues that genome sequencing is raising today:

- Who should have their genomes sequenced?
- Who will pay for sequencing?
- How should genome information be used, and not abused?
- Should relatives be told about mutations they could have?
- How can privacy be protected, especially with the ability to store genome information in clouds?

This chapter explores how single-gene tests have laid the groundwork for today's exome and genome-level tests. The technology of genetics may be changing, but the issues of access and fair use began more than half a century ago, when young parents noted the odd smell of their sick children's urine and took it to a biochemist. They unknowingly started the field of genetic testing.

THE FIRST GENERATION OF GENETIC TESTS: PROTEINS

The earliest genetic tests did not analyze DNA, but examined proteins, the products of gene action. These tests inferred that enzymes were missing from buildups of the biochemicals they would otherwise break down, or they detected proteins with altered shapes that impair their function. A test for an inherited disease given to a population group, and not just people with a family history of the condition, is termed a screen. The first three single-gene screens were for phenylketonuria (PKU), Tay–Sachs disease, and sickle cell disease.

PKU—AN EARLY SUCCESS STORY

Pearl S. Buck (1892–1973) was a prize-winning novelist from the US who wrote about living in China. She is best known for her book

The Good Earth. Less familiar is her 1950 book *The Child Who Never Grew*, about her daughter Caroline, born in 1921. Caroline had a very limited attention span, and her eyes were oddly blank and unresponsive. She made purposeless movements. By the time the Bucks brought their daughter to the US for evaluation when she was 3 years old, she was clearly mentally retarded, and worsening. ("Mental retardation" is now called a type of "intellectual disability".)

Caroline Buck had PKU, a classic "inborn error of metabolism." Absence of an enzyme (phenylalanine hydroxylase) causes buildup of its substrate (the amino acid phenylalanine), as well as accumulation of an abnormal breakdown product (phenylpyruvate) (Figure 4.1). The fact that the accumulated substance is a normal component of dietary protein presents an opportunity to treat the condition with dietary restriction. The ability to screen for the disease at birth was a medical milestone because treatment could begin in time to prevent the symptoms.

For PKU, as for cystic fibrosis (CF; discussed in Chapter 2), incidence and mutation types vary among populations (Table 4.1). Also like CF, PKU may be common in parts of Europe because carriers have enjoyed a survival advantage and, therefore, increased in the population. PKU carriers have elevated blood phenylalanine levels that are not high enough to cause PKU symptoms, but are sufficient to inactivate a poison found in rotting potatoes. The poison is deadly to a fetus. In the great famines that swept Ireland and Scotland in

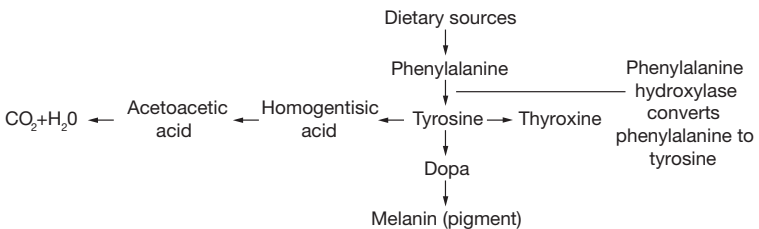


Figure 4.1 PKU pathway. In the inborn error of metabolism PKU, the amino acid phenylalanine cannot be broken down. Eating a special low-phenylalanine "medical food" from birth can prevent the intellectual disability that the condition causes.

Source: Adapted from *Genetics for Healthcare Professionals*, Figure 2, p. 122

the 1800s, might PKU carriers have better survived by eating moldy potatoes that did not make them sick, nor poisoned their unborn children? According to this hypothesis of balancing selection, with time, the incidence of this protective mutation would have increased and spread eastward as the Vikings took wives and slaves from the Celtic lands.

Caroline Buck was in her teens when PKU began to be understood. In 1934, the parents of two young children with “feeble-mindedness,” in Norway, noticed a musty odor to the children’s urine. Six-year-old Liv walked strangely and spoke few words. Her 4-year-old brother Dag seemed stalled in infancy, unable to walk, talk, or feed himself, and he still wore nappies. The parents, Harry and Borgny Egeland, were perplexed, as both children had seemed healthy at birth. Harry told a colleague about his children’s problems, who mentioned them to a physician and biochemist named Asbjörn Fölling. Intrigued, he analyzed the urine in a makeshift biochemistry lab in an attic at University Hospital in Oslo.

First, Fölling ruled out infection. Then, he added ferric chloride to the children’s urine to rule out diabetes, which would have turned urine purple instead of the reddish-brown that normal urine turns. Liv’s and Dag’s urine turned green! Astonished, Fölling asked the parents not to give the children any medications for a week, and then he tested the urine again. It was still green! Using standard organic chemistry techniques, Fölling identified the substance responsible for the green color, and likely the odor, as phenylpyruvic acid. Could it cause intellectual disability? He wondered where else he could find the anomaly.

Fölling next tested urine from 430 people institutionalized for intellectual disability and found phenylpyruvic acid in eight, including two pairs of siblings. He named the condition *imbecillitas phenylpyruvica*. The affected individuals shared other characteristics: fair skin, rashes, stooped posture and spastic gait, and broad shoulders. Fölling published his findings on these 10 patients (Fölling 1934: 169; 2008: 4), and, shortly afterwards, another researcher in the UK described three other families. The autosomal recessive mode of inheritance was clear: both sexes were affected, and the parents were healthy. The clincher was the fact that, for three sets of parents, husband and wife were blood relatives. They were passing on a recessive mutation inherited from shared ancestors.

Table 4.1 Incidence of PKU in selected populations

<i>Population</i>	<i>Incidence</i>
Global	1/15,000
China	1/16,000
Finland	<1/100,000
Ireland	1/4,500
Japan	1/119,000
Norway	1/13,000
Sweden	1/30,000
Turkey	1/2,600
US Caucasian	1/10,000

Next, Fölling invented a better way to detect phenylpyruvic acid in urine, which became known as the “wet diaper test.” It used bacteria that produce a colored compound when exposed to phenylpyruvic acid. The test could spot PKU, but not until the child was 6 weeks of age.

In 1937, George Jervis, the first director of the Institute for Basic Research in Developmental Disabilities, in Staten Island, New York, renamed Fölling’s disease *phenylketonuria*, for “phenylalanine in the urine.” Jervis worked with 50 patients at Letchworth Village State School in Thiells, New York, adding 135 more patients from other state institutions (Jervis 1937: 944). He also noted that all children of women with PKU had small heads, intellectual disability, low birth weight, and heart abnormalities.

By 1952, biochemists had discovered how PKU arose (Undenfriend and Cooper 1952: 503). When phenylalanine can’t be broken down, its breakdown product, the amino acid tyrosine, becomes depleted. As tyrosine is a precursor of melanin, the pigment molecule, people with PKU have fair skin and hair. The biochemical block affects other metabolites, causing hyperactivity, autistic-like behavior, and seizures.

In 1953, German physician Horst Bickel suggested that a low-protein diet might prevent intellectual disability by decreasing the phenylalanine level (Bickel et al. 1953: 812). One of the first patients treated was a 2-year-old at Birmingham Children’s Hospital, in the UK, who could not walk or talk, cried piteously, and repeatedly

banged her head (Kayt Marra, personal communication and PhD thesis 2010). Fed a formula very low in phenylalanine, but with a tiny amount of whole milk, eating only a few fruits, vegetables, and gluten-free grains, the girl started to crawl, haul herself up, and try to walk. Her head-banging stopped, and her hair darkened. When the diet accidentally lapsed, symptoms returned.

By 1958, the first commercial formula for PKU became available. However, a problem of timing emerged. The diet had to start very soon after birth to prevent intellectual disability, yet the wet diaper test didn't work until 6 weeks of age. A blood test, called the Guthrie test, would prove much more accurate. It is still done today, and founded the field of newborn screening.

In 1957, Robert Guthrie was a microbiologist at the State University of New York at Buffalo. He and his wife had a son with intellectual disability and a niece who had been diagnosed with PKU at 15 months, using the wet diaper test—much too late to prevent symptoms. At a meeting of the New York State Association for Retarded Children, Guthrie chatted with a physician working with people who had PKU, and he had an idea. Guthrie had been investigating biomarkers in cancer patients' blood—biochemicals that would observably change a bacterial culture and so could be used to diagnose or monitor the cancer. Guthrie redirected his cancer project to detect elevated phenylalanine in the blood of newborns. The test used a strain of a common bacterium (*Bacillus subtilis*) that could grow only in the presence of phenylalanine, phenylpyruvic acid, and a related acid. If a newborn's blood had these substances, the bacteria would grow—and the child had PKU.

In 1961, Guthrie described his test at a New York State Association for Retarded Children conference in Jamestown, NY. Soon, two hospitals began sending him filter papers with blood drops from the heels of newborns. Then Guthrie ran his test on 3,118 institutionalized patients with intellectual disability; 21 had PKU (Guthrie and Susi 1963: 338). A larger study validated the test in 400,000 newborns (Guthrie and Whitney 1964). At about this time, photographs from affected families began to show disabled older siblings and younger ones who had benefited from the diet. But following the strict protocol wasn't easy. The formula and drink, carefully manufactured to avoid phenylalanine, were rather unpalatable, and coaxing young

children to consume them without wailing at mealtime when they couldn't eat other food was an issue.

The Guthrie test caught on, fast. Circumstances helped, too, for President John F. Kennedy's mentally disabled sister, Rosemary, had drawn attention to the problem. Rosemary did not actually have intellectual disability. At the age of 23, in 1941, a prefrontal lobotomy, performed supposedly to treat her mood swings, instead damaged her brain. The Kennedy family has since championed treatment of those with mental deficiencies in honor of Rosemary.

The first mass newborn screening program for PKU in the US was in the Kennedy's home state, Massachusetts, in 1963. It spread quickly, and today all states require it. Newborn screening, teamed with treatment, was an economic success. In the early 1960s, the cost of screening all newborns in the US and treating those with PKU was about \$3.3 million per year; the cost of not screening and then institutionalizing people with PKU was \$189 million per year. PKU is one of eight inherited conditions (all autosomal recessive) plus a hormonal condition (congenital hypothyroidism) that the "newborn blood spot test" offered in the UK detects (Table 4.2). States in the US screen for dozens of diseases.

In 1983, Savio Woo and colleagues at the Baylor College of Medicine discovered the *PAH* gene, which encodes phenylalanine hydroxylase, the enzyme deficient in PKU (Woo et al. 1983: 151). Since then, researchers have identified more than 450 mutations in the *PAH* gene, associating different severities of PKU with different allele combinations.

Continuing investigation of the effects of the PKU diet led to important discoveries. Unlike the initial protocol of following the

Table 4.2 Newborn screening tests in the UK

CF

Glutaric aciduria type 1

Homocystinuria

Isovaleric acidemia

Maple syrup urine disease

Medium-chain acyl-CoA dehydrogenase deficiency

Phenylketonuria

Sickle cell disease

diet only through childhood, and then only through adolescence, the regimen now is to do so for life, with careful selection of only very low phenylalanine foods and prepared drinks considered a “medical food.” The PKU community learned the importance of maintaining the diet during pregnancy, when women with PKU who had stopped the diet in childhood gave birth to children with severe intellectual disability, even children who had not inherited PKU. So much phenylalanine had accumulated in the mothers’ systems, even though they were too old for their brains to be noticeably affected, that all offspring were essentially poisoned in the womb.

PKU newborn screening became a model for other genetic screening programs for several reasons:

- The test is easy, cheap, and accurate.
- Consequences of *not* screening are costly and severe.
- Early detection improves outcome.
- The test does not alarm parents, nor harm children.
- Genetic testing can predict severity.
- Treatment is available.

The Guthrie test for PKU developed in the 1960s is still performed in many nations on newborns, unless parents decline testing. The 1970s brought tests for other single-gene disorders. The most publicized programs differed from PKU newborn screening in one key way: they targeted specific populations. Promoting a genetic test for a particular group of people, no matter how biologically sound, can open a door towards discrimination, stigmatization, and misunderstanding. Two genetic testing programs of the 1970s were especially telling. One was a success, and the other a disaster. The difference wasn’t in the science, but in the way the tests were delivered.

SUCCESSFUL SCREENS—TAY–SACHS DISEASE AND BETA THALASSEMIA

In the spring of 1973, I took a carrier test for Tay–Sachs disease at my university on New York’s Long Island. Tay–Sachs is a devastating degeneration of the nervous system that begins in infancy and is invariably fatal within a few years. Tay–Sachs screening in the US during my university days was part of a public education program

to test students of Ashkenazi Jewish background, whose ancestors came from central or eastern Europe. Before widespread genetic screening, Tay–Sachs disease was 100 times more common among the Ashkenazim than among most other groups, and carriers were ten times more prevalent.

Tay–Sachs disease is autosomal recessive. I was planning to marry another Ashkenazim, so I knew that, if the test found I was *not* one of the one in thirty Ashkenazim who is a carrier, then he need not be tested—according to the laws of inheritance, we couldn't have a child with Tay–Sachs. The test detected the half-normal concentration of an enzyme, hexosaminidase A, in carriers, who are symptomless. Each child of a carrier couple faces a one in four chance of inheriting the disease. Fortunately, my blood had the normal level of the enzyme.

Thanks to worldwide population screening, today fewer than twenty infants are born each year with Tay–Sachs disease, and most or all of them are *not* Ashkenazim. The disease is so severe, and untreatable, that couples who learn that they are both carriers nearly always choose not to have children together, use egg or sperm from a third person, adopt children, or have prenatal diagnosis and end affected pregnancies. Before genetic testing, about 1 in 3,600 Ashkenazi newborns had the condition.

A baby with Tay–Sachs disease appears normal for the first few weeks. However, deep in the cells, in tiny sacs called lysosomes where powerful digestive juices degrade debris, the missing enzyme causes a buildup of fatty material. Nerve cells in the brain and spinal cord slowly drown in fat, and the body weakens. By six months, the child stops trying to crawl or stand. Just sitting becomes difficult, as nerve messages to muscles grow silent. The child no longer pays attention when read to, and the eyes move strangely. Sudden sounds provoke an odd startle, perhaps even a seizure. At the backs of the child's eyes, "cherry red spots," visible in an ophthalmologic exam, mark the fatty buildup.

By the first birthday, a child with Tay–Sachs disease withdraws. Vision fades. By two years, movement ceases, although the arms may suddenly strike out, a sign of profound brain damage. Swallowing becomes difficult, seizures worsen, and responses ebb. Children with Tay–Sachs disease rarely live past their fourth birthdays.

Mutations do not have a particular fondness for certain types of people. We all have several mutations that, if present in two copies, would cause disease. A mutation becomes associated with a group of people if the genetic change originates within the group, and then the people partner only among themselves, neither diluting the strength of the mutant allele in their gene pool as other people join, nor sending it out as people leave. Such inclusiveness may be by choice, or forced upon a people by the vagaries of history, as it was for persecuted people like the Jews.

The most common allele for Tay–Sachs disease among the Ashkenazim adds four bases to the gene. It and two other mutations account for 98 percent of “Jewish” mutations in this gene. The mutations likely arose shortly after 70 AD, when the Jews were evicted from Rome and wandered, arriving in Russia and Poland by 1100 AD. Throughout history since then, as Jewish people were herded into ghettos and many perished, population bottlenecks (see Chapter 2) sampled, sequestered, and ultimately amplified the most common Tay–Sachs mutations. In addition, like carriers of CF and sickle cell disease, carriers of Tay–Sachs disease may have had an advantage that favored the mutant allele, such as resistance to tuberculosis or even a quick wit that enabled them to survive human-wrought horrors. A few other populations at elevated risk for Tay–Sachs disease have their own mutations—the Irish, Cajuns of Louisiana, the Old Order Amish in Pennsylvania, and French-Canadians.

The first clinician to describe Tay–Sachs disease, in early 1887, was Warren Tay, a dermatologist, ophthalmologist, and surgeon in London. Later that year, a New York City neurologist, Bernard Sachs, published a similar description. Sachs, an Ashkenazi who worked in the Jewish community, noted that cases were “almost exclusively observed in Hebrews.” Tay reported, three years later, that his original case had since acquired two siblings, who were also sick. Four years after that, he noted the disease in families where the parents were related, suggesting autosomal recessive inheritance of an allele from a shared ancestor, such as a common great-grandparent. It wasn’t until 1968 that the enzyme was discovered to be missing in children with Tay–Sachs disease, and in 1970 that carrier parents have a reduced level (O’Brien et al. 1971: 61). Could enzyme levels be used to identify carriers *before* they conceived affected children

together? Michael M. Kaback, then a professor of pediatrics at Johns Hopkins University Hospital in Baltimore, devised such a test.

Carrier screening programs for Tay–Sachs disease soon began, targeting young married couples at synagogues and community centers, and students at colleges with large Jewish populations. Education was a major part of the effort. Students could meet with a genetic counselor at any time during the testing process, or years later, when carrier couples wanted to have children. Genetic counseling at that time was a brand new discipline.

The scope of the Tay–Sachs screening in the US in the early 1970s was staggering: 1.2 million young adults tested, 36,000 carriers identified, 1,056 carrier couples found, and 2,416 prenatal tests performed. As a result, disease incidence plummeted: 99 percent of identified carrier couples had children free of Tay–Sachs disease (Kaback et al. 1993: 2307). In the UK, the National Health Service (NHS) began carrier screening in 1999. The program focuses on schools and community centers in London and Manchester, where most of the UK's 270,000 Jewish citizens live.

In the early 1980s, an orthodox Jewish community in Boro Park, a section of Brooklyn, New York, took Tay–Sachs testing to a new level, with a program called *Dor Yeshorim* (“generation of the righteous” in Hebrew). Rabbi Joseph Ekstein founded it after losing four of his own children to Tay–Sachs disease. *The idea*: to test all teens in the community, with privacy protected, revealing carrier status when a couple begins seriously dating. *The goal*: to avoid carrier couple marriages, or to inform them so that they do not have children together. *Dor Yeshorim* is in accordance with the Jewish law not to end pregnancies. The “informed mate selection” strategy expanded both beyond the Brooklyn community and in the number of autosomal recessive conditions it covers.

By the 1990s, it became possible to switch from enzyme-testing for Tay–Sachs carriers to DNA testing. This new approach, however, actually introduced problems, because the types of mutation and their frequencies vary among populations. A test for the top three Jewish mutations, for example, would catch 98 percent of carriers among the Ashkenazim, but only 20 percent of carriers in other groups, where other mutations are more common.

Another complication in administering DNA-based Tay–Sachs disease tests was marriage outside of the community, resulting in

"admixture" of alleles. During the 1970s, admixture of Ashkenazim with many other groups increased from 13 to 43 percent. The "pure" Ashkenazi couples who had inspired Tay-Sachs testing were becoming rarer, and, as a result, DNA testing had decreased sensitivity in "mixed marriages." For these reasons, Tay-Sachs testing returned to tracking the phenotype—enzyme level—rather than the genotype—specific mutations (Gross et al. 2008: 54). However, for Ashkenazi couples, testing for specific mutations still makes sense.

Several companies offer "Jewish genetic disease" panels. A person with four Ashkenazi grandparents has a one-in-four chance of being a carrier for at least one such disease. "Targeted genetic screening" of specific population groups is not discriminatory. Rather, it is based on the fact that some inherited diseases are more prevalent in certain populations, because of human behavior.

Another example of successful targeted genetic screening is for beta thalassemia. This autosomal recessive anemia is caused by mutations in the beta globin gene that are different from the one that causes sickle cell disease, discussed next. Beta thalassemia is treated with blood transfusions and drugs that counter the buildup of iron from the transfusions. This blood disorder is much more common in certain Mediterranean populations, such as in Greece, Sardinia, and Cyprus, than elsewhere in the world. In Sassari, northern Sardinia (part of Italy), 1 in 10 adults is a carrier (Longinotti et al. 2008: 238).

Several screening programs have offered carrier screening and genetic counseling, and helped carrier couples deal with avoiding or treating beta thalassemia in their children. A program at high schools in Montreal, Canada, offered testing for Tay-Sachs disease to Ashkenazi students and for beta thalassemia to students with Mediterranean ancestry. From 1972 through 1992, the program tested 14,844 Ashkenazim for Tay-Sachs disease, identifying 521 carriers, and 25,274 students of Mediterranean descent, identifying 693 carriers of beta thalassemia. Carrier couples identified when they were 16 years old returned for prenatal diagnosis years later. Incidence of both diseases fell by 90 percent over the two decades, and, today, most cases are outside the original high-risk groups (Mitchell et al. 1996: 793; McCabe 1996: 762). Critical to the success of the Montreal screen was that Canada has nationalized health care, and so people did not fear discrimination based on genotype.

A LESSON IN GENETIC DISCRIMINATION—SICKLE CELL DISEASE

Targeted population screening for Tay–Sachs disease nearly eliminated the illness. In contrast, targeted population screening for sickle cell disease caused confusion, fear, and stigmatization.

A specific, single-base mutation in the beta globin gene causes sickle cell disease, resulting in anemia and other symptoms. Recall from Chapter 1 that the hemoglobin molecule that transports oxygen in the blood is built of two alpha subunits and two beta subunits, encoded by two different genes. Under low-oxygen conditions, the abnormal hemoglobin proteins in a person with sickle cell disease align in a way that bends the red blood cells containing them into crescent shapes (Pauling et al. 1949: 543). These cells lodge in tiny blood vessels, blocking circulation. A stabbing or crushing pain hits in the body parts robbed of oxygen. An attack is called a “crisis.”

Until the early 1980s, sickle cell disease was often fatal in childhood, from complications that included infection, stroke, pulmonary embolism, and kidney, liver, or heart failure. At that time, the name was changed from sickle cell *anemia* to *disease* to acknowledge additional symptoms. However, like many single-gene conditions, sickle cell disease is highly variable—a person might be only mildly ill, or so sick that hospital becomes a second home.

The sickle cell mutation arose independently multiple times in Africa, where the protection against malaria that it offered led to increased prevalence, as discussed in Chapter 2. The mutation also arose in India or Saudi Arabia and is more prevalent among Mediterranean peoples. In the UK, one in nine people in the high-risk population groups is a carrier—Africans, African-Caribbeans, Asians, and people from the eastern Mediterranean and Middle East. Among northern Europeans, sickle cell disease is very rare. In the US, 1 in 400–600 African-Americans has sickle cell disease, and 1 in 10–12 is a carrier.

Reports from Africa of what was probably sickle cell disease date back to the 1600s. The first recorded case was Walter Clement Noel, a dental student in Chicago who had come from Grenada (Herrick 1910: 517). In 1904, Noel went to a hospital, complaining of great fatigue. A young intern, Ernest E. Irons, examined the student’s urine and blood, did a physical exam, and took a medical history. The red blood cells in the smear were “peculiar, elongated and sickle-shaped,”

so odd that Irons sketched them and showed them to his superior, James B. Herrick. The two followed the dentist-in-training for the next few years as he suffered several sickle cell crises. Noel died at age 32 of pneumonia, and Herrick published his case without including his intern (Haller et al. 2001: 889). Another early case report described a 7-year-old girl with a cough; night sweats; pain in her abdomen, legs, and joints; and fatigue, noting that her red blood cells sickled (Scriver and Waugh 1930: 375).

From 1970 to 1972, the US government began pursuing testing for the mutation, despite the lack of prenatal diagnosis, prevention strategies, or treatment for sickle cell disease. The goal was to identify carriers, who were unfortunately said to have “sickle cell trait.” People with the disease would presumably already know it from symptoms. Testing was mandatory in 12 US states and indirectly targeted African-American children and young adults. For example, New York State law required sickle cell carrier testing of anyone applying for a marriage license not “of the Caucasian, Indian, or Oriental races” and of children who lived in cities—the centers of African-American communities. Ironically, this was the time of the Civil Rights movement.

The term sickle cell “trait” was misleading (Fost and Kaback 1973: 742; Hampton 1974: 58). Carriers only rarely experience symptoms, which may happen when they are under low-oxygen conditions, such as at great altitudes. Many physicians did not know the difference between “trait” and “disease.” “Laws identified the diseases of ‘sickle cell anemia’ and ‘sickle cell trait.’ *Trait* is not the same as *disease*. The Air Force Academy wouldn’t take recruits who had the trait. It became a real stigma,” said Dr. Kaback, who led the Tay–Sachs disease screening program in the US and was heavily involved with sickle cell disease screening and testing.

In the US, a major medical lab sold a test directly to consumers. A person would prick a finger, drip blood into a tube, shake it, and then hold the tube against a card that had lines on it. Obscured lines meant the person had some sickled hemoglobin molecules, but the test could not distinguish “trait” from “disease”—that is, the symptom-free heterozygote from the sick homozygote. “That little test spelled disaster. The Black Panthers and ministers, who meant well, bought boxes and boxes of the test and went door-to-door, testing little black kids. They would say, ‘Mrs. J., *your baby has sickle*

cell,'" recalls Dr. Kaback. Unconfirmed reports cited women committing suicide after learning that their children had "sickle cell," when the children were actually healthy carriers.

In 1972, the US Congress passed the National Sickle Cell Anemia Control Act, which tempered the discrimination. Prenatal testing became possible a few years later (Hollenberg et al. 1971: 698; Kan and Dozy 1978: 910). By the 1980s, mortality from sickle cell disease dropped, as affected newborns started a 5-year course of antibiotics and received immunizations to prevent deadly infections. Some patients benefited from bone marrow transplants or an antisickling drug, hydroxyurea.

In 2004, the NHS in the UK began testing pregnant women, and then their partners if the women were carriers, and offering genetic counseling, prenatal diagnosis, and pregnancy termination to carrier couples. By 2006, the NHS offered wider carrier testing, preconception genetic counseling, and newborn screening, focusing on areas in London and the West Midlands, home to population groups at highest risk.

The population screens for Tay–Sachs disease and sickle cell disease in the US differed profoundly, both in how the programs were conducted and in the nature of the two disorders. Tay–Sachs was the less variable and more severe condition—death always came in early childhood. At the time of the screen, prenatal diagnosis was possible, and the education campaigns stressed that carriers are symptomless. In contrast, sickle cell disease varies in severity, and, at the time of the screen, prenatal diagnosis was not possible. As Dr. Kaback said,

The reason sickle cell shouldn't have been done is that there was no intervention (at that time). Find carriers, then the people were left hanging. The reason Tay–Sachs screening worked is that people had an option—not to have children, or prenatal diagnosis.

Public education was vastly different, as Dr. Kaback explains: "For 14 months we planned and developed the educational component for people to understand the idea of testing for Tay–Sachs carriers. The sickle cell programme was a disaster because there was no careful educational plan, and people misunderstood."

Table 4.3 Criteria for a population genetic screen

-
- Excellent sensitivity (proportion of people diagnosed with a condition who actually have it; few false negatives)
 - Excellent specificity (proportion of people found not to have the condition who actually do not have it; few false positives)
 - Screening is less costly than not screening and paying for treatment
 - Preventive measures and/or treatments available
 - Pre-test and post-test genetic counseling offered
 - Ethical review board approves the testing protocol
 - Privacy maintained
 - Informed consent required
 - Outcomes clearly explained
 - Program regularly evaluated
-

The historical and societal backdrop also differed profoundly: In the US at that time, Jewish people were better educated and had had more opportunities than African-Americans, and were less recognizable. As a result, sickle cell disease and trait became stigmatized; Tay–Sachs disease did not, and quietly vanished.

Testing for Tay–Sachs disease and sickle cell disease helped pave the way for today’s genetic tests and genome sequencing, revealing starkly that DNA information cannot be given to people without the explanation that genetic counseling provides. Table 4.3 lists current criteria for population genetic screens.

THE SECOND GENERATION OF GENETIC TESTS: DNA

The genetic screens of the 1970s detected specific proteins. By the late 1980s, when researchers had sequenced the genes behind several common single-gene disorders, DNA testing became possible (Lewis 1993: 14). However, because knowing the genotype (mutations) does not always predict the phenotype (traits or symptoms), results of a DNA test can cause undue anxiety and precipitate life-altering decisions. One reason for the disconnect between phenotype and genotype is penetrance, introduced in Chapter 3. Penetrance is the proportion of individuals inheriting a mutant genotype who actually have symptoms. Two genetic diseases—Huntington’s disease (HD)

and hereditary hemochromatosis (HH)—illustrate how penetrance affects the utility of a DNA-based test. For HD, penetrance is “complete”: anyone who inherits a mutation and lives long enough will develop the disease. For HH, penetrance is “incomplete”: only a small percentage of people who inherit a mutant genotype develop the condition.

HUNTINGTON'S DISEASE: MOVEMENT AND MORE

Before the 1970s, HD was called “Huntington’s chorea,” which focused on its most obvious symptom, uncontrollable, dancelike movements. “Chorea” became “disease” with the realization that subtle behavioral and cognitive (thinking) difficulties are part of the clinical picture. There isn’t any curative treatment.

HD begins gradually and almost imperceptibly, usually in one’s late thirties. The earliest signs may be an odd positioning of facial muscles, or unusual eye movements. The first symptom can be difficulty multitasking. One woman, a police dispatcher, could no longer answer the phone and listen to the police radio at the same time. She burst into tears at her helplessness in following a familiar cake recipe, covering her kitchen in flour, and froze in utter confusion at the supermarket when faced with a wall of cereals to choose from. The woman realized that her difficulty in decision-making and her depression echoed her father’s illness, and that she would probably experience personality changes, such as becoming angry and irritable.

Motor symptoms of HD usually begin shortly after cognitive clues. Fidgeting becomes more frequent and progresses to larger, repetitive movements. Hands wave and feet tap, in endless, apparently purposeless patterns. Loss of the perception of the body’s location in space triggers tripping, falling, and colliding, as if intoxicated. Symptoms worsen. At first, the brain suddenly blocks grasping movements, and reached-for objects crash to the floor; later on, a grasp may morph into a powerful kick or lurch. Woody Guthrie, the famed folksinger, who died of HD in 1967, was overcome with twitches and staggers, constantly whirling about, injuring himself with his flailing limbs.

Like the woman who could no longer multitask, Woody Guthrie saw in himself the signs he knew all too well from his mother. That is the essence of autosomal dominant inheritance: it affects both

genders, and every sick person has a sick parent. About 8,000 people in the UK and 30,000 in the US have HD. Many more people are "at risk," facing the one in two chance of having inherited the mutation from an affected parent.

HD varies in age of onset, the sequence of symptoms, and their severity, even within a family. What doesn't vary is the inevitability of the disease. Therefore, a DNA test is highly accurate and predictive, as symptoms usually do not begin until adulthood. A healthy person who knows he or she has inherited the HD mutation is said to be "pre-manifest." A person "at risk" does not yet know whether she or he has inherited the mutation.

The HD mutation is unusual. It is extra copies of a three-base DNA sequence in a gene that encodes a protein called huntingtin. The more extra copies, the earlier the onset, and the more severe the symptoms. The number of repeats can predict age of onset. The abnormally elongated huntingtin protein affects spiny-shaped neurons in a part of the brain that controls movement, preventing them from sending signals. This initiates a complex cascade of malfunction that kills the cells, lifting controls over movement.

An interesting history lies behind the development of the DNA test for HD. The earliest references to the disease concern women burned at the stake in Salem, Massachusetts, in 1692, after their peculiar movements led people to believe that they were witches. The first clinical description came in 1872, from "horse and buggy doctor" George Sumner Huntington (Huntington 1872: 317). As a boy in eastern Long Island, New York, George accompanied his father and grandfather, also physicians, as they visited patients, including two rail-thin women who contorted constantly, and men who walked as if drunk. Huntington presented his observations on "hereditary chorea" at a medical meeting, noting the late onset and mental disturbance. Stating that the "thread is broken" after a generation does not have the disease, he described autosomal dominant inheritance three decades before Mendel's laws were rediscovered. He and others reported on more than 1,000 cases, over 12 generations of the extended family, tracing the mutation to two brothers from Suffolk, England.

Other large families helped researchers in the quest to discover the HD gene. In 1979, Columbia University psychologist Nancy Wexler, whose mother had HD, began leading yearly visits to a

village where the homes are built on stilts, along the shores of Lake Maracaibo, Venezuela. Here, hundreds of people have HD. They are all descendants of a Portuguese sailor who brought the mutation to the settlement 10 generations ago. The people traded Dr. Wexler blood samples for jeans and sweets. Searching the DNA in the blood samples for sequences that only the affected individuals shared led the investigators to a genetic marker—a sequence linked on the same chromosome as the HD mutation (Gusella et al. 1983: 234). All the sick Venezuelans had the marker. Because a single gene causes HD, the marker would help other families, too.

By 1986, the genetic marker test became available to other families with HD, with limitations. Several relatives had to participate to track the inheritance pattern, because, in some families, the marker would not lie on the same copy of chromosome 4 as the HD mutation itself. This is why accuracy of the marker test was 96 percent, and not 100 percent. Finding the gene itself, a decade later, made the marker test obsolete, increased accuracy, and made a test possible for anyone—no relatives required.

Paramount in planning and delivering the HD test was consideration of the complex emotional and bioethical issues associated with predicting disease in a healthy, young person. The test could tell whether or not an individual could pass the disease to their children. In the past, at-risk individuals had children without knowing this, because symptom onset is usually in one's late thirties, after many people have become parents. Intensive, months-long genetic counseling determines whether a person is psychologically prepared to handle the consequences of testing. The HD test is always given in a medical setting, such as the UK's NHS's genetics centers. Uptake of the test has been less than researchers expected—in many nations, only 15–25 percent of at-risk individuals take it. Fortunately, most people who have dealt with the information that the test provides have done well. Perhaps those who wouldn't be able to handle the information choose not to take the test. People take the test to end the uncertainty and to help with making decisions about having children, choosing education and careers, financial planning, and other actions. The test is reliable because virtually everyone who has a mutation eventually becomes ill, unless they die of something else first. This is very unusual for a single-gene disease.

HEREDITARY HEMOCHROMATOSIS: IRON OVERLOAD

In the best of genetic tests, genotype predicts phenotype, as it does for HD. A genetic test is less useful if penetrance is low, leading to false positives—mutations are there, but the person may not get sick. Perhaps mutations in other genes are protective. “Incomplete penetrance” is the situation for Hereditary Hemochromatosis.

HH causes “iron overload.” The bloodstream absorbs too much iron from food and, over many years, deposits the excess in vital organs, such as the liver, pancreas, and heart. Skin color changes, as noted in the initial description of the disease in a 28-year-old: “From the time this man came into the hospital, I was struck by the almost bronzed appearance of his countenance, and the blackish color of his penis” (Trousseau 1865: 663). If symptoms are untreated, the disease is deadly. Fortunately, treatment is easy: remove blood periodically to keep the iron level down. Menstruation takes care of this naturally for women, who typically do not show symptoms until after menopause.

Two common mutations in a gene called *HFE* cause most cases of HH, which is autosomal recessive. Before the gene was sequenced in 1996 (Feder et al. 1996: 399), the diagnostic test measured how strongly a protein called transferrin binds iron. Adding DNA testing would presumably add precision, but, owing to incomplete penetrance, the opposite occurred. As was the case for Tay–Sachs disease, detecting the phenotype (extra-strong transferrin) was more meaningful than determining the genotype. One study found that the percentage of people who inherit mutant genotypes and who actually develop HH symptoms is less than 1 percent (Beutler et al. 2002: 210)! Penetrance varies by population, but for HH it is always low—and that greatly limits the utility of genetic testing.

HH is worth testing for, however, because it is common, and complications are preventable. In most European populations, 10 percent of people are carriers; in Ireland it is 20 percent. Tests for HH offered direct-to-consumer on the Internet are confusing, because they provide little or no information about penetrance. For example, the American Hemochromatosis Society webpage urges DNA testing: “All it takes is a . . . DNA test kit . . . and you can confirm, or rule out, Hereditary Hemochromatosis in a matter of days.” However, most people with the mutation do not actually have the disease. The genetic test can cause unnecessary anxiety.

The testing programs for all of these single-gene disorders have taught us that it is important to understand the biological basis of a disease to best use genetic information. This is not often the case for mail-in DNA tests offered on the Internet, and consumers unfamiliar with genetics may be misled.

THE THIRD GENERATION OF GENETIC TESTS: EXOMES AND GENOMES

People take single-gene tests in three circumstances:

- screenings conducted on a general population (CF in all pregnant women; PKU for all newborns);
- screenings of populations at elevated genetic risk due to ancestry (Tay–Sachs disease and sickle cell disease);
- relatives have been diagnosed with or have symptoms of a known single-gene condition (Huntington disease).

In some families, an individual has symptoms that do not match any known single-gene (Mendelian) disorders. As mentioned at the opening to this chapter, until the advent of exome and genome sequencing, such families would embark on “diagnostic odysseys” to find an answer, consulting different physicians about different symptoms, with few practitioners considering the whole clinical picture. Today, exome or genome sequencing can reveal whether such an individual has an “atypical presentation” of a known condition, has a disorder so rare that practitioners have simply never seen it before, or has a collection of symptoms and test results that has never been described in the medical literature.

In addition to revealing mutations, determining the exome or genome sequence of a patient and the parents can add even more information. If both parents have the telltale mutation, then the disease was passed down from carriers. In this case, the parents have the mutation too, and each additional offspring faces a 25 percent risk of inheriting the family’s condition. If, however, the parents do not have the mutation, then it is new. Such a “*de novo*” mutation in the child is usually dominant, because it takes just one copy of the mutation to cause the condition.

The first case of exome or genome sequencing to diagnose a condition that had baffled physicians was Nicholas Volker (Worthy et al. 2011: 255). Born in 2004, Nic developed “leaking intestines” by the age of two. Holes in his rectum and large intestine released fecal material, causing severe infections. The symptoms looked like inflammatory bowel disease, but were much more severe. A feeding tube and more than 100 surgeries to remove parts of the large intestine could not stop the relentless onslaught of whatever it was that the boy had. Genetic tests for anything resembling the symptoms were all negative. A clue, at first overlooked, was that blood transfusion seemed to help. By age 4, he weighed under 1.4 stone (20 pounds).

The answer to the mystery of Nic’s leaking digestive tract came from exome sequencing. The Medical College of Wisconsin had planned to use the fledgling technology in 2014, but moved up the date to try to save Nic’s life. At the time, exome sequencing cost about £52,000 (\$75,000); today it costs under £700 (\$1,000). However, interpreting the information adds to the expenses. Analysis of Nic’s exome sequence in 2009 finally revealed that he had inherited a single-gene condition passed on the X chromosome, called XIAP (for “X-linked inhibitor of apoptosis”). His mother’s exome sequence had the mutation too, so she is a carrier. XIAP was known—but not for causing leaky intestines. Instead, it was recognized as an immune system disorder. The gene encodes a protein that normally prevents the immune system from attacking the intestines when it is fighting viral infection there. Nic’s gene had one different DNA base, and that was enough to block the protection. His intestines were indeed dissolving. The transfusions were a reprieve because they supplied the protein.

Fortunately, XIAP is treatable with a cord blood stem cell transplant, which Nic received from an anonymous donor in 2010. He has recovered, but still suffers many seizures a day from a long-ago infection, has to live with a partial digestive tract, and suffers from post-traumatic stress disorder. The newspaper that covered his ordeal won a Pulitzer prize and introduced exome sequencing, and the brave Nicholas Volker, to the world (Johnson and Gallagher 2015).

Genome sequencing is replacing exome sequencing as the cost of deciphering DNA plummets—why not get a full picture? In the UK, the 100,000 Genomes Project is using genome sequencing to diagnose children whose symptoms have not fit any disease descriptions (see the introduction to Chapter 3). Sometimes the answer leads to treatment, as it did for Nic, and sometimes it only provides information that might someday become practical.

When Georgia Walburn-Green was born in 2012, physicians at first thought that she had only an enlarged head. But as she grew into toddlerhood, she still couldn't talk, although she clearly understood speech. She also had poorly functioning kidneys, lagged far behind in growth, and had tumors in her eyes. The family went on a 4-year “unknown journey slowly finding problems,” said her mother, Amanda. Genome sequencing at Great Ormond Street Hospital as part of the 100,000 Genomes Project led to detection of a mutation in a gene called *KDM5b*. The gene encodes an enzyme that removes methyl groups from certain genes. The mutation causes multiple symptoms, because it controls the expression of several genes.

There aren't any treatments yet for Georgia's condition—interfering with the function of a “demethylase” might have adverse effects by inappropriately removing methyl groups from where they're needed in the genome. But another little girl being treated at Great Ormond, named Jessica, did find a happy ending to her diagnostic odyssey thanks to genome sequencing. Her seizures were due to GLUT1 deficiency syndrome, which impairs glucose transport into the brain. A special high-fat diet offered an alternative energy source, and she was able to sharply reduce her anti-seizure medications.

Every human genome sequenced contributes data to the emerging portrait of the diversity of the human species at the DNA level. As genome sequencing becomes part of the diagnosis for diseases with unfamiliar symptoms, and also part of population-wide projects, many of us, sick as well as healthy, will have the opportunity to learn about our own genes and genomes (Lewis 2015: 1301). Will you be among them?

WHAT WOULD YOU DO?

The 6-year-olds line up, set to race 100 meters so that the watching coaches can determine who is likely to be a better soccer player. The kids aren't nervous—they're excited and having fun—but some of the parents are.

"I'm not worried at all. Jared's guaranteed to be a great forward," said one mother, smugly, to another.

"Guaranteed? How can you know that?" asked the other parent, looking skeptical.

"A genetic test. Jared was tested when he was a baby, so we knew he'd be a superior sprinter. Pretty great, right?"

The other mother was puzzled. "Why couldn't you just wait until he got a little older, to see if he runs fast? And if he even *likes* running?"

"You just don't understand."

Actually, the mother who had a genetic test performed on her young son didn't understand. The test was for an allele of a gene called *actinin 3* (*ACTN3*). Two alleles of the gene are known, *R* and *X*. After a study reported that people with two *R* alleles (*RR*) are overrepresented among sprinters, and people with two *X* alleles (*XX*) are more common among endurance athletes, a company invented a test, guessing that parents anxious for their children to be the best would race to it. They were right.

Theoretically, the test is based on the fact that the *R* allele encodes a protein that functions with fast twitch muscle fibers, powering sprinting. When the gene is deleted (the *X* allele), the protein is not made, and the slow twitch muscles instead have an advantage, promoting endurance. Through evolutionary time, either genotype had an advantage, depending upon whether it was more adaptive for our ancestors to be able to sprint past a cheetah and leap into a tree, or outrun a carnivore over a long distance. Both alleles, therefore, have stood the test of time.

Genes do not act in a vacuum. Other genes, as well as environmental influences, greatly impact athletic ability. One geneticist remarked of the test, "Why not simply line kids up and see who runs the fastest?"

Would you have the *ACTN3* test performed on your young child?

SUGGESTED READING

Genetic testing unfolded over decades. “A Brief History of Genetic Testing” provides an overview of the early single-gene tests (2008; Science Progress; [www.scienceprogress.org/2008/05/a-brief-history-of-genetic-testing/]). Lisa Genova’s 2015 novel *Inside the O’Briens* (2015; Simon & Schuster) paints a compelling portrait of the slow progress of Huntington disease and the struggle to decide whether or not to take the predictive genetic test. *One in a Billion: The Story of Nic Volker and the Dawn of Genomic Medicine*, by Mark Johnson and Kathleen Gallagher (2016, Simon and Schuster), is the amazing story of Nicholas Volker, the boy who was one of the first whose diagnostic odyssey ended thanks to exome sequencing. The authors won the Pulitzer Prize for their series in the Milwaukee Journal Sentinel (www.jsonline.com/news/health/111224104.html). Ricki Lewis tells the story of another boy who got an answer following exome sequencing in “Exome Sequencing Comes to the Clinic” (2015, *The Journal of the American Medical Association* 313(13): 1301–1303) and in “Vincent’s Story: Discovery of a New Autism Spectrum Disorder” at Rare Disease Report (www.raredr.com/contributor/ricki-lewis-phd/2016/02/vincent-s-story-discovery-of-a-new-autism-spectrum-disorder).

GENE THERAPY AND GENOME EDITING

Imagine a recipe that has an error. It calls for too much or too little of an ingredient, leaves one out, or perhaps includes something not quite right. Correcting the instructions can perhaps salvage the dish. Compensating for or correcting errors is the idea behind the two biotechnologies discussed in this chapter. **Gene therapy** adds a working copy of a gene that is missing or mutant in a cell, but it does not remove the existing gene. **Gene editing** is more precise than gene therapy because it can add, remove, or replace a gene. **Genome editing** refers to editing more than one gene at a time. Gene therapy and gene/genome editing are performed on the cells that are affected in a particular disease, or alter other cell types in a way that counters symptoms.

PROTEIN THERAPY CAME FIRST

The idea to treat genetic diseases at the DNA level arose shortly after Watson and Crick built their models of the genetic material, from deducing how DNA replicates and encodes protein. However, understanding gene structure, function, and regulation was an enormous task that took thousands of researchers decades. It isn't surprising that it took until 1990 for the first clinical trial to test a gene therapy in people. However, treatments at the protein level

preceded and then proceeded in parallel to the development of DNA-based therapies. These protein-based strategies work in three basic ways:

- *Enzyme replacement* restores functional levels of missing or deficient enzymes.
- *Substrate reduction* counteracts the buildup of the substance that a missing or deficient enzyme normally acts upon.
- *Pharmacological chaperone therapy* re-folds a misfolded protein.

Fabry disease illustrates the first two types of protein-based therapies. The disease is due to a mutation in the gene that encodes an enzyme called alpha-galactosidase A (*GLA*) that breaks down a fat-sugar molecule called globotriaosylceramide. Fabry disease is a lysosomal storage disease, similar to Tay–Sachs disease, discussed in Chapter 4. *GLA* is on the X chromosome, and therefore Fabry disease is passed from carrier mothers, who aren't affected or have mild cases, to sons, who are severely affected. About 10,000 people globally have Fabry disease.

The enzyme deficiency behind Fabry disease causes globotriaosylceramide to accumulate in many types of cells. Symptoms appearing in childhood include pain in the feet and hands, poor sweating ability, eye clouding, ringing in the ears, dark red spots, stomach upset, and hearing loss. Later symptoms include stroke, heart attack, and kidney damage.

Enzyme replacement therapy for Fabry disease is an intravenous infusion of the enzyme. Substrate reduction therapy intervenes in the production of globotriaosylceramide, in pill form. (The dietary treatment for PKU, discussed in Chapter 4, is another example of substrate reduction therapy.) An example of pharmacological chaperone therapy is the drug ivakaftor (Kalydeco), used to treat cystic fibrosis and described in Chapter 2. It enables CFTR protein to fold correctly, forming ion channels that permit salt components to enter cells that line the respiratory tubes and other organs.

Protein-based treatments for genetic diseases will continue to be used and developed, but they must be given repeatedly, like any drug, and they can be very costly. Gene therapy and gene/genome editing, however, promise more permanent solutions to the problem of treating genetic disease. In clinical applications, both gene therapy and

genome editing are done on somatic cells, so that they correct genetic instructions only in the body parts that the disease affects. Gene therapy or genome editing performed on eggs, sperm, or fertilized ova would affect the **germline**—the formation of a new individual. We return to this controversial point at the end of the chapter.

GENE THERAPY

Gene therapy introduces a wild type (“normal”) copy of a gene into a cell where that gene is absent or malfunctioning. The DNA may be injected into a cell, sent in within fatty bubbles called liposomes, or, most commonly, delivered as parts of the genetic material of viruses, which naturally enter cells. Viruses used in this manner are called vectors. Researchers select a particular type of virus based on the virus’s natural target and remove the parts that would cause disease and evoke a human immune response. Some viruses infect only certain cell types, and some infect only cells that are dividing. The two most widely used gene therapy viral vectors are adeno-associated viruses (AAV) and lentivirus (HIV, disabled to not cause AIDS). Were the genetic code not universal—all species and viruses manufacture the same amino acids using the information in the same RNA triplets—gene therapy would not be possible.

THE FIRST APPROVED GENE THERAPIES

The challenges in gene therapy are daunting. The introduced genes must:

- enter enough of the correct cell types, in the targeted part of the body, to exert a measurable or noticeable effect;
- be transcribed into mRNA and then translated into protein in the targeted cells;
- function long enough to prevent or alleviate symptoms and keep them from worsening for an extended time or indefinitely;
- not harm other cells they might enter;
- not disrupt genes in a way that causes cancer.

In order to conceptualize and develop a gene therapy, the mechanism of the pathology must be understood, causative mutations known,

and the affected body part accessible. The gene therapies that are approved or are farthest along in development are for diseases that affect only a few patients, so they may seem odd choices. However, those ultra-rare diseases went first because their mechanisms are the best understood, and therefore the gene therapy is the most likely to be effective. Most common diseases are *not* caused by mutation in one gene.

Not all genetic diseases are good candidates for gene therapy. Treating cystic fibrosis, for example, requires reaching cells that aren't coughed up in a matter of weeks. Treating muscular dystrophy requires getting the therapy to enough muscle cells to make a difference in the clinical course of the disease, such as enabling boys to take more steps. With such compelling criteria for a gene therapy to be effective, it's little wonder that the first protocol took 26 years to be approved. The European Medicines Agency approved that first gene therapy, alipogene tiparvovec (Glybera), to treat the painful disorder lipoprotein lipase deficiency (LPLD) in late 2012.

The enzyme deficiency behind LPLD causes triglycerides to accumulate in and clog the pancreas, liver, skin, and spleen. The research path to the gene therapy began in the laboratory of Michael Hayden at the University of British Columbia. In September 1986, the researchers met a 19-year-old with sky-high triglycerides, painful pancreatitis, and skin lesions, and identified his *LPL* mutation (Kastelein et al. 2013: 472). A severely fat-restricted diet didn't help the young man to overcome the enzyme deficiency by reducing the substrate. So the team developed a gene therapy that delivers the wild type version of the gene, working with cats that naturally get the disease and a mouse model. Then a few severely affected patients were given the gene therapy, delivered in AAV. Their abdominal pain diminished, triglyceride levels fell, and they could eat more, including foods they couldn't tolerate before. One patient even had a baby.

Glybera is extremely expensive, at \$1.4 million for a one-time series of 42 injections into various muscles. Only about 1,000 individuals in Europe and about 340 people in the US have LPLD. But the gene therapy is serving as a model for others.

For gene therapy, the path from theoretical idea to practical treatment has been like a roller coaster, with tantalizing highs followed by plunging lows. After initial successes, too many clinical trials ended in short-lived improvement, or none at all. Then, in 1999,

a death due to gene therapy stalled the field for a decade, until a handful of promising results revived interest. Below are the stories of some of the children who have helped to put gene therapy on the map. Table 5.1 summarizes their conditions.

A ROCKY ROAD

The first gene therapy clinical trial in the US began at the National Institutes of Health, on a September afternoon in 1990, when 4-year-old Ashanthi DeSilva received an infusion of her own helper T cells. (Recall from Chapter 3 that these cells activate the B cells that produce antibodies.) Ashi had a rare form of inherited severe combined immune deficiency (SCID) called *adenosine deaminase (ADA) deficiency* (ADA-SCID). Her T cells had been removed, given normal ADA genes, and then reinfused into her body. Children with ADA deficiency typically spent their short lives in and out of hospital, receiving blood transfusions and bone marrow transplants to temporarily bolster their failing immunity.

The gene therapy that dripped into Ashi's arm worked, but only transiently because T cells have a short lifespan. Ashanthi and another little girl improved, but they still had to take enzyme supplements to maintain their immunity and stay healthy. Soon the researchers repeated the experiments but with two important changes: they tried the gene therapy on babies, and gave them their own umbilical cord stem cells that had functioning ADA genes added. Because these cells continually give rise to T cell precursors, more cells have the added gene and immunity lasts longer. Since then, several dozen children born with ADA deficiency have undergone gene therapy and are well. The clinical trials have been conducted in the UK, Italy, the US, and France. In April 2016, the European Medicines Agency recommended approval of gene therapy for children with ADA-SCID. Approximately 15 children are born in Europe with the disease each year. The treatment is called Strimvelis and full approval is expected by 2017. It was developed at the San Raffaele Telethon Institute for Gene Therapy in Milan and GlaxoSmithKline.

The promising start to gene therapy for ADA deficiency slowed with two serious setbacks. The first happened in September 1999, when 18-year-old Jesse Gelsinger underwent gene therapy for

Table 5.1 Gene therapies discussed in the chapter

<i>Disease</i>	<i>Symptoms</i>	<i>Gene therapy</i>
Lipoprotein lipase (LPL) deficiency	Triglyceride buildup causes abdominal pain, skin lesions	AAV delivers LPL injected into muscles
Adenosine deaminase (ADA) deficiency	Disabled immunity cannot fight recurrent severe infections	Retrovirus delivers patient's fixed genes into stem cells introduced into bloodstream
Ornithine transcarbamylase (OTC) deficiency	Ammonia buildup in brain causes abdominal pain, coma	AV delivered wild type genes into liver, caused death
Severe combined immune deficiency X1 (SCID-X1)	Disabled immunity cannot fight recurrent severe infections	Retrovirus delivering wild type gene instead activated oncogene. New safe retroviral and lentiviral vectors developed
Leber congenital amaurosis (LCA) due to RPE65 deficiency	Night blindness and progressive visual loss	AAV introduces wild type <i>RPE65</i> genes into retinal pigment epithelium, restoring vision
Giant axonal neuropathy	Progressive loss of mobility as motor neurons become unable to stimulate muscles	AAV introduces wild type <i>gigaxonin</i> genes into motor neuron axons in spinal cord

ornithine transcarbamylase (OTC) deficiency. The disease is recessive and inherited on the X chromosome. It impairs the ability to digest protein, stripping off the nitrogen atoms from the constituent amino acids and combining them with hydrogen atoms to form ammonia. Normally ammonia would exit the body in urine. In OTC deficiency, excess ammonia travels in the bloodstream to the brain, where it does great damage. Most babies born with the mutation in every cell become comatose the first day and die by the second day. Jesse had volunteered for the gene therapy trial to try to “save the babies.” He had a very mild case because the mutation was in only some of his cells, and a very-low-protein diet and medications that bind ammonia had helped him. When he was a senior in high school, a lapse in self-care and an ensuing coma that lasted a few weeks led him to volunteer for the clinical trial to test the gene therapy. It would provide a permanent solution, an alternative to taking dozens of pills a day and so carefully monitoring diet.

On Monday, September 13, 1999, Jesse became the eighteenth person in the clinical trial to receive billions of a type of virus called an adenovirus (AV) in his liver. Each virus had, as part of its DNA, a copy of a normal human OTC gene. But, in addition to entering the type of liver cell that would compensate for the genetic defect, the viruses also entered immune system cells in Jesse’s liver. That triggered a massive, body-wide immune response, apparently to a previous infection by a related virus. Within days, Jesse suffered multi-organ failure. On the Friday after the treatment, Jesse’s father, in a room packed with the tearful medical staff at the University of Pennsylvania hospital, turned off life support (Gelsinger 2001).

Jesse’s death prompted the redesign of many vectors and clinical trial protocols, and many experiments were delayed, some for years. Children with inborn errors of metabolism stopped receiving gene therapies that were promising. Adults in some trials for cancer had to stop treatment. Then in 2000 came the second blow to the fledgling field. Five of 20 boys who had received experimental gene therapy to treat severe combined immune deficiency X1 (SCID-X1) (“bubble boy disease”) developed leukemia and one boy died of the white blood cell cancer—although 18 of the boys had become able to mount an immune response. The genetic disease had been conquered, but at what cost? It turned out that the viral vectors—a type called a retrovirus—had integrated into oncogenes (discussed in Chapter 3).

Because the inherited disease had responded, an international team altered the retroviral vector so that it would “self-inactivate” and not stimulate the oncogene. At the same time, another group of researchers began a clinical trial using a different vector entirely, lentivirus. Gene therapy for SCID-X1 now works, without causing leukemia.

The deaths of Jesse Gelsinger, the boys with leukemia, and then, in 2007, of a young woman being treated with a gene therapy injected into one knee to fight rheumatoid arthritis, sidelined many promising gene therapy trials. The picture changed in late 2009.

RESTORING VISION

In 2007, three groups of researchers, one in the UK and two in the US, treated three patients each for a form of hereditary visual loss then called Leber congenital amaurosis type 2. Today “caused by deficiencies in *RPE65*” is tacked onto the name, to indicate the gene behind the condition and distinguish it from 22 other forms of the hereditary blindness. Patients make too little of an enzyme that enables the eye to use vitamin A. The active form of the vitamin is a crucial component of a biochemical pathway called the visual cycle, which changes the energy from incoming light into signals from firing neurons that the brain’s visual cortex recognizes as an image. The enzyme works in a very thin layer just beneath the retina called the retinal pigment epithelium, or RPE. The gene, delivered in AAV, is gently slipped into the eye with a needle while the patient is anesthetized.

Results with this gene therapy were and continue to be spectacular, both in the first patients treated and today. More than 300 individuals destined for blindness are now able to see. A report from one of the patients from 2007 illustrates the dramatic effect. Eighteen-year-old Steven Howarth, treated by the team at University College London/Moorfields Eye Hospital, appeared on BBC shortly after, demonstrating his newfound ability to see the frets on his guitar. “I used to be shy and silent. Now I have confidence to walk around on my own,” he said. Other patients reported being able to see an illuminated alarm clock and the lights on the dashboard of a car for the first time, and another enjoyed seeing a starry sky for the first time. Said eye surgeon James Bainbridge, MD, of Moorfields,

“It’s tremendously exciting to see that this technique is safe in an extremely fragile tissue and can improve vision in a condition previously considered wholly untreatable.” The Moorfields group also included geneticist Robin Ali, PhD, and retinal specialist Tony Moore, MD.

Over the years, the ability to see following the gene therapy has persisted, but researchers are still investigating different viral vectors and determining exactly where to place the genes in the eye to achieve the longest lasting and best results (Bainbridge et al. 2015: 1887). Regulatory approval is expected in the US in 2017. One of the early participants there, Corey Haas, was 8 years old at the time his first eye was treated in 2008 and continues to enjoy an astonishing brightening of his world (Lewis 2012). He even goes turkey hunting!

THE FIRST GENE THERAPY INTO THE SPINAL CORD

When Lori and Matt Sames saw newborn Hannah’s blond corkscrew curls on March 5, 2004, they were astonished, because they and their other two daughters have straight hair. Hannah smiled, sat, crawled, scooted, and walked on time, but her footsteps were halting and hesitant. She slowly grew clumsy, as the strength ebbed from her legs. Many physicians that her parents took her to were either flummoxed or accused the distraught Lori of overworrying. Yet already, unusually abundant filaments of a protein called gigaxonin were filling and distending the long axons of the motor neurones running down Hannah’s legs, blocking messages from reaching her muscles. Her kinky hair was a sign too—it came from abnormal keratin protein filaments filling her hair follicles.

Phone video of Hannah at age 3 showing her distinctive hobbling alerted a friend who was a physical therapist and thought the child might have a form of muscular dystrophy. That led, finally, to a pediatric neurologist. “He took out a huge textbook and showed us a photo of a skinny little boy with kinky hair, a high forehead, and braces that went just below the knee—he looked exactly like Hannah,” Lori recalled. The boy had a very rare disease called giant axonal neuropathy (GAN). Three days of tests at a children’s hospital in New York City confirmed the diagnosis. Hannah was one of only about 40 children known at the time to have the disease.

A genetic counselor pronounced GAN a death sentence, but Lori and Matt, instead of collapsing in despair, instead founded an organization, Hannah's Hope Fund. Over the next 8 years, they assembled a research team led by Steven Gray, PhD, at the Gene Therapy Center at the University of North Carolina at Chapel Hill. The team developed a gene therapy that is being given to children at the National Institutes of Health Clinical Center, the same place where Ashi DeSilva received her groundbreaking gene therapy for ADA deficiency in 1990.

GAN is autosomal recessive; Lori and Matt are each carriers. The family's mutations are deletions, with DNA missing from chromosome 16. That posed a problem at first for Hannah. Because her cells do not make any gigaxonin at all, the sudden flood of the protein following gene therapy could appear foreign to her body, evoking a deadly immune response like the one that killed Jesse Gelsinger. The doctors and scientists had learned from his case and Hannah received immunosuppressant drugs to prepare her body for the new genes.

The gene therapy, delivered using AAV, is the first to be introduced into the spinal cord. The viruses carry functional gigaxonin



Figure 5.1 Hannah Sames has giant axonal neuropathy (GAN). The tireless work of her family is making gene therapy possible. © Dr. Wendy Josephs

genes into the roots of motor neurones there, entering the nuclei and enabling the cells to produce the missing protein. If the GAN gene therapy works, it could lead to ways to treat other inherited diseases, such as spinal muscular atrophy, but also motor neurone disease (amyotrophic lateral sclerosis) and even spinal cord injuries.

By the time Hannah received 100 trillion viral vectors in 2016, she was 12 years old. She was wheelchair-bound but could still curl her toes, and beginning to lose her voice and vision. Today she is stronger. Without the gene therapy, Hannah would have lost her battle in her early twenties (Figure 5.1).

GENE AND GENOME EDITING

Genome editing is a new technology that cuts and pastes genetic material much like word processing software makes it possible to easily edit text. Several techniques are used to add, remove, or replace specific DNA sequences. Gene and genome editing introduce an exquisite precision to the older techniques of gene therapy, which add functional genes but do not delete the nonfunctional ones. Genome editing is especially powerful because it can simultaneously target multiple genes; gene therapy targets single genes.

DEPLOYING MOLECULAR SCISSORS

The techniques of gene and genome editing borrow from natural systems that repair double strand breaks in DNA. The “molecular scissors” that genome editing uses are enzymes, called restriction enzymes, that bacteria use to protect their own genomes from viral infections. The enzymes are special in that they cut *both* strands of the DNA double helix, which triggers the cell’s natural DNA repair pathways to heal the damage. Clever packaging of desired DNA sequences with the bacterial enzymes introduces the intended genetic change.

Although the first genome editing tools began being developed in 2009, the idea was formulated in the late 1980s. It led to the 2007 Nobel Prize in Physiology or Medicine awarded to Mario Capecchi, Martin Evans, and Oliver Smithies for their harnessing of homologous recombination, which is one of two DNA repair pathways activated by double strand breaks. The trio used the strategy to

create “knockout” mice missing key genes. Such mice have since served as models for many human genetic diseases.

The three major genome editing tools used today trace their conceptual origins to the bacterial restriction enzymes (Table 5.2). These first “molecular scissors” became the basis of recombinant DNA technology in the 1970s, discussed in Chapter 1. Restriction enzymes, and the enzymes used in genome editing, are endonucleases, which means that they cut nucleic acids (DNA and RNA) at specific base sequences that are not at the ends of the molecules. These enzymes consist of a part that binds DNA at a certain short sequence (like the controlling end of scissors positioned where paper is to be cut) attached to the endonuclease part that snips the double helix across both strands (the dual blades of the scissors).

CRISPR-CAS9

CRISPR-Cas9 is the genome editing technique that has exploded into public consciousness since the first paper describing it was published in late 2011, because it is easier to use than the other two techniques, zinc fingers and TALENS. CRISPR-Cas9 is the newcomer, but its speed, ease, and lower cost is catapulting it into prominence, with global market prediction for 2019 topping \$3.5 billion.

Table 5.2 Genome editing techniques

<i>Technique</i>	<i>How it works</i>
Zinc finger nucleases (ZFNs)	Part of a protein that folds into finger shape, discovered in African clawed frog in 1987. Joined to restriction enzyme FokI, a ZFN cuts DNA at a specific sequence.
TALENs (transcription-activator-like effector nucleases)	A DNA-binding protein from the plant pathogen <i>Xanthomonas</i> joins the restriction enzyme FokI to cut DNA at a specific sequence.
CRISPR-Cas9	Bacteria integrate viral DNA near repeats (CRISPRs), and transcribe both viral DNA and repeats into RNA. The RNA associates with a protein (Cas9) to induce a double strand break that removes the viral DNA and activates DNA repair.

The acronym “CRISPRs” stands for “clustered regularly interspaced short palindromic repeats” and “Cas9” stands for “CRISPR associated protein 9.” The technology arose from a collaboration between two women who met at a conference in 2011, although others had similar ideas at about the same time. Since 2006, microbiologist Emmanuelle Charpentier, PhD, director of the Max Planck Institute of Infection Biology in Berlin, had been investigating how bacteria resist disease, such as from viruses, which she terms “invading genomes.” “The host has a defense against the invaders that is considered the innate immune system of bacteria,” she said.

Meanwhile, Jennifer Doudna, PhD, professor of molecular and cell biology and chemistry at the University of California, Berkeley, was using the gene knockout technology from the 1980s. She was interested in the short repeated DNA sequences of CRISPRs. What do they do? If they guide DNA-cutting, could they provide a way to easily manipulate genomes?

Dr. Doudna explained how CRISPR-Cas9 works.

Bacteria defend against viral infection by acquiring little bits of DNA from viruses into their genomes. They make RNA copies of viral sequences, and incorporate them into one or more proteins used to target the viral DNA. Then the RNA-protein complex finds double-stranded regions of DNA, unwinds them, and positions itself so two active sites can cut the double-stranded DNA at a precise, targeted sequence. Cells recognize double strand breaks and repair them using two pathways that add new sequence or heal the old.

This molecular machine—the CRISPRs and Cas9—enables the bacteria to “remember” viral genomes they have encountered and remove them if they appear again—much like how an immune system functions in more complex organisms.

The CRISPR-Cas9 system can rapidly search extensive sequences of DNA and recognize and home in on a specific sequence. “Bacteria use CRISPR-Cas9 to cut a viral DNA sequence, but scientists harness it to make double strand breaks where we might like to introduce a small change in the genome,” said Dr. Doudna. Dr. Charpentier added, “The RNA-programmable CRISPR-Cas9 allows precise surgery in the cells of many organisms, including mice, plants, monkeys, and humans.” Guide RNAs are short, synthesized

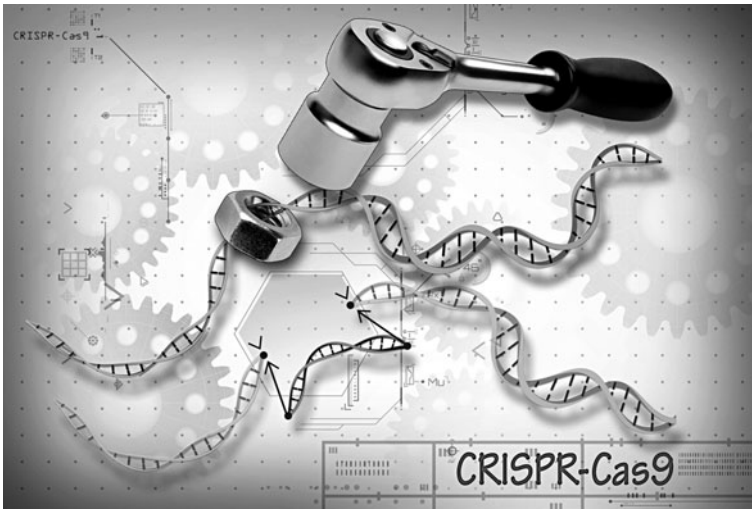


Figure 5.2 CRISPR-Cas9 is a gene and genome editing tool that enables researchers to add, delete, or replace selected genes.

National Human Genome Research Institute (NHGRI)

molecules created to target and cleave a specific DNA sequence at two strands.

The women and their co-workers described the new gene-cutting technique in late 2011 and published a report in *Science* in early 2012 (Jinek et al. 2012: 816). The guide RNA introduces the versatility. It can be designed to add a DNA sequence, replace a sequence, or remove a sequence—just like word processing software. Dr. Doudna compares ZFNs and TALENs to hardware, requiring labor-intensive synthesis of key proteins, but CRISPR-Cas9 is more like software, requiring just a change to the guide RNA to target different addresses in the genome (Figure 5.2).

ECLECTIC APPLICATIONS

CRISPR-Cas9 and variations on it have inspired discussion, excitement, fears, and at least one lawsuit for patent infringement. Feng Zhang, PhD, and his team at the Broad Institute at the Massachusetts

Institute of Technology and Harvard University published a report about using CRISPR-Cas9 on mammalian cells in 2013 (Cong et al. 2016: 819). Drs. Doudna and Charpentier filed their patent first, but Dr. Zhang's patent was put into an expedited program and was granted first. While the fighting over credit continues, the list of applications of gene and genome editing, mostly using CRISPR-Cas9, is growing. It includes:

- correcting diseases by removing or replacing mutant genes;
- rapidly creating animal models of human diseases;
- creating hypoallergenic chicken eggs that do not have the egg white protein to which some people are allergic, enabling them to tolerate vaccines grown in eggs;
- fashioning pigs that have had genes removed that enable them to be used as organ donors to people;
- using "de-extinction" to introduce genes of ancestral species into modern genomes, such as engineering a mammoth from an elephant, a passenger pigeon from a pigeon, and a small mammal called a jerboa from a mouse;
- introducing traits into show animals or pets faster than can be done with traditional breeding. This approach can potentially endow a lethargic dog breed such as a basset hound with the natural sprinting ability of a sheepdog, create goldfish with interesting color patterns, and fashion pet mini-pigs;
- providing resistance to diseases. Dogs given warthog genes resist African swine fever, cattle resist sleeping sickness, and people whose cells have the *CCR5* mutation cannot become infected with HIV (discussed in Chapter 3);
- controlling "vector" species such as ticks and mosquitoes that spread infectious diseases to humans;
- creating genetically modified organisms that cannot escape their controlled environments. Farmed catfish have their reproductive hormone genes removed, so that they can only survive in water to which the hormones are supplied;
- adding hornlessness genes from elite cattle breeds to other breeds to eliminate having to remove horns to ship the animals.

Gene and genome editing are being developed to treat several types of diseases in humans. Testing in humans is years behind gene

therapy, but the newer technologies are much faster to develop. Here is a closer look at the evolving role of gene editing in tackling inherited disease, cancer, and infectious disease.

SICKLE CELL DISEASE AND BETA GLOBIN

The challenge in using gene editing to treat single-gene (Mendelian) diseases is the same as for gene therapy—delivering the intervention to enough of the correct cell type to prevent worsening of symptoms or even to restore normal function. Returning to the example of Duchenne muscular dystrophy, any gene-editing-based treatment would have to be injected into enough affected muscles to enable a boy to walk. Blood is different. Blood-forming (hematopoietic) stem cells introduced into the bloodstream can divide to give rise to daughter cells that can differentiate as any blood cell type, theoretically enabling the stem cells to combat many diseases.

Sickle cell disease is an especially good candidate for gene editing because the implicated beta globin gene, which is part of the molecule hemoglobin that carries oxygen, is right next to several “shadow” copies of similar genes that function under the different oxygen requirements of the embryo and fetus. Drugs are already being used that “turn on” quiescent fetal hemoglobin genes, enabling the fetal version of beta globin to compensate for the sickle beta globin. (Chapters 2 and 4 discuss the disease further.)

CRISPR-Cas9 approaches to combating sickle cell disease are both direct and indirect.

In the direct approach, researchers from Johns Hopkins University teamed CRISPR-Cas9 with induced pluripotent stem (iPS) cells made from patients who have sickle cell disease. Because iPS cells are derived from somatic cells, such as skin fibroblasts, they can be exposed to chemicals that coax them to differentiate as specific needed cell types without destroying embryos. Human iPS cells can recapitulate a disease’s start in a lab dish.

To tackle sickle cell disease, researchers began with skin fibroblasts from patients and reprogrammed the cells in a lab dish to develop into red blood cells. At the same time, they used CRISPR-Cas9 to alter one copy of the single-base mutation that causes the disease, making the manipulated cells able to encode functional adult beta globin (Huang et al. 2015: 1470). Although mature red blood cells

normally do not have nuclei and therefore do not have genes, the gene-edited red blood cells indeed contained normal hemoglobin, indicating that the change had worked on the precursor stem cells and carried through the developmental pathway to reticulocytes, which extrude their nuclei and thereby become mature red blood cells, or erythrocytes. Lacking a nucleus enables these highly specialized cells to be packed with hemoglobin. “Our results represent a significant step toward the clinical applications of genome editing using patient-derived iPSCs to generate disease-free cells for cell and gene therapies,” the researchers wrote.

In an indirect approach using gene editing to treat sickle cell disease, researchers from Dana-Farber/Boston Children’s Cancer and Blood Disorders Center, the Broad Institute, and Harvard manipulated a variant of an “enhancer” gene, called *BCL11A*, to inhibit the activity of the molecular switch from fetal to adult hemoglobin. This action enables some molecules of fetal hemoglobin to persist, stepping in for adult beta globin (Canver et al. 2015: 192). The change is restricted to exactly where it is needed—red blood cells. The researchers used CRISPR-Cas9 to engineer the change, via small deletions, in hematopoietic stem cells, which give rise to all blood cell types. Red blood cells derived from those starting cells indeed manufactured fetal beta globin. Daniel Bauer, MD, PhD, a pediatric hematologist/oncologist at Dana-Farber/Boston Children’s said:

There was no efficient way of conducting this kind of experiment until now. Our goal was to break the enhancer, rather than fix the hemoglobin mutation, but to do so in very precise ways that are only possible since gene editing technologies like CRISPR became available.

CANCER AND CAR TECHNOLOGY

Treating white blood cell cancers—leukemias—are among the first clinical uses of gene editing. A strategy called **chimeric antigen receptor** (CAR) technology arms cells with hybrid cell-surface proteins whose components form naturally in the body, but not together.

A CAR joins two arms of the immune system. It is part T cell receptor, binding to and activating T cells (a type of white blood cell) to signal the production of immune chemicals called cytokines.

A CAR is also part antibody, which guides the doctored cell. Like a drone or missile, the antibody part of the CAR directs the T cell to a specific target—such as cancerous B cells or HIV-infected cells. In this way, CAR technology, which uses gene editing to mix and match the DNA encoding the two parts, can make diseased cells “visible” to a patient’s immune system by teaming an antibody with a signal.

The applications of gene editing to leukemias use TALENS and zinc finger nucleases, because the research began before development of CRISPR-Cas9. Like the idea of gene editing itself, the idea to arm T cells to attack cancerous B cells arose in the late 1980s, from Zelig Eshhar, PhD, and colleagues at the Weizmann Institute in Israel (Gross et al. 1989: 10024).

The first case of gene editing to treat leukemia made headlines in January 2013, with the story of 6-year-old Emma Whitehead in the US. She had been near death when she received the CAR treatment 9 months earlier. Zinc finger nucleases were used to fashion a billion CAR T cells, engineered to bind an antigen called CD19 that festooned her cancerous B cells. By September 2013, Emma was well enough to return to school, and the effect has persisted so far (Maude et al. 2014: 1507). Three years out, she was “doing fantastic. This year so far she has met President Obama, Tom Brokaw, and Heidi Klum!” said Bruce Levine, PhD, the Barbara and Edward Netter Professor in Cancer Gene Therapy at the Perelman School of Medicine and a pioneer of CAR technology, who cared for Emma.

Emma’s own cancer cells were targeted to make them more visible to her immune system. A few years later, another little girl with leukemia was treated with gene editing using TALENS. She received doctored *donor* cells—a hugely important step for making the technology available to many, especially patients too sick to spare their own cells to develop the treatment. That was the case for the second patient, Layla Richards, who was born in June 2014 and diagnosed with very aggressive acute lymphoblastic leukemia at 14 weeks. Chemotherapy and a bone marrow stem cell transplant could not stop her cancer cells, so just before her first birthday, Waseem Qasim, MD, Consultant Immunologist at Great Ormond Street Hospital in London and Professor of Cell and Gene Therapy at University College London, asked the company developing the treatment, Cellectis, for a special license to try its experimental

product on his very sick patient. The company agreed, and Layla received the only vial of the cells that the hospital had.

The infusion took just minutes to administer, which the child didn't even notice, said her dad. TALEN enzymes had snipped out the genes in the donor cells whose activity would have alerted Layla's immune system to reject the cells, and at the same time rendered the incoming cells resistant to a powerful chemotherapy, which could then be used to mop up any lurking cancer cells. So the treatment had a dual effect on the little girl's immune system: keeping rejection at bay while activating a response against the cancer cells.

Two weeks later, Layla developed a mild rash as her immune system began to respond to the introduced cells. Over the next two weeks, fixed T cells showed up in her circulation, and their numbers climbed, there and in her bone marrow too. At the 2-month mark, declared "cancer-free," Layla received another bone marrow transplant to replace her immune system, and a month later, she went home. ("Cancer-free" means cancer cells are undetectable, not necessarily that they are all gone.) Dr. Qasim said:

We have only used this treatment on one very strong little girl, and we have to be cautious about claiming that this will be a suitable treatment option for all children. But, this is a landmark in the use of new gene engineering technology and the effects for this child have been staggering.

MALARIA AND GENE DRIVES

Gene and genome editing can be used to render harmless "vector" organisms that spread infectious diseases to humans. These include deer ticks that transmit the bacteria that cause Lyme disease, snails that harbor the parasitic worms that cause schistosomiasis, and perhaps most familiar, mosquitoes. *Aedes aegypti* mosquitoes spread yellow fever and dengue, *Anopheles* species spread malaria, and *Aedes aegypti* and *Aedes albopictus* carry zika virus disease.

The idea to use genetics to address vector-borne diseases isn't new. In 1940, Soviet geneticist Aleksander Serebrovskii proposed intentionally using chromosomal translocations, in which different chromosome types switch parts, to control population sizes. Translocations interfere with meiosis (sex cell formation) in a way that

makes about half of a population sterile. Much more recently researchers have engineered transgenic mosquitoes, which have a selected genetic change in every cell. But transgenic mosquitoes are very expensive to breed, and can target only one gene at a time. CRISPR-Cas9 is faster, more specific, and capable of multiplexing so it can hit several targets in a mosquito genome. TALENs and ZFNs have two limitations. They are too big to easily get into insect genomes on a population scale, and they introduce DNA repeats that can cause errors in how the DNA-cutting enzymes function.

To change or eliminate a mosquito vector population, genome editing introduces a gene that makes the insect unable to carry the infectious virus or organism, plus a second genetic factor that gives a reproductive advantage to the insects that have the alteration. Replacing a specific genotype *en masse* like this is called a **gene drive**. One version of this strategy utilizes a natural phenomenon called “homing,” in which the introduced change is copied onto the partner chromosome, where it replaces the existing allele. As a result, all offspring inherit the change, not just the half that do so according to Mendel’s law of segregation. Traits that can be engineered into disease vectors and cause gene drives can do so by:

- biasing the sex ratio or shortening life so that the vector population dwindles and eventually disappears;
- blocking the vector’s ability to transfer the pathogen to the host;
- directly blocking the pathogen such as by producing a poison.

Malaria, which kills more than a million people a year, illustrates how gene editing works. The protozoan parasite *Plasmodium falciparum* spends about 2 weeks in the midgut of an *Anopheles* mosquito, then goes to the salivary glands, from which it travels to a human via a bite. However, introducing a mutation in a gene that is part of a cell signaling pathway called Akt disrupts development of the mosquito so that 75–99 percent of the insects cannot support the parasite. Insects that still carry the parasite tend to have fewer of them for shorter times. That is the situation for mosquitoes that are heterozygotes, having one copy of an *Akt* mutation. Using “homing” creates mosquitoes that are homozygous for the mutation, armed with two copies, and they do not carry any parasites at all. The situation is a pest control trifecta: fewer mosquitoes overall, fewer

parasites per infected mosquito, and a greatly shortened period of infectivity (Hammond et al. 2016: 78).

A different approach to modifying mosquito populations uses CRISPR-Cas9 to disrupt three genes that make females sterile, using a different guide RNA to each gene. This intervention generates females that are heterozygous for the three hits, each of which is recessive. When the females mate, more than 92 percent of their offspring are sterile, and that is too few to effectively spread malaria. Instead of creating a mosquito population that cannot support the parasite, this approach simply wipes out selected groups of mosquitoes.

Use of gene and genome editing and other DNA-based technologies to control infectious disease vectors is very different from applications in Mendelian diseases and cancer in humans, because of the involvement of the environment. While eliminating disease-spreading mosquitoes is beneficial and may reduce reliance on pesticides and herbicides, several issues come into play. Will the approach cause extinctions? Will reducing the vector's ecological niche open up space to another, perhaps invasive or disease-carrying species? What are the unexpected ecological consequences? Can the engineered change spread to other species? Can the approach be subverted as a bioterror weapon?

Modifying DNA using gene therapy and gene and genome editing promises new types of treatments that might even provide cures by correcting the source of a disease. Yet at the same time, such technologies are the subject of heated discussions when they are used on the first cell of an organism, creating an individual with a selected genetic change effectively knit into each of its cells.

WHAT WOULD YOU DO?

Fear of novel biotechnologies is not new. Louise Joy Brown, the first “test-tube” baby, made headlines in 1980 and was regarded as somewhat of a freak until it became clear that she was a normal little girl. More than a million people have been born following *in vitro* fertilization (IVF) since Louise hit the headlines. A decade later, the first couples to select disease-free embryos after IVF using preimplantation

genetic diagnosis (PGD) were harshly criticized. When recombinant DNA technology was first discussed in 1975, it, too, was feared, but went on to become the source of dozens of life-saving medicines. In fact, some of the leaders of the recombinant DNA movement in the 1970s are today involved in easing public fears of genome editing.

The biggest fear, particularly about CRISPR-Cas9 because it is so easy to do, is altering the human germline. The term refers to changing the genetic material of a fertilized ovum so that the change is perpetuated in every cell of the new organism. This is a much different scenario than using gene therapy or gene/genome editing to alter only the cells of the body part affected in a particular disease, such as blood or muscle.

Altering the human germline is possible, but many researchers have agreed not to attempt it. Monkey germlines were altered in 2014. In 2015, researchers from Sun Yat-sen University used CRISPR-Cas9 on fertilized human ova that intentionally had too much genetic material to divide beyond a few days (Liang et al. 2015: 363). The attempt was inefficient and led to “off-target effects” when the enzymes cut where they were not intended to, but the highly controversial experiment demonstrated that it is possible to edit the human germline. Francis Collins, MD, PhD, director of the National Institutes of Health in the US, said in response to the paper from China, “The concept of altering the human germline in embryos for clinical purposes has been debated over many years from many different perspectives, and has been viewed almost universally as a line that should not be crossed.” But it has been crossed.

Besides editing the human germline just to see if it is possible, doing so is not practical. It is easier to use the existing technologies of PGD and IVF to select mutation-free embryos. Bioethicists and others are having intense debates over human germline modification.

Do you think that human germline modification should be attempted, and if so, in which situations? If the consensus among researchers is not to do it, should those who do so be punished? What are the possible dangers of altering the human germline?

SUGGESTED READING

The Forever Fix: Gene Therapy and the Boy Who Saved It, by Ricki Lewis, traces the field's beginnings as an idea in the 1950s until current clinical trials (2012, St. Martin's Press). A short version, "Gene Therapy's Second Act," appears in the March 2014 issue of *Scientific American*. Many of her DNA Science blog posts at the Public Library of Science (<http://blogs.plos.org/dnascience/>) update the progress of gene therapy. "Biology and Applications of CRISPR Systems: Harnessing Nature's Toolbox for Genome Engineering," by A.V. Wright, J.K. Nuñez, and Jennifer Doudna (one of the inventors of CRISPR-Cas9), discusses the promise of the technology (2016, *Cell* 164(1–2): 29–44). Alison Abbott wrote "The quiet Revolutionary: How the Co-discovery of CRISPR Explosively Changed Emmanuelle Charpentier's life" (2016, *Nature* 532: 432–434). Ricki Lewis interviewed both Drs. Doudna and Charpentier in "A Conversation with CRISPR-Cas9 Inventors Charpentier and Doudna" at <http://blogs.plos.org/dnascience/2015/12/03/a-conversation-with-crispr-cas9-inventors-charpentier-and-doudna/>.

FROM ANCESTRY TO DESTINY

We are all, at the core of our genetic selves, African. Anatomically modern humans have lived in Africa for at least 200,000 years, leaving tantalizing but scant clues in the form of bone fragments, footprints, stone tools, and evidence of tool use. Linguistics provides other clues, because languages followed some of our ancestors as they left Africa, in trickles and surges, to settle the rest of the world. At the same time, human evolution continued in Africa.

DNA AS CLOCK AND COMPASS

The ability to understand where groups of present-day people came from relied first on human remains and the objects found near them—shards of a skull, images painted on cave walls, chipped rock axes. The 1970s brought the ability to sequence DNA. When researchers realized that a DNA sequence mutates at a known rate, they also realized that the changes in sequence over time could be used as a molecular clock. Following time using a DNA-based clock would lessen reliance on artifacts and preserved body parts, because the vestiges of mutational events from the past remain in the genomes of people living today.

A DNA clock considers each DNA base in a gene sequence as a bit of information. The more of a gene's DNA sequence that two

present-day people share, the more closely related they are. Put another way, the more likely it is that they shared a recent ancestor. An alternative explanation for genetic similarity is that two individuals have many DNA sequences in common simply by chance. That is much less likely than that they inherited their identical genotypes from a shared ancestor.

Genetic ancestry, the subject of this chapter, is an area of genetics that seeks similarities at the DNA level, and interprets them to reconstruct our origins and identify present-day relatives. Geography is important in making sense of our genetic diversity. Our distribution on the planet has had a profound effect on our genetic ancestry, establishing distinct variations of the human genome in people who remain in the same area for many generations. Such groups that have children only among themselves and retain a unique culture in one place over long periods of time are termed “indigenous.” The partitioning of collections of gene variants in indigenous populations is dissipating and blurring right now, as human gene pools mix as people travel and meet others or are visited by others. **Gene pool** refers to all of the gene variants in a population; it is not a characteristic of an individual. Before the echoes from our past fade as genomes homogenize, researchers—in a sense genetic archeologists—are comparing the DNA sequences of modern people to discover where we all likely came from.

Information about our past can be deduced from how our DNA sequences vary at specific places in the 3.2 billion building blocks of a human genome. Those sequence differences become meaningful when they are plotted as tree diagrams, with new mutations depicted as branchpoints. Once a mutation occurs, it is presumably passed to descendants.

Groups of mutations come to define specific branches, or lineages, of human family trees. Genetic sleuths have meticulously tracked these inferred branchpoints by comparing DNA sequences among diverse people living today and have reconstructed the major lineages of humankind. The same type of tree diagram can go even farther back in time to show relationships among species. **Hominoids** are animals that are ancestral to apes and humans only. **Hominins** are ancestral to humans only. That is, a hominin appears on the branch of the evolutionary tree that gave rise to humans only, after the split from the most recent common ancestor of us and the chimps and

bonobos, which occurred about 6 to 7 million years ago. Cartoons depicting humans following chimps in a line are not accurate. Our two species branched from a shared ancestor. In fact, because fossil evidence is so rare and we have no genetic material to analyze from millions of years ago, evolutionary trees are actually very bushy—we are missing a great deal of information about species that may have formed and then died out, leaving no evidence of their existence. Evolutionary tree diagrams that indicate species divergences are based on the same concept as diagrams that depict the relationships of family members—shared ancestry (Figure 6.1).

In 2005, the National Geographic Society began the Human Genographic Project, which is sampling DNA from people all over the world. The project started with indigenous peoples who, by definition, have remained in their homelands for many thousands of years. The portals to the deepest past lie in them, because their unique combinations of DNA sequences have not mixed with those from other groups. However, their numbers are plummeting.

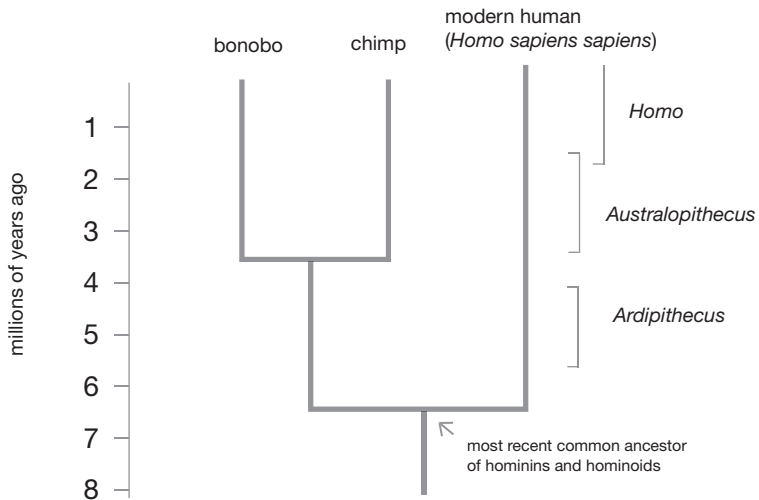


Figure 6.1 An evolutionary tree diagram depicts hypothetical branching points that indicate the last common (most recent shared) ancestor of two groups. This illustration omits other apes. Figure 6.2 illustrates one possible tangled relationship among the archaic humans (Neandertals and Denisovans) and us.

Of today's world population of about 7.5 billion, indigenous peoples number only about 370 million. The groups are scattered among 5,000 groups, and are on all continents in 90 nations (Gracey and King 2009: 65). The Genographic Project has since expanded to include anyone wishing to send in a DNA sample and a low fee, with the goal of tracking at least 100,000 modern human genomes. Recent efforts have analyzed DNA from:

- ancient skeletons discovered in Patagonia, at the southern tip of Chile, from about 12,000 years ago;
- men who migrated from southeast Asia to India about 8,000 years ago;
- Roma (gypsy) people who spread from southeast Asia to Iran, Armenia, Turkey, and the Balkans before reaching Europe, where they became widespread.

Until we had the ability to compare single nucleotide polymorphisms (SNPs) and other markers such as copy number variants across genomes, most of what we knew about genetic ancestry came from two easy-to-study parts of the genome—the Y chromosome, found only in males, and the DNA sequences in mitochondria, which are passed from women to all of their offspring. Y chromosome DNA sequences are used to trace male ancestry, and mitochondrial DNA sequences are used to trace female ancestry. Today researchers have information from thousands of complete human genome sequences, including the autosomes, to flesh out data from Y chromosomes and mitochondrial DNA that have been compiled for years. Later in the chapter, we consider compelling genome-wide information on the people of the British Isles that aligns perfectly with history.

The Y chromosome and mitochondrial DNA sequences reveal only two of the many lineages that contribute to the genome of any one individual. To envision this, consider that you have 4 grandparents, but 8 great-grandparents, 16 great-great-grand-parents, and so on. The farther back you go, the less any one ancestor contributed to your genome. This concept becomes important when considering that some modern population groups have low percentages of ancestors who were archaic humans such as Neandertals—which is not the same as having a low percentage of their genome sequences from Neandertals.

DNA sequence comparisons usually are consistent with evidence from archeology and fossil evidence from paleontology. Clues from these very different sources converge with our oldest “modern” ancestors, who dwelled in Africa about 170,000 years ago. But the full story begins much earlier, with our closest non-human relatives. For them, although we lack DNA evidence, the few clues that we have provide a context to consider what transpired to make us human. Table 6.1 summarizes major stops on the road to humanity, discussed in the next sections.

Table 6.1 Timeline of humanity

<i>Years ago</i>	<i>Major events</i>
6–7 million	Common ancestor of chimp and humans lived in Africa
5.8 million	<i>Ardipithecus</i> dwelled in Africa
2–4 million	Several species of <i>Australopithecus</i> lived in Africa
2.5 million	<i>Homo</i> emerged in and spread beyond Africa
2 million	Human-like characteristics arose and persisted
254,000	Most recent common male ancestor (Y haplogroup)
200,000	Modern humans in Africa; Neandertals left Africa for Europe
170,000	Mitochondrial Eve (most recent common female ancestor)
156,000	Herto people lived in Ethiopia
100,000	Bones in caves in Israel and China from modern people. Denisovans, Neandertals, and others interbred
50,000–55,000	Y chromosomes exploded in diversity, indicating widespread migration and population growth
45,000–50,000	Land bridges arose. People left Africa for southern India, Australia, New Guinea, Middle East
40,000	Oase1 fossil indicates 6–9 percent Neandertal ancestry
35,000	14 ancestral populations lived in Africa, where Bantu-speakers replaced click-speakers. People migrated from Asia to Europe
25,000–15,000	People reached Siberia, escaping ice age in areas called “refugia”
10,000–15,000	Agriculture arose and spread. Native American gene pool established in North America

THE FIRST HOMININS

Between 6 and 7 million years ago, among the dozen or more types of ape that dwelled on the African continent, lived chimp-like animals that were distant ancestors of humans. Over the millennia, mutations happened. Such genetic changes that altered appearance, behavior, or perhaps disease resistance in ways that improved the odds of survival in the particular environment enabled their owners to pass on the helpful gene variants. In this way, over time, the populations of chimp-like animals diversified, perhaps in different ways in different places in response to natural selection in the context of specific environmental challenges. We have glimpses of hominins who preceded us from fossilized remains found in an area of Ethiopia called the Afar Triangle, where the Red Sea, Gulf of Aden, and Main Ethiopian Rifts meet. Since 1981, Tim White, a paleontologist from the University of California, Berkeley, has led teams of students and scientists to the area to meticulously unearth pre-human specimens and the species that lived with them. The area is a mile-deep triangle of roughly 500 kilometers (310 miles) on a side, some 225 kilometers (140 miles) northeast of Addis Ababa, near the Awash River.

The oldest remains of human ancestors from the Awash River valley are from 4.4 million years ago, according to radioactive dating of rocks from a nearby volcano. These fossils, of animals named *Ardipithecus ramidus*, may have been the first hominin (White et al. 2009: 75). One partial skeleton was from a woman who stood about 120 centimeters tall and weighed about 50 kilograms. The shapes and enamel thickness of the teeth suggest a diet rich in fruit, consistent with the remains of figs and berries in the area. Her hand and foot bones indicate that she stood erect and used her palms to traverse the trees. Males and females were about the same size, and the skull was about the same size as that of a chimpanzee. They lived in the woodlands and perhaps visited scraps of nearby forest, but *not* in the savannah (grasslands), where later hominins lived.

Fossil evidence is more abundant from about 4 million years ago, when *Australopithecus* (also called australopithecines) lived. They were a little less chimp-like than *Ardipithecus*, yet clearly more simian than modern humans. For a few million years, Africa was home to at least five species of these australopithecines, some overlapping in time. Because they could not travel far, their

populations remained isolated and genetically distinct. But the traits that would carve humanity from those long-ago gene pools were already beginning to emerge.

For the australopithecines, an increasingly erect stance made life on the open grasslands possible. A larger brain fueled the cooperative and cognitive skills that would one day vastly improve nutrition, even permitting the trial-and-error tastings of plants that would become the first medicines. Rarely, an accident would preserve a scene from this distant past. A flash flood 3.3 million years ago, in what is now Ethiopia, buried a young girl, leaving intact a skeleton revealing long legs, like those of a human, topped by a stockier upper body reminiscent of a chimp. Her head was larger than that of a chimp, her toes indicated she walked mostly erect, and her teeth were a mix of those of her ancestors (Johanson and White 1979: 321).

By about 2.5 million years ago, the australopithecines were still hunting and gathering, but some of them had become different enough for scientists to give them their own name—*Homo* (Asfaw et al. 2002: 317). The transition from *Australopithecus* to *Homo* is murky in the fossil record, but, generally, the newcomers exhibited more sophisticated tool use, family structure, use of fire, and better control of the environment. The australopithecines faded from the fossil record as *Homo* debuted and thrived. Tim White envisions *Homo's* emergence by looking at the fossils that festoon his lab and office:

It all started about 2.5 million years ago, when guys started banging rocks together. That allowed the niche to expand, and the start of culture. Tool making, utilizing stone, probably began in *Australopithecus*. These bipeds were small-brained, and they weren't busy becoming human, but being australopithecines. At some point, a population of that highly intelligent, bipedal generalist—*Australopithecus*—began to exhibit behaviors that we see in the chimps, which hunt monkeys but lack tools. That particular sect of early hominin that didn't lack those tools anymore formed the beginning of the lineage that would ultimately diverge from other australopithecines that kept on being australopithecines, and that lineage would go on to become early *Homo*.

Researchers are uncertain whether to classify *Homo* groups from different parts of Africa as the same or different species. Classifications

can change as fossils are discovered. *Homo erectus* came closest in time to *Homo sapiens*, which includes ourselves and the archaic humans (Neandertals and **Denisovans**). *Homo erectus* lived from about 1.9 million years ago to as recently as 35,000 years ago. We know that they could migrate because their fossils have been found beyond Africa, in Java, Europe, China, and Southeast Asia. The positioning of their fossilized remains indicate that *H. erectus* could use tools and fire, cooperate, and maintain social relationships, and preserved jaws and teeth suggest that they were carnivores. They lived in families of male–female pairs (most primates have harems), and females nurtured offspring while males hunted. They may have had some form of verbal communication, because they were the first hominins to have an angled skull base that would have made it possible to produce more varied sounds.

About 2 million years ago, some of the qualities that make us human began to appear and, when helpful, persist. Vocalization became language. Walking improved. Stronger necks, larger leg muscles, and bigger butts made us born to run, easing migration and improving diet as hunting skills sharpened. A mutation in the gene encoding a type of muscle protein called “room for thought” shrunk the jaw, enabling the brain to expand (Stedman et al. 2004: 415). Perhaps the gene duplications that today distinguish our genomes from those of our closest non-human relatives appeared at this time, providing a new canvas for the gene variants to arise that would propel our abilities to remember and learn.

DNA EVIDENCE OVERLAPS FOSSIL EVIDENCE

Paleontologists date fossils according to location among rock layers, coupled with chemical clues, such as the timetable of the breakdown of a radioactive isotope into a different element. Genetic genealogists, on the other hand, look from the present back in time, using DNA sequences as their guideposts rather than fossils and footprints. The two approaches are most exciting when they tell the same story. That has happened in the tale of our “deep ancestry”—“deep” compared with traditional genealogy that uses family lore, documents, and photographs, yet shallow when compared with our legacy from an ancestor shared with chimps, as shown in Figure 6.1.

Fossil evidence meets, or at least overlaps, genetic evidence about 160,000 years ago. In 1997, Tim White's team discovered pieces of three human-like skulls sticking out of the dried mud roads of a temporarily abandoned town in Ethiopia, Herto, where floodwaters had chased away the living residents (White et al. 2003: 742). Said White, "What to a nonpaleontologist might have looked like a cow pie, to us was a fossil hominin cranium." It took years to reassemble the skull fragments into something resembling the original heads, and to use potassium-argon dating to arrive at the approximate time of 156,000 years ago. These people, which the researchers named *Homo sapiens idaltu* (*idaltu* means "elder" in the local language), had large brains, faces, and teeth, and probably resembled Australian aborigines.

The Herto fossils are important for what they lack as well as for what they have. The skulls were found alone and undamaged—the rest of the bodies were nowhere to be seen. This argues against cannibalism, which would have damaged the skull. It may even be evidence of mortuary practice or reverence for the skull minus the torso and limbs, suggesting culture.

The idea of the Herto people being the first, or one of the first, modern humans dovetails nicely with the idea of a "mitochondrial Eve," the term used to denote the most recent female ancestor that all of us living today share. This is where DNA sequences enter the picture.

In the mid-1980s, another Berkeley researcher, graduate student Rebecca Cann, working with mentors Allan Wilson and Mark Stoneking, had the revolutionary idea to use as a clock mitochondrial DNA (mtDNA), which mutates fast (Cann et al. 1987: 31). They would look at differences in the mtDNA sequences of as diverse a group of modern people as they could assemble. As they did so, it quickly became clear that the most diverse present-day human genomes come from Africa. Genomes from elsewhere include parts of that ancestral base set of DNA variants, as well as other mutations that occurred later, marking migratory paths when superimposed on geographical maps.

Recall that mitochondria in cells house the chemical reactions that extract energy from food nutrients. The thousands of mitochondria in each cell have their own DNA, where Cann and her colleagues looked for clues to the past. Because she knew the mutation rate

of key mitochondrial genes, Cann could note the DNA sequence differences among African genomes and count backwards to approximate when the genomes diverged. This extrapolation placed the time of emergence of our most recent common maternal ancestor at 100,000–200,000 years ago. Subsequent estimates from genetic anthropologist Sarah Tishkoff, then at the University of Maryland in College Park and today at the University of Pennsylvania, and her colleagues, added mtDNA data from hundreds more modern Africans (Tishkoff et al. 2003: 208). These data refined the date for the approximate start of modern humanity to about 170,000 years ago—an estimate remarkably close to the time of the Herto people, according to their remains.

Comparing Y chromosome sequences traces movements of males. Like the ease of following mitochondrial DNA sequences because the mitochondrial genome is small, the Y chromosome has few genes. Because the Y chromosome exists alone—a male cell has only one—the genes do not have anything to cross over with, and therefore the order of gene variants persists from one generation to the next. That is, they cannot be separated by crossovers. Such a group of tightly linked genes is called a **haplogroup**. A comparison of 456 Y chromosomes from all over the world dates the most recent common male ancestor to 254,000 years ago in Africa (Karmin et al. 2015: 459). Other major haplogroups emerged as males left Africa and natural selection molded their genomes in response to pressures introduced from new environments.

EXPANSION OUT OF AFRICA

Several species of australopithecines overlapped in time 2 to 4 million years ago in Africa largely because challenging geography and climate kept them apart. The earliest *Homo* species, too, were distinct, co-existing with and then outliving the australopithecines. About 400,000 years ago, *Homo*, and perhaps other groups who didn't leave evidence that has been discovered, began to exit Africa, reaching Europe and western Asia. Anatomically modern humans first appeared about 200,000 years ago, in Africa, Asia, and Europe.

People continued to leave Africa when the climate cooperated, traversing a temporarily wetter Sahara Desert to reach the colder north. We don't know why they went, but an ebb and flow of

pre-humanity probably pulsed out of Africa, with most surges squelched by extreme environments. Over the millennia, many small groups probably attempted to leave Africa, dying out along the way. Hard times in rough climes must have wiped out many a settlement, enabling only the hardiest, luckiest, or cleverest individuals to survive. The forces behind population change listed in Table 2.5 were at play: mutation, migration, genetic drift, non-random mating, and the most important, natural selection.

The earliest evidence we have of people leaving Africa is from about 100,000 years ago, in bones found in caves in Israel and a jawbone from a cave in southern China. We have no evidence of events over the next 30,000 years. A brutal ice age dawned about 60,000 years ago, but, by 50,000 years ago, as land bridges emerged, more people were able to leave Africa. They ultimately colonized southern India (at the time joined to Malaysia and Sri Lanka) and then Australia and New Guinea, which were one land mass. They reached Siberia by 25,000 years ago and over the ensuing 10,000 years, due to the isolation, their gene pool diverged from that of the ancestral Asians. Then some of them traveled south into the Americas over a land bridge that had been present since 28,000 years ago. The journey took 5,000 years. The descendants of these migrants established the gene pools of Native Americans (Tackney et al. 2015: 13833).

When the land bridges over which humanity had journeyed flooded, the distribution of surviving populations made it look as if people had somehow jumped from place to place, such as from Africa to Australia, and Asia to the Americas, rather than trekking across transient tongues of linking lands. Human skeletons found in New South Wales attest to this colonization, as do distinctive Y chromosome DNA sequences shared today between people from India and Australian aborigines.

By 50,000–45,000 years ago, others had left Africa, populating the Middle East. Meanwhile, in eastern Africa, as the forests gave way to savannahs, people spread out over the continent. By 35,000 years ago, 14 ancestral populations had become established in Africa. Gradually, Bantu-speaking people from the west began to replace older peoples who spoke “click” languages, such as the Khoisan and Pygmies, isolating them in pockets of the Sudan, South Africa, and Ethiopia.

People began to arrive in Europe from central Asia around 35,000 years ago. Some of them met and perhaps even mated with residents who had left Africa starting 200,000 years ago—the Neandertals.

FROM ARCHAIC HUMANS TO MODERN HUMANS

We distinguish ourselves, *Homo sapiens sapiens*, as anatomically modern humans, compared to other groups that are considered archaic humans. We have identified two types of archaic humans: **Neandertals** (*Homo sapiens neanderthalensis*) and **Denisovans** (*Homo sapiens denisova*). We are all subspecies.

Neandertals spread throughout Europe and Asia, while the Denisovans stayed mostly in Siberia and southeast Asia—at least according to evidence. The Denisovans predate the Neandertals, but evidence is so scant that the beginnings have yet to be deciphered, if they ever will be. Evidence so far suggests that the Neandertals diverged from the Denisovans about 466,000 years ago. However, we can deduce from shared DNA sequences that on occasion, more recently than 100,000 years ago, members of the groups met, encounters happened, and genomes mixed (Figure 6.2). The three subspecies of humanity share more than 99 percent of our DNA sequences, and certain gene variants that originated in either of the other two subspecies are present in some people today. Researchers are tracing shared DNA sequences to work out when and where interbreeding occurred. However, any conclusions are really only hypotheses. The story of our ancestry changes when DNA becomes available from newly discovered remains and when DNA is resequenced as technology improves.

NEANDERTALS

The Neandertals take their name from a limestone cave in Neander Valley, Germany, where quarry workers uncovered the first bones in 1856. That initial partial skeleton suggested that the Neandertals were stooped, large-boned human-like individuals with large jaws, jutting faces, and curiously tiny chins under large teeth. Their image improved somewhat with the discovery of bones from 10 Neandertals in a cave in Iraq, collected between 1957 and 1961. These bones

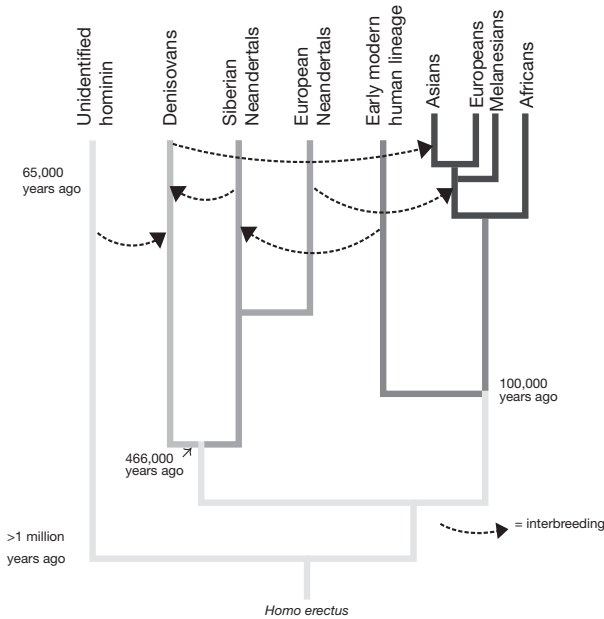


Figure 6.2 Comparison of genome sequences indicates that interbreeding took place among at least four subspecies of *Homo sapiens*. Darker branches are more recent.

lay among tools, and the fact that some of the bones were from 50-year-old individuals suggested that the Neandertals had been in enough control of their environment to live until what was then a very old age.

Unlike the fossilized bones from steaming hot Africa, so baked that the DNA degraded, some Neandertal bones have yielded DNA that remained intact in the cooler climate. In 1994, people exploring a cave, called El Sidrón, in northwestern Spain found bone fragments from 12 Neandertals, with preserved DNA within. By 2005, researchers had sequenced some of the mitochondrial genes from the bone fragments. Mitochondrial DNA does not degrade in the environment as rapidly as does DNA from the nucleus, and so it can be extracted from very old or damaged specimens.

The first full Neandertal genome sequence was published in 2010. The remains were discovered four years earlier in Vindija Cave in

Croatia (Green et al. 2010: 710). The researchers, from the Max Planck Institute for Evolutionary Anthropology in Leipzig, Germany, sequenced the genome from cells in a hip bone of a female who lived 38,000 years ago. They carefully removed a bit of material with dental drills so as to preserve the precious specimen for further analysis. In 2014 came a “higher coverage” more accurate Neandertal genome sequence from a female who lived 50,000–100,000 years ago in the Altai mountains of Siberia (Prüfer et al. 2014: 43).

The second genome sequence confirmed what the earlier genome had suggested—that Neandertals and the ancestors of modern humans had mated. In addition, the large number of homozygous sites in the genome of the ancient woman indicates that her parents had been half-siblings, and that consanguinity was common (relatives having children with relatives). Breeding with relatives may have been unavoidable when people lived in isolated small groups. The analyses so far indicate that limited “gene flow” (sex) occurred among our ancestors, Neandertals, and Denisovans, about 125,000 years ago, and perhaps others that we do not know about were included. Comparing archaic genome sequences to modern human DNA sequences also reveals gene variants that are unique to us.

The human and Neandertal genomes are 99.7 percent identical in DNA sequence. David Reich, a researcher from Harvard University and the Boston Evolutionary Genomics Supergroup, who has done much of the DNA research on the archaic humans, explains how our genomes can be nearly identical, yet some of us have parts of their genomes.

You get 50 percent of your ancestry from your mother, and 50 percent from your father. And yet you are identical in all but 0.000001 percent of your DNA to each of them. The statement we are making is about what proportion of one’s ancestors, going back in time say 100,000 years ago, were Neandertals.

(Sankararaman et al. 2016)

People of African ancestry living today do not have any ancestors who were Neandertals, because African ancestry precedes the exit of Neandertals from Africa. For Europeans, however, 1.5 percent of their ancestors were Neandertals, and for Asians, approximately 1.8 percent were Neandertals (Wall et al. 2013: 199).

DNA sequencing of remains from a 40,000-year-old human from Romania called “Oase1” indicates that he had 6 to 9 percent Neandertal ancestors from about 100,000 years ago (Fu et al. 2015: 216). He was therefore intermediate in degree of ancestry between Neandertals and us. The pieces of Oase1’s genome that descended from Neandertals are so extensive that researchers estimate the Neandertal genes entered just four to six generations previously. Entry of DNA sequences into the genome of a different species or subspecies is termed **introgression**.

It is tempting to try to imagine what might have happened between bands of Neandertals that met up with small groups of our direct ancestors, and how a few Neandertal genome regions persisted into some modern human genomes. Comparing genome regions provides compelling clues. For example, the Neandertal Y chromosome is distinct from those of modern humans, and is about twice as old as the oldest known modern Y chromosomes, those of the Mbo people from the Bantu tribe in Cameroon (Mendez et al. 2016: 728). The observation that variants of at least three genes on the Neandertal Y function in the immune system suggests one reason why our genomes aren’t more blended. Perhaps these gene variants coming in with Neandertal sperm were incompatible with a female *Homo sapiens sapiens*’ immune response to being pregnant, and caused her to miscarry. The representation of the Neandertal genome in our own genomes is so small that until Neandertal genomes were sequenced, fossil evidence alone suggested that the two subspecies did not mate, or if they did, no offspring survived.

Researchers are analyzing specific genes that have different variants in modern human genomes and from the few Neandertal genomes that have been sequenced. In one clever approach, researchers consulted a database that includes the genome sequences and electronic medical records and disease diagnostic codes for 28,416 patients. These data were compared to 6,000 Neandertal haplotypes (groups of linked genes). Were any modern disease diagnoses associated with specific Neandertal gene variants? In fact, a dozen Neandertal gene variants seem to affect our health. They are quite varied, and include:

- urinary tract disorders;
- three genes that direct an immune response against fungi, bacteria, and parasites;

- a gene variant that increases risk of developing precancerous skin lesions;
- vitamin B1 transport;
- a gene variant that accelerates blood clotting. Today this increases risk of stroke and heart attack, but may have protected Neandertals from injury;
- tobacco addiction;
- depression.

It is possible that gene variants that helped Neandertals today harm us (Gibbons 2016). For example, the gene variant that accelerates blood clotting may have enabled Neanderthals to survive injury. In today's calmer world, fast clotting blood increases the risk of stroke and heart attack.

DENISOVANS

Several Neandertals have had their genomes sequenced, as have a few Denisovans. We have fairly complete Neandertal skeletons, enough to make detailed museum reconstructions. Our knowledge of the Denisovans, however, comes only from a piece of a pinky finger and two teeth (albeit large ones!).

The first Denisovan fossil, the finger, was discovered in Denisova Cave in the Altai Mountains of southern Siberia in 2008. At first, paleontologists assumed the remains came from Neandertals, who had also occupied the cave. However, by 2010, geneticists had sequenced the mitochondrial DNA from the finger, and it indicated something that was not a Neanderthal, and not a modern human (Reich et al. 2010: 1053). Within another two years, researchers at the Max Planck Institute for Evolutionary Anthropology had sequenced the genomes of cells from two molars from other Denisovan individuals found in different rock layers, indicating that they had been left at different times (Meyer et al. 2012: 222). The differences between the genomes from the teeth suggest that the Denisovans had lived long enough to successfully reproduce and become somewhat genetically diverse.

The pinky bone came from a young girl or woman, whom the researchers named Denise. She lived between 32,000 to 50,000 years ago, according to both DNA and fossil evidence, and had brown eyes and hair and dark skin, according to DNA evidence alone. The closest

living relatives of the Denisovans appear to be people from Northern Island in Melanesia, in Papua New Guinea (Vernot et al. 2016). Genome sequences suggest that about 3.5 percent of their ancestors were Denisovans, going back about 100,000 years. Some Polynesians and aboriginal Australians also have partial Denisovan ancestry. These DNA signposts in modern genomes of people whose families have not moved indicate where the Denisovans lived.

The Denisovan cave, where the fossils are from, must have been a busy place, at least at some times over the millennia. In addition to the Denisovans, the cave was home to a group we call the Altai Neandertals. Their genomes have many “runs of homozygosity,” places where the DNA sequence is identical on both copies of the chromosome. This pattern indicates long-ago extreme consanguinity, with siblings becoming parents of the same offspring, repeatedly. There is even evidence of cannibalism between neighboring Neandertal groups, in the Denisovan cave and among the El Sidrón Neandertals. Like consanguinity, cannibalism is abhorrent today but might have been a necessity to survival long ago.

The Neandertals and Denisovans may represent the tip of a very large iceberg, for we can only investigate the DNA of peoples for whom we have actual remains, with intact DNA. Despite these limitations, a picture is emerging of bands of peoples, archaics or moderns depending on what we call them, who met and mingled over a span of at least 100,000 years in Asia and Europe (Figure 6.2). Comparing the genomes of the two archaic humans that we know about to our own reveals that the Neandertals and Denisovans were more closely related to each other than either was to us, and that Neanderthal genes joined older Denisovan genomes. In addition, the Denisovan genomes indicate that they bred with at least one other type of archaic human. Genes also flowed from our ancestors who were neither Denisovan nor Neandertal into their genomes (Kuhlwilm et al. 2016). Our efforts to reconstruct what happened from so few clues is a little like trying to understand the plot of a novel from reading only a few scattered sentences.

INTO EUROPE

Following Y chromosome DNA sequences reveals the migrations that eventually populated Europe, at least of the males. Cataloging Y

chromosome haplogroups among modern men enables researchers to deduce where the men came from. One investigator called the Y chromosome “a unique record of human history.”

Y chromosome haplotype analysis indicates that people who left Africa went to the Arabian Peninsula and then to south and southeast Asia, where genetic diversity increased. From there, about 55,000 to 50,000 years ago, males (perhaps with females and children too) began migrating into Europe. A sudden diversity in Y chromosome sequences dating to that time indicates widespread migration and population growth.

SURVIVING ICE AGES

By about 35,000 years ago, the ice descending on northern Europe forced the pockets of peoples living there to migrate toward the warmer and drier Mediterranean region. The areas they moved to are known as “refugia,” because they provided escape from the advancing glaciers and cold. By 16,000 years ago, people resumed their westward journeys in climate-driven migrations that paralleled similar episodic journeys in Africa. Again, the migrating Europeans found others already there, descendants of the people who had left Africa 20,000 years earlier via the Arabian Peninsula.

The little groups of people that emerged from the ice ages began to approach each other, and some even to coalesce, by about 10,000 years ago. Agriculture helped to fuse humanity, enabling people to stay in one place. As the changing climate came to favor grasses, people were learning how to collect, store, and plant seeds. Hunter-gatherers no longer had to hunt or gather. With a reliably static and consistent food source, populations grew explosively and spread. Communities began to mix and mingle, rather than replace each other. Genetic lines blurred. Still, small groups who had managed to survive the Ice Age in the Balkans, Iberia, and Italy left distinctive genetic markers. Today, the men of Europe fall into seven major groups, each defined by a distinctive Y chromosome (Karafet et al. 2008: 830). They represent the legacy of seven men and their relatives who ventured westward thousands of years ago and were fortunate, or savvy, enough to have left fertile descendants.

Over time, European human genomes diverged from African and other human genomes, in response to natural selection favoring gene

variants that best suited the European environment. This may be how skin lightened. Variants of a gene called *MC1R*, which encodes a cell-surface protein called the melanocortin 1 receptor, stimulate cells in the skin to produce a dark pigment called eumelanin. A mutation that occurred sometime, somewhere in Europe might have disabled the gene so that instead it produces a different pigment, pheomelanin, which confers lighter skin, freckles, and reddish-blond hair. As people reached northern latitudes and the generations passed, natural selection may have favored those with lighter skin because a fair complexion let in more ultraviolet radiation from sunlight, needed to manufacture vitamin D (Laleuza-Fox et al. 2007: 1453). These paler people didn't need the high eumelanin levels necessary in the tropics to protect against the higher exposure to ultraviolet radiation in the stronger sunlight. And so, in the north, people with lighter skin were better suited to their environment. People with *MC1R* mutations have double the risk of developing the skin cancer melanoma, so they wouldn't have survived the more intense sun of equatorial lands.

THE PEOPLING OF THE UNITED KINGDOM

British geneticist Sir Walter Bodmer of the University of Oxford led a brilliant study published in 2015, "the fine-scale genetic structure of the British population" (Leslie et al. 2015: 309). His team identified the DNA base at more than half a million places in the genome—SNPs—for 2039 "ordinary people" living in the UK today, in cities as well as rural areas. Each participant had four grandparents who had lived all their lives within 80 kilometers of the grandchild, whose genome therefore represented all four grandparents. The researchers analyzed how the genetic information alone, not considering geography, varied among the participants, and found that it fell into 17 very distinct ancestry clusters. Remarkably, the 17 clusters closely match what we know of the peopling of the UK from history—with a few surprises (Figure 6.3).

Three of the seventeen genetically defined groups represent the Orkney Islands. These are the people most different from all the others, which is consistent with the conquering and dominance of the Norse Vikings in this archipelago off northeastern Scotland from 875 AD until 1472 AD. Many of the gene variants in people living on the islands today are also found among modern Norwegians.

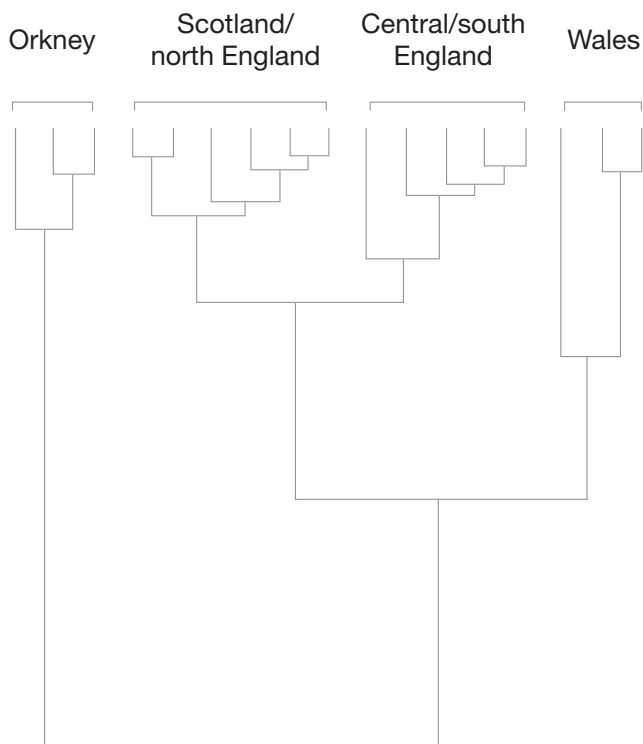


Figure 6.3 Genome-wide comparisons of single-base differences among 2,039 British people whose grandparents lived within 80 kilometers of them were used to build this evolutionary tree diagram showing 17 population “clusters.” The gene-derived tree tracks exactly with what is known about the history of the people of the British Isles. The Orkney Islands separated first when the Norse Vikings replaced native tribes. Then Wales split off, later establishing distinct groups in the north and south. Then England split, with the division to central and south England happening first, and then north England and Scotland.

Other genetic distinctions uncannily follow geographical borders, such as differences between neighbors in Cornwall and Devon, and in the English/Welsh borders compared to surrounding regions. The researchers also genotyped people from 10 other European nations,

noting those that account for 2.5 percent or more of the ancestors of modern people who live in the UK. This analysis found that the earliest migrations to still be represented to a large extent in modern genomes in the UK were from west Germany, Belgium, and north-west France, and less commonly but still significant, from southern France and Spain, Denmark, Norway, and Sweden.

In general, the study found that the UK slowly became populated as many physically isolated groups formed and grew, rather than from the sweeping migrations depicted in historical accounts. According to genomes, there was no single, dominant Celtic population in the non-Saxon parts of the UK, but instead, many clusters of Celtic settlements. Genetic data are slightly rewriting history in that the Saxons have apparently contributed 10 to 40 percent to ancestry in Central and South England, not the 50 percent based on evidence from language, pottery styles, and types of cereal crops. The Saxons originated from Germanic tribes living near the North Sea in late Roman times.

ÖTZI THE TYROLEAN ICEMAN

Y chromosome data indicate that from 5,900 to 4,800 years ago, people called the Yamnaya, from the Eurasian Steppe, moved westward and replaced many European groups. We have a remarkable peek at one man whose remains were preserved in the ice on a mountain more than 10,000 feet high 5,200 years ago, in the Ötztaler Alps of northern Italy. Hikers discovered Ötzi the Ice Man, also known as the Neolithic Tyrolean Iceman, in 1991. His DNA offered one set of clues, and his remains and circumstance, another.

Ötzi had a Y chromosome haplogroup that is today seen among men living in the Mediterranean area, specifically the Sardinians and Corsicans (Coia et al. 2016). His mitochondrial genome no longer exists, and probably died out about 5,000 years ago. Both his Y haplogroup and mitochondrial genome suggest that his ancestors had arrived in Europe about 8,000 years ago. The fact that his Y haplogroup is found elsewhere in Europe, but his mitochondrial genome died out, could mean that some men from his group left the isolated Alps, but the women bearing his mitochondrial genome died out in the local, very harsh environment. The rest of Ötzi's genome indicates that he had type O blood, he was lactose intolerant, and he might have

died of heart disease had he not landed in a ditch and been quickly covered with snow after a brutal fight.

The hikers discovered Ötzi clutching a knife. The ancient man had obviously been beaten up. He had cuts and bruises, and an arrowhead was embedded in his left shoulder, piercing a vital artery. The wound contained blood from two other people, and his cape held the blood of a third individual. Mosses adorning his body may have been wound dressings. Two smidgeons of brain tissue held blood-clotting proteins.

Ötzi was rather stylishly dressed. He wore leather suspenders, furry leggings, a loincloth, a fanny pack, a bearskin cap and cape, and snowshoes that resembled sandals. His ears had dents that suggested earrings, and his skin was stained, possibly tattooed. Mushrooms found in his satchel may have been used as antibiotics. The season was late summer or early fall, according to berries found nearby, and his last meal had been meat, ibex, and venison.

AFRICAN GENETIC DIVERSITY CONTINUES

The “out of Africa” movement populated the rest of the world, but humans continued to live and their genomes diverge in Africa. *Homo sapiens idaltu* from Ethiopia was one such group, from 156,000 years ago. At least one group of *Homo* that left Africa eventually came back.

African genetic diversity reverberates today in the great variety of skin tones, hair textures, facial features, heights, and body forms of modern Africans. Even people living in the same communities can have very different genomes, in the parts where people vary. Researchers sequenced the genome of Desmond Tutu, the civil rights activist and a Bantu (the majority population group), and that of a modern-day hunter-gatherer member of the Khoisan people, who live in the Kalahari Desert in Namibia (Schuster et al. 2010: 943). A tiny minority, the Khoisan are probably the most ancient people, genetically speaking, alive today. The genomes of Desmond Tutu and the Khoisan differed at 1.3 million single-base sites, and none of these sites had the same DNA base as several dozen other genomes sequenced, mostly from Europeans. The Khoisan had gene variants that reflect natural selection favoring adaptations to desert life (Table 6.2).

Table 6.2 Khoisan gene variants that show adaptation to life in the desert

<i>Gene</i>	<i>Adaptation</i>
<i>ACTN3</i>	Sprinting ability
<i>TAS2R38</i>	Ability to taste bitter substances, enabling detection of poisons in plants
<i>CLCNKB</i>	Chloride channel that conserves water
<i>DARC</i>	Absence of variant that protects against malaria, not needed because environment is not wet enough to sustain malaria mosquitoes

GENETIC GENEALOGY AND ANCESTRY TESTING

Most people who delve into their DNA to illuminate their ancestry are not interested in a mitochondrial Eve of 160,000 years ago, or a European ancestor from 50,000 years ago. They want to find cousins, living now or recently. “Genetic genealogy” uses the same techniques that evolutionary biologists use—tracking shared DNA sequence variants or numbers of short repeated sequences. The reasoning in finding a great-aunt is the same as tracking the early expansions from Africa—the more markers (DNA sequences) two people share, the more recently we infer that they shared an ancestor. The rarer the markers, the stronger the match. Note the resemblance of the evolutionary trees shown in the three figures in this chapter. A family tree stretching back only a few generations would look similar. Evolution is divergence from shared ancestors.

To satisfy curiosity about genetic ancestry, companies sell tests for mtDNA and Y chromosome DNA haplogroups, and SNPs on the autosomes, which serve as markers, or signposts, in the genome. The companies identify SNPs in the DNA in cells in a submitted sample, typically saliva or cheek scrapings, and search for population groups that match. They then send the customer geographical maps with color-coded references to major parts of the world where one’s ancestors might have lived. Usually, the information is not a surprise. Most people who order the tests are already in contact with others who share their surnames, memories, or information in documents such as birth, death, marriage, and baptismal certificates. However,

genetic genealogy data can be more quantitative than memories and oral histories. For example, two men with other similarities who share 65 of 67 Y chromosome markers are extremely likely to be close relatives. If they share only 54 of the 67 markers, then they shared an ancestor farther back. Their more distant match reflects input from genomes that they do not share.

Genetic genealogy has limitations. For example, the same mutation can arise in unrelated families by chance, leading to wrong conclusions unless other types of information back up the link. Testing several parts of the genome minimizes this risk by providing more points of comparison. Another reason that DNA evidence can be disappointing is that it identifies lineages, and not individuals, and often cannot pinpoint the exact generation in which a shared ancestor lived.

THE FUTURE: WILL HUMANITY HOMOGENIZE OR DIVERGE?

When our descendants 100,000 or more years from now look back, as we now look back at the Denisovans and Neandertals, what will they see?

If geographical separation, environmental conditions, and natural selection fueled our evolution in the distant past, what will happen in the future, now that we can easily travel anywhere? What will be the repercussions of our ability to treat diseases so that mutations that would have been culled in our ancestors remain in the gene pool, and to choose our children by glimpsing their genomes at various stages of prenatal development? Will our ability to eliminate other species using gene drives (see Chapter 5) have unforeseen effects on human populations?

Envisioning the future human gene pool has long piqued our imaginations (Table 6.3). Common themes and scenarios include:

- “dystopian” societies that choose who can reproduce;
- humans diverge into separate species;
- evolution rewinds and we come under the domination of apes;
- our species restarts from a select few individuals in the face of a population-whittling disaster such as a meteor strike, alien

Table 6.3 Future views of humanity

<i>Time</i>	<i>Novel/film</i>	<i>Plot</i>
soon	<i>GATTACA</i>	An “invalid” man, whom society keeps from fulfilling his dreams because he is destined to die young and is myopic, buys DNA from a “valid” individual, only to be exposed by the DNA in an errant eyelash.
2274	<i>Logan’s Run</i>	In a domed city following a global holocaust, a computer in control allows people to live until age 30. Sex is free, everyone is young and beautiful, but the protagonists escape and find an old man.
2540	<i>Brave New World</i>	In the World State, everyone lives a happily drugged existence, with no individuality or families. Five castes are gestated, their intelligence controlled by chemicals, not genes, in Hatcheries and Conditioning Centres.
3978	<i>Planet of the Apes</i>	Over centuries of people using apes for menial tasks and medical experimentation, people get lazy, and the apes smart. After DNA damage and intense selection by a nuclear war, humans go wild and become slaves of their hairy cousins.
4028	<i>Wayward Pines</i>	“Aberrations” (“Abbies”) have taken over the world. They are “violent humanoid predators” who eat the remaining humans. They resulted from mutations.
802,701	<i>The Time Machine</i>	A Time Traveler visits a time after humanity has split. The above-ground Eloi are blonde, thin, meek, placid, and unthinking. The below-ground Morlocks are dark, stocky, aggressive, and use technology to cultivate the Eloi as food.

attack, climate change, pandemic, exploding sun, deployment of a weapon of mass destruction, or a devastating war.

A book and TV series have even explored a world without people.

What will we look like thousands of years hence? Will we have mixed our species into a genomic homogeneity so that everyone has the same skin color and the same facial features? Will a uniform, global human gene pool obliterate the distinctive collections of gene variants that once defined so many ethnic populations? Or will our ideological and economic differences continue to separate us, eventually altering our genomes, as geography did so effectively in the distant past?

Joshua Akey, who leads one of the largest groups in the world that explores the Neandertals and Denisovans, at the University of Washington in Seattle, offered his view of the future, with the caveat that evolutionary change is not predictable.

Even relatively small amounts of migration is a potent force in breaking down population structure, so I suspect the populations of the future will be much different than today, and in general be more homogeneous. However, given that we occupy such a broad geographic range, there may always be some structure in patterns of global human diversity.

He also suggests that 100,000 years hence, perhaps we will have colonized Mars, “which would create new opportunities for genetic divergence to accumulate.”

In these days of division, terrorism, discrimination, and conflict, it can be calming to think back to that long-ago cave in southern Siberia where the seeds of our present selves took root—and to the future.

WHAT WOULD YOU DO? *EUGENICS*

Historically, we have objected to intentional, organized attempts to alter future gene pools, which is a practice called **eugenics**. In positive eugenics, government policies or subtler societal pressures encourage

people to reproduce if they have traits that are valued and thought to be inherited. Negative eugenics prevents people from reproducing if they have traits thought to be inherited and considered undesirable. The Nazi's efforts to eliminate the Jewish people and other groups during World War II is the classic example of a eugenic program, but it has happened in many times and places.

In 1910, the US kept meticulous records on institutionalized individuals who were deemed unfit to reproduce, at the Eugenics Record Office at Cold Spring Harbor, New York. Some of these helpless and unknowing people were sterilized. From 1934 until 1976, the government in Sweden forcibly sterilized people with mental illness, and then did the same to impoverished, unmarried mothers. Efforts from the women's movement in the 1970s finally ended the practice. In 1994, China passed the Maternal and Infant Health Care Law to outlaw reproduction if partners have a "genetic disease of a serious nature." However, the list of conditions included many that are not inherited.

Wars can have a eugenic effect if the battling populations have different gene variants. A conflict of very long duration can affect gene pools. In recent years in the Congo, Rwanda, and Darfur, soldiers have raped women with the intent to disseminate their own genes into the population that they take over. That might not be as methodical as creating concentration camps, but it is still an attempt to control reproduction in a way that favors individuals with particular genotypes.

Eugenics is sometimes confused with *genetics*, but the goals are different. The goals of medical genetics are to predict, prevent, diagnose, and treat diseases. But long-term consequences of medical genetics may mimic eugenics if they eliminate individuals with certain genotypes and encourage others to reproduce.

It is now possible to sequence the genome of a fetus. Will the ability of parents-to-be to learn the genetic make-up of their children-to-be have eugenic effects? Should governments control which inherited diseases or traits should be eliminated and which not? Return to Chapter 4 and identify possible eugenic effects of widespread use of any of the described genetic tests.

SUGGESTED READING

Many books provide fascinating views of what we know of our direct forebears. *Lucy: The Beginnings of Humankind*, by Donald Johanson and Maitland Edey (1990; Simon and Schuster) remains a classic look at the australopithecines. *Neanderthal Man: In Search of Lost Genomes* (2014; Basic Books) is by Svante Pääbo, the pre-eminent investigator of the archaic humans. DNA modifies what we know from history about the populations of Great Britain, discussed in *Four Flags: The Indigenous People of Great Britain: DNA, History and the Right to Existence of the Native Inhabitants of the British Isles*, by Arthur Kemp (2015; Ostara Publications), and in *Saxons, Vikings, and Celts: The Genetic Roots of Britain and Ireland*, by Bryan Sykes (2006; W.W. Norton). *Sapiens: A Brief History of Humankind* by Yuval Noah Harari (2015; HarperCollins) provides an historian's view of all who came before us.

GLOSSARY

Adenine: One of the four nitrogenous bases in DNA. Complementary base pairs to thymine (T).

Allele: A variant of a gene.

Amino acid: A protein building block.

Antibodies: Proteins produced in B cells and built of Y-shaped subunits that bind non-self antigens, signaling other parts of the immune response.

Antigen: A molecule, or part of one, that stimulates an immune response.

Autosomal dominant: Mode of inheritance in which an allele on a non-sex chromosome that causes the phenotype is present in one copy. May affect either sex.

Autosomal recessive: Mode of inheritance in which an allele on a non-sex chromosome that causes the phenotype must be present in two copies. May affect either sex.

Autosome: A chromosome that does not include a gene that determines sex.

Balancing selection: Maintenance of a mutant allele in a population because the carrier state has a survival or reproductive advantage over individuals who do not have the mutant allele.

Cell: The structural and functional unit of a living organism.

Cell cycle: The sequence of events that a cell undergoes, including division (mitosis) and the time spent in preparation for division (interphase).

Chimeric antigen receptor (CAR) technology: Targeting cancer cells with a laboratory-manufactured hybrid molecule consisting of part antibody and part T cell receptor.

Chromosome: A structure in a cell's nucleus, built of DNA and associated proteins, that contains the genes and appears rod-like during cell division. A human cell has 23 pairs of chromosomes.

Cline: An allele frequency that increases or decreases across a geographical area.

Codon: An RNA triplet that encodes a specific amino acid.

Cytokines: Protein products of T cells that have protective functions, including signaling bone marrow to manufacture lymphocytes, detecting virally infected cells, causing fever, and destroying bacterial toxins.

Cytoplasm: Gel-like substance in a cell, outside the nucleus, in which organelles are suspended or move.

Cytosine: One of the four nitrogenous bases in DNA. Complementary base pairs to guanine (G).

Denisovans: A subspecies of *Homo*; is a type of archaic human

Dihybrid cross: Breeding individuals who are heterozygous for the same two genes.

Diploid: A cell that has two copies of each chromosome type.

DNA: The genetic material, deoxyribonucleic acid. DNA is a long molecule built of sequences of four nitrogen-containing bases: adenine (A), guanine (G), cytosine (C), and thymine (T). The bases are held in pairs by two backbone structures, forming a double helix.

DNA microarray: A small glass or plastic chip that bears short DNA sequences, which serve as probes for complementary DNA sequences in patient samples.

Dominant: An allele that affects a phenotype in one copy.

Epigenetics: Changes to DNA that affect its expression, other than changing the base sequence.

Eugenics: Attempts to alter the future gene pool.

Exome: The part of the genome that encodes protein.

Expressivity: Variability in symptom severity among people with the same genotype.

Gene: A DNA sequence that encodes a protein's amino acid sequence.

Gene drive: Using a genetic change, such as genome editing, to eliminate a certain genotype in a generation.

Gene editing: Use of any of three major technologies (zinc fingers, TALENs, and CRISPR-Cas9) to add, delete, or exchange a gene into a genome.

Gene pool: All of the alleles in a population.

Gene therapy: Addition of a gene into a cell type that is affected in a particular genetic disease.

Genetic ancestry: Comparing DNA sequences to infer familial relationships.

Genetic drift: A subpopulation sequesters certain allele combinations from the larger group.

Genome: One set of the DNA information in a cell. A genome is species-specific and also varies among individuals of the same species to a small extent.

Genome editing: The use of DNA cutting technologies (zinc fingers, TALENs, and CRISPR-Cas9) to add, delete, or replace more than one gene in a genome.

Genome-wide association study: A comparison of many sites in the genome between two groups of individuals, who differ in an important way, but are matched in others. Used to identify genes that contribute to the risk of developing a particular condition.

Genotype: The specific alleles for a gene or genes in an individual.

Germline: Cells that give rise to a new individual; gametes (sperm and egg) and a fertilized ovum.

Guanine: One of the four nitrogenous bases in DNA. Complementary base pairs to cytosine (C).

Haplogroup: Marker sets on the Y chromosome or mitochondrial DNA, used to trace ancestry.

Haploid: A cell with a single set of chromosomes; egg or sperm.

Haplotype: The alleles on a segment of a chromosome, which tend to be inherited together as a unit.

Heterozygote: An individual with two different alleles of a gene.

Hominin: An individual on the branch of the evolutionary tree that gave rise to humans only. Also called hominid.

Hominoid: A species that is ancestral to apes and humans.

Homozygote: An individual with two identical alleles of a gene.

Independent assortment: The inheritance pattern of a gene on one chromosome does not affect the inheritance of a gene on another chromosome.

Introgression: Entry of DNA sequences into the genome from one species or subspecies into another.

Karyotypes: A chart of chromosome pairs ordered by size.

Liquid biopsy: Detecting cancer from DNA that cancer cells have shed into the bloodstream.

Lymphocyte: A type of white blood cell; a T or B cell.

Meiosis: Form of cell division that gives rise to sex cells (sperm or egg), halving their chromosome number.

Messenger RNA: Type of RNA that includes the information for a specific amino acid sequence.

Microbiome: The collection of microorganisms and viruses on and in a human body.

Mitosis: Form of cell division that maintains the chromosome number in somatic (body) cells.

Monohybrid cross: Breeding two individuals who are heterozygous for the same gene.

Mutant: An unusual or abnormal expression of a gene.

Mutation: A change in a DNA sequence, natural or induced.

Natural selection: The differential survival of individuals whose inherited trait variants better adapt them to reproduce successfully in a particular environment.

Neandertals: A subspecies of *Homo*; a type of archaic human.

Nucleotide: A DNA building block, consisting of a phosphate group, a sugar, and a nitrogenous base.

Nucleus: The genetic headquarters of a cell, which encloses DNA.

Oncogene: A gene that causes cancer when overexpressed.

Organelles: Structures in cells that carry out specific functions.

Pedigree: A diagram that uses squares (males), circles (females), and vertical and horizontal lines to depict the pattern of inheritance of a specific trait or illness in a family.

Penetrance: The percent of individuals with the same genotype who express the associated phenotype.

Phenotype: The expression of an allele combination (genotype).

Pleiotropy: The association of several symptoms with a genotype.

Protein: A long molecule built of amino acids that provides or contributes to an inherited trait.

Recessive: An allele that affects a phenotype when present in two copies.

Recombinant DNA technology: Placing DNA from an organism of one species into the genetic material of an organism of a different species, where the cell produces the encoded protein.

Ribosomal RNA: A component of a ribosome.

Ribosome: A structure in a cell built of RNA and protein molecules on which translation of mRNA into protein occurs.

RNA: Ribonucleic acid; an informational molecule that is an intermediate language between DNA and protein.

Segregation: Separation of two copies of a gene into separate gametes.

Sex chromosome: A chromosome that includes a gene that determines sex.

Single nucleotide polymorphism (SNP): A single-base site in the genome that differs in more than 1 percent of a population.

Thymine: One of the four nitrogenous bases in DNA. Complementary base pairs to adenine (A).

Telomere: A chromosome tip.

Transcription: RNA synthesis.

Transcription factor: A protein that turns on specific groups of genes.

Transfer RNA: Type of RNA that brings a specific amino acid to a particular mRNA codon (three-base sequence).

Translation: Protein synthesis.

Translocation: An event in which chromosomes exchange parts, or one chromosome attaches to another.

Tumor suppressor: A gene that, when deleted or inactivated, causes cancer.

Wild type: The most common allele in a population; the “normal” allele.

X-linked recessive: Mode of inheritance in which a male expresses a trait whose gene is on the X chromosome because he does not have a second allele to mask it; a female requires two copies of the allele to express the trait.

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